Abstract

Geometry of Anti-Poisson Involutions in the Deformations and Resolutions of Kleinian Singularities

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This dissertation investigates the geometry of anti-Poisson involutions in the deformations and resolutions of Kleinian singularities. It combines the author's previous work [Hu25] with ongoing research.

Kleinian singularities are quotients of $\mathbb{C}^2$ by finite subgroups of $\mathrm{SL}_2(\mathbb{C})$. They are in bijection with the simply-laced Dynkin diagrams via the McKay correspondence. Anti-Poisson involutions and their fixed point loci appear naturally in the classification of irreducible Harish-Chandra modules over Kleinian singularities. The dissertation addresses three main objectives:

(1) Classify anti-Poisson involutions of Kleinian singularities up to conjugation by graded Poisson automorphisms and compute their fixed point loci ([Hu25]).

(2) Describe the scheme-theoretic preimages of these fixed point loci under minimal resolutions of Kleinian singularities, whose irreducible components are $\mathbb{P}^1$'s and $\mathbb{A}^1$'s ([Hu25]).

(3) Analyze the restriction of anti-Poisson involutions to formal neighborhoods of singular points in filtered Poisson deformations of Kleinian singularities.

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Contents

Chapter 1

Introduction

1.1 Background and goals

Kleinian singularities are quotients of $\mathbb{C}^2$ by finite subgroups $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$. They can be described as surfaces in $\mathbb{C}^3$ with isolated singularities at 0. The McKay correspondence states that the finite subgroups of $\mathrm{SL}_2(\mathbb{C})$ are in bijection with the simply-laced Dynkin diagrams. Let $\pi\colon \widetilde{X} \to X := \mathbb{C}^2/\Gamma$ denote the minimal resolution of a Kleinian singularity. The reduced exceptional fiber is of the form

$$\pi^{-1}(0)_{\mathrm{red}} = C_1 \cup \cdots \cup C_n,\ C_i \simeq \mathbb{P}^1. \tag{1.1.1}$$

The C_i has self-intersection $C_i \cdot C_i = -2$ and pairwise transversal intersection according dually to a Dynkin diagram of types ADE.

The algebra $\mathbb{C}[u,v]$ is a graded Poisson algebra with respect to the usual bracket and grading. The regular functions on a Kleinian singularity $\mathbb{C}[X] = \mathbb{C}[u,v]^\Gamma$ is a graded Poisson subalgebra of $\mathbb{C}[u,v]$. Let θ be an *anti-Poisson involution* on X, which means that θ is a graded involution on $\mathbb{C}[X]$ such that $\theta(\{\cdot,\cdot\}) = -\{\theta(\cdot),\theta(\cdot)\}$. We would like to study the *scheme-theoretic fixed point locus* $X^\theta := \operatorname{Spec}\mathbb{C}[X]/(\theta(f) - f | f \in \mathbb{C}[X])$ and its scheme-theoretic preimage $\pi^{-1}(X^\theta)$ under the minimal resolution.

The filtered Poisson deformations of a Kleinian singularity X is parametrized by $\mathfrak{h}/W$, where $\mathfrak{h}$ is the Cartan space and W is the Weyl group corresponding to the ADE type labeling Γ. Let X_λ denote the filtered Poisson deformation of X with parameter $\lambda \in \mathfrak{h}/W$. Recall that θ denotes an anti-Poisson involution of X. For compatible choices of θ and λ, there exists a unique anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$ such that $\operatorname{gr}\theta_\lambda = \theta$. The singular surface X_λ may have more than one singular points, but the completion of X_λ at each singular point is isomorphic the completion of a Klenian singularity at the unique singular point 0

[Slo80, Section 6.5].

Let $p \in X_\lambda$ be a singular point fixed by θ_λ, and consider the restriction of θ_λ to the formal neighborhood

$$\widehat{\theta}_{\lambda,p} \colon \widehat{X}_{\lambda,p} \to \widehat{X}_{\lambda,p}. \tag{1.1.2}$$

Then $\widehat{\theta}_{\lambda,p}$ is an anti-Poisson involution of the Kleinian singularity $\widehat{X}_{\lambda,p}$. The dissertation addresses three main objectives:

(1) Classification of anti-Poisson involutions: Classify anti-Poisson involutions of X up to conjugation by graded Poisson automorphisms and compute the scheme-theoretic fixed point loci X^θ (Chapter 3).

(2) Study of the preimage under the minimal resolution: Describe the scheme-theoretic preimage $\pi^{-1}(X^\theta)$ under the minimal resolution. Specifically, determine its irreducible components, study how different components intersect with each other, and figure out if the scheme is reduced or not (Chapter 4).

(3) Analysis of anti-Poisson involutions in formal neighborhood of deformations: Determine the Type of the anti-Poisson involution $\widehat{\theta}_{\lambda,p}$ in the formal neighborhood $\widehat{X}_{\lambda,p}$ in terms of the classification of anti-Poisson involutions discussed in Objective (1) (Chapter 5).

1.2 Motivations

We discuss the motivation to study anti-Poisson involutions in the deformations and resolutions of Kleinian singularities. They are related to the classification of certain unipotent Harish-Chandra $(\mathfrak{g}, K)$-modules. Roughly speaking, the fixed point loci X^θ are associated varieties of Harish-Chandra $(\mathfrak{g}, K)$-modules, and the preimages $\pi^{-1}(X^\theta)$ are the singular supports of their Beilinson-Bernstein localizations, upon intersection with transversal slices to codimension 2 orbits.

In Section 1.2.1, we review key results on nilpotent orbits and symmetric pairs, which will be essential for the subsequent discussion on Harish-Chandra modules in Section 1.2.2. In Section 1.2.3, we briefly discuss why we are interested in the Poisson deformations of Kleinian singularities, as well as the restriction of anti-Poisson involutions to formal neighborhoods.

1.2.1 Nilpotent orbits and symmetric pairs

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$ and $\mathbb{O} \subset \mathfrak{g}$ a nilpotent orbit. Then $\mathbb{O}$ can be identified with a coadjoint orbit in $\mathfrak{g}^*$ using the Killing form, and therefore carries the Kostant-Kirillov symplectic 2-form.

Let $\overline{\mathbb{O}}$ be the closure of $\mathbb{O}$ in $\mathfrak{g}$ and $\mathbb{O}'$ an orbit of codimension 2 in $\overline{\mathbb{O}}$. Take $e \in \mathbb{O}'$ and let $\{e, f, h\}$ be an $\mathfrak{sl}_2$-triple containing e. Consider the *Slodowy slice*

$$S := e + \ker(\operatorname{ad} f), \tag{1.2.1}$$

which is a special transversal slice ([Slo80, Section 7.4]) to the orbit through e. Assume that e is a normal point in $\overline{\mathbb{O}}$. Then $S \cap \overline{\mathbb{O}}$ is a normal Goreinstein (due to Panyushev [Pan91, Theorem 1]) singularity of dimension 2, hence isomorphic to a Kleinian singularity X because Kleinian singularities are the only normal Gorenstein singularities in dimension 2. Note that $\overline{\mathbb{O}}$ is always normal in type A [KP79, Theorem], and we can pass to the normalization in the general situation.

Anti-Poisson involutions appear naturally from involutions of Lie algebras. Let $\sigma\colon \mathfrak{g} \to \mathfrak{g}$ be a Lie algebra involution. We can decompose $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, where $\mathfrak{k}$ is the eigenspace with eigenvalue 1 under σ, and $\mathfrak{p}$ is the eigenspace with eigenvalue -1. Set $\theta := -\sigma$. Then θ is an anti-Poisson involution of $\mathfrak{g}$.

If we further assume that the $\mathfrak{sl}_2$-triple $\{e, f, h\}$ is *normal*, i.e., $\sigma(e) = -e,\ \sigma(f) = -f,\ \sigma(h) = h$, then S is σ-stable, hence θ-stable. As long as $\mathbb{O}' \cap \mathfrak{p} \neq \emptyset$, such normal triple always exists [CM93, Theorems 9.4.1, 9.5.1]. Assume further that σ preserves $\mathbb{O}$, which is always satisfied for all regular and subregular nilpotent orbits in any $\mathfrak{g}$, as well as for all nilpotent orbits in type A. Then $\theta := -\sigma$ restricts to an anti-Poisson involution on $S \cap \overline{\mathbb{O}} \simeq X$. We have $S \cap \overline{\mathbb{O}} \cap \mathfrak{p} = (S \cap \overline{\mathbb{O}})^{\theta} \simeq X^{\theta}$, where the fixed point locus shows up.

1.2.2 Harish-Chandra modules

Let $\mathfrak{g}, \mathfrak{k}, \sigma, \theta$ be as in Section 1.2.1. Let G be the simply connected algebraic group with Lie algebra $\mathfrak{g}$ and $K \subset G$ the connected algebraic subgroup with Lie subalgebra $\mathfrak{k}$. Let $\mathcal{U}(\mathfrak{g})$ be the universal enveloping algebra of $\mathfrak{g}$, equipped with the PBW filtration. By a *Harish-Chandra* (shortly HC) $(\mathfrak{g}, K)$-module, one means a finitely generated $\mathcal{U}(\mathfrak{g})$-module M such that $\mathfrak{k}$ acts locally finitely and the action of $\mathfrak{k}$ integrates to that of K. A *good* filtration on a HC $(\mathfrak{g}, K)$-module M is a K-stable filtration that is compatible with the filtration on $\mathcal{U}(\mathfrak{g})$ and such that $\operatorname{gr} M$ is finitely generated over $\operatorname{gr} \mathcal{U}(\mathfrak{g})$. Pick a good filtration on M, we can define the *associated variety* $\operatorname{AV}(M) := \operatorname{Supp}(\operatorname{gr} M)$ as a set. Note that $\operatorname{AV}(M)$ is independent of the choice of the good filtration (c.f. [Vog91, Section 2]). When M is irreducible, we have $\operatorname{AV}(M) \subset \mathcal{N} \cap \mathfrak{p}$, where $\mathcal{N} \subset \mathfrak{g}$ is the nilpotent cone.

We focus on a special class of irreducible HC $(\mathfrak{g}, K)$-modules, namely those annihilated by unipotent ideals. Let $\mathbb{O} \subset \mathfrak{g}$ be a nilpotent orbit. The algebra of regular functions $\mathbb{C}[\mathbb{O}]$ is a graded Poisson algebra. It makes sense to speak about filtered quantizations of $\mathbb{C}[\mathbb{O}]$, and they are parameterized by $H^2(Y^{\mathrm{reg}}, \mathbb{C})$, where Y is a $\mathbb{Q}$-factorial terminalization of $\mathrm{Spec}(\mathbb{C}[\mathbb{O}])$ [Los22, Theorem 3.4]. Consider the *canonical* (in the terminology of [LMBM21, Section 1.4]) quantization $\mathcal{A}_0$, its parameter is $0 \in H^2(Y^{\mathrm{reg}}, \mathbb{C})$. The corresponding kernel of the quantum comoment map $\mathcal{I}_0(\mathbb{O}) := \ker(\mathcal{U}(\mathfrak{g}) \to \mathcal{A}_0)$ is the *unipotent ideal associated with* $\mathbb{O}$ in the terminology of [LMBM21, Section 1.4]. All such ideals are maximal. (For classical groups, see [LMBM21, Theorem 8.5.1]; for exceptional groups, see [MBM23, Theorem 5.0.1].) Let M be an irreducible HC $(\mathfrak{g}, K)$-module annihilated by $\mathcal{I}_0(\mathbb{O})$. Then we have $\mathrm{AV}(M) \subset \overline{\mathbb{O}} \cap \mathfrak{p}$. Let us further assume that

$$\mathrm{codim}_{\overline{\mathbb{O}}} \partial\mathbb{O} \geqslant 4. \tag{1.2.2}$$

A result by Vogan states that $\mathrm{AV}(M)$ is irreducible hence is the closure of a single K-orbit in $\mathbb{O} \cap \mathfrak{p}$ [Vog91, Theorem 4.6]. Let $\mathbb{O}_K$ denote the open K-orbit in $\mathrm{AV}(M)$. Losev and Yu prove that there is a bijection between the set of irreducible HC $(\mathfrak{g}, K)$-modules annihilated by $\mathcal{I}_0(\mathbb{O})$ with associated variety $\overline{\mathbb{O}}_K$ and the set of irreducible K-equivariant twisted (by half-canonical twist) local systems on $\mathbb{O}_K$ [LY23, Theorem 1.3.1]. The bijection sends a HC module M to the twisted local system $(\mathrm{gr}\, M)|_{\mathbb{O}_K}$.

However, when $\mathrm{codim}_{\overline{\mathbb{O}}} \partial\mathbb{O} \geqslant 2$, the classification of irreducible HC $(\mathfrak{g}, K)$-modules annihilated by $\mathcal{I}_0(\mathbb{O})$ is unknown. One difficulty is that $\mathrm{AV}(M)$ may not be irreducible. For example, consider the Lie algebra $\mathfrak{sl}_2(\mathbb{C}) = \mathbb{C}\langle e, h, f\rangle$ with the inner involution $\sigma := \mathrm{Ad}(\mathrm{diag}\{1, -1\})$. Then we have $\mathfrak{k} = \mathfrak{g}^\sigma = \mathbb{C}h$ is the Cartan subspace. Let $\mathbb{O}$ be the regular nilpotent orbit in $\mathfrak{sl}_2(\mathbb{C})$. The unipotent ideal $\mathcal{I}_0(\mathbb{O})$ is the two-sided ideal generated by the Casimir element $C = (h+1)^2 + 4fe$. Let M be an irreducible $\mathfrak{sl}_2(\mathbb{C})$-module such that h acts semisimply with even integral eigenvalues and the Casimir element C acts by 0. Then M has neither a highest weight nor a lowest weight vector. Equip M with a good filtration. Then $\mathrm{AV}(M)$ is a union of two lines. Another difficulty is that the condition $\mathrm{codim}_{\overline{\mathbb{O}}}(\partial\mathbb{O}) \geqslant 4$ is essential to show that non-isomorphic HC modules correspond to non-isomorphic twisted local systems in the proof of [LY23, Theorem 1.3.1].

The geometric data needed in the codimension 2 is therefore much more complicated than a twisted local system on an open K-orbit in $\overline{\mathbb{O}} \cap \mathfrak{p}$. Assume that $\overline{\mathbb{O}}$ admits a symplectic resolution given by the Springer resolution $\pi\colon T^*(G/P) \to \overline{\mathbb{O}}$, where $P \subset G$ is a parabolic subgroup. This holds for regular nilpotent orbits $\mathbb{O}_{\mathrm{reg}}$ in any types, and the Springer resolution is given by $T^*(G/B) \to \mathcal{N} = \overline{\mathbb{O}}_{\mathrm{reg}}$, where $B \subset G$ is a Borel subgroup and $\mathcal{N}$ denotes the nilpotent cone. It also holds for any nilpotent orbits in type A. For other orbits in other types, further analysis is required (c.f. [Fu03, Sections 3.3–3.4]). We change our viewpoint to K-equivariant $\mathcal{D}$-modules on the flag variety G/P, whose global sections are HC $(\mathfrak{g}, K)$-modules by the Beilinson-Bernstein localization theorem [BB81]. It is expected that the $\mathcal{D}$-modules are more manageable and exhibit refined geometric properties since they live on the smooth variety G/P. Recall that the associated varieties of HC $(\mathfrak{g}, K)$-modules annihilated by $\mathcal{I}_0(\mathbb{O})$ are contained in $\overline{\mathbb{O}} \cap \mathfrak{p}$. Thus, their Beilinson-Bernstein localizations have singular supports contained in $\pi^{-1}(\overline{\mathbb{O}} \cap \mathfrak{p})$.

To proceed further with the classification in codimension 2 case, it is necessary to understand the structure of the variety $S \cap \overline{\mathbb{O}} \cap \mathfrak{p}$ as well as their preimage $\pi^{-1}(S \cap \overline{\mathbb{O}} \cap \mathfrak{p})$ under the Springer resolution, where S is the Slodowy slice to a point in a codimension 2 orbit $\mathbb{O}'$, defined in eq. (1.2.1). As we explained at the end of Section 1.2.1, $S \cap \overline{\mathbb{O}}$ is isomorphic to a Kleinian singularity X, and $S \cap \overline{\mathbb{O}} \cap \mathfrak{p} \simeq X^\theta$ is the fixed point locus. The restriction of the Springer resolution $\pi\colon T^*(G/P) \to \overline{\mathbb{O}}$ to $S \cap \overline{\mathbb{O}} \simeq X$ gives rise to the minimal resolution of the Kleinian singularity X, and we have $\pi^{-1}(S \cap \overline{\mathbb{O}} \cap \mathfrak{p}) \simeq \pi^{-1}(X^\theta)$, where the preimage of fixed point locus shows up.

1.2.3 Quantizations and deformations

In addition to unipotent Harish-Chandra $(\mathfrak{g}, K)$-modules, we are also interested in Harish-Chandra modules over all filtered quantizations of nilpotent orbits. The images of these modules under the restriction function $\bullet_\dagger$, constructed in [Los15], correspond to Harish-Chandra modules over filtered quantizations of Kleinian singularities whose associated gradeds have supports contained in X^θ. Since the filtered quantizations and Poisson deformations of X are both parametrized by $\mathfrak{h}/W$, we hope that the families of varieties $X_\lambda^{\theta_\lambda}$ and their preimages under simultaneous resolutions $\pi^{-1}(X_\lambda^{\theta_\lambda})$, as λ varies, can serve as geometric analogues to a family of Harish-Chandra modules over different quantization parameters. The description of $X_\lambda^{\theta_\lambda}$ and $\pi^{-1}(X_\lambda^{\theta_\lambda})$ requires a local analysis of the anti-Poisson involution θ_λ in formal neighborhoods around the singular points of X_λ (c.f. the second project in Section 1.3.3).

1.3 Main results and future directions

1.3.1 Main results

This thesis provides complete solutions to the three Objectives (1)-(3) proposed in Section 1.1. We summarize our main results below.

Objective (1)

Recall that X denotes a Kleinian singularity. Proposition 1.3.1 gives a classification of the anti-Poisson involutions θ.

Proposition 1.3.1. *There are finitely many anti-Poisson involutions on $X := \mathbb{C}^2/\Gamma$ up to conjugation by Poisson automorphisms. They can all be explicitly written in terms of generators of $\mathbb{C}[X]$.*

The finiteness of the conjugacy classes of anti-Poisson automorphisms is treated in Corollary 3.2.3. The explicit forms of the anti-Poisson involutions are studied in Propositions 3.3.1, 3.3.2, 3.3.5, 3.3.8, 3.3.11, 3.3.14 and 3.3.17. Proposition 1.3.2 below answers gives a description of the fixed point locus X^θ.

Proposition 1.3.2. *Let θ be an anti-Poisson involution on X. The fixed point locus X^θ is reduced. If X^θ is not a single point, each irreducible component of X^θ is either $\mathbb{A}^1$ or a cusp.*

This is a case by case calculation based on the classification result in Proposition 1.3.1. The proofs are given in Corollaries 3.3.4, 3.3.7, 3.3.10, 3.3.13, 3.3.16 and 3.3.19.

Objective (2)

Recall $\pi\colon \widetilde{X} \to X$ denotes the minimal resolution. Theorem 1.3.3 is crucial to the description of $\pi^{-1}(X^\theta)$ as a subvariety, which constitutes the major part of the solution to Objective (2).

Theorem 1.3.3. *There exists a unique anti-symplectic involution $\tilde{\theta} : \widetilde{X} \to \widetilde{X}$ such that $\pi \circ \tilde{\theta} = \theta \circ \pi$. It can be constructed explicitly via involutions of quiver varieties. We call $\tilde{\theta}$ a* lift *of θ.*

Note that the lift $\tilde{\theta}$ always exists due to the definition and uniqueness of the minimal resolution $\pi\colon \widetilde{X} \to X$. However, to analyze the action of $\tilde{\theta}$ on $\widetilde{X}$, especially the action on the exceptional fiber $\pi^{-1}(0) = \bigcup_{1\leqslant i\leqslant n} C_i$ with $C_i \simeq \mathbb{P}^1$, we need an explicit construction of the lift $\tilde{\theta}$. The proof of Theorem 1.3.3 is based on a case by case study in Propositions 4.3.1, 4.3.3, 4.3.6, 4.3.8, 4.4.4, 4.4.6, 4.4.8, 4.4.10, 4.5.3, 4.5.5, 4.6.2 and 4.7.1. The key ingredient is to realize both $\widetilde{X}$ and X as Nakajima quiver varieties explicitly. The technical details in types D_n, E_6 are dealt with in Propositions 4.4.1 and 4.5.1. The proofs in types E_7, E_8 can be simplified with the help of Lemma 4.6.1. With Theorem 1.3.3, we can describe the variety structure $\pi^{-1}(X^\theta)_{\mathrm{red}}$ following the recipe given in Section 4.2.

Denote the 1-dimensional irreducible components of X^θ by $L_1, \cdots, L_m$. Set

$$\tilde{L}_j := \overline{\pi^{-1}(L_j \setminus \{0\})},\ 1 \leqslant j \leqslant m. \tag{1.3.1}$$

Then we have $\tilde{L}_j \simeq \mathbb{A}^1$ (c.f. Proposition 4.2.2). Define $b_i := \#\{1 \leqslant j \leqslant m \mid \tilde{L}_j \cap C_i \neq \emptyset\}$. Recall that $\mathcal{C}$ denotes the Cartan matrix of the Dynkin diagram labeling Γ. Proposition 1.3.4 allows us to determine the multiplicities of each component in $\pi^{-1}(X^\theta)$ as a divisor, which provides the remaining solution to Objective (2).

Proposition 1.3.4 (4.2.8)**.** *If $X^\theta \subset X$ is a principal divisor, then $\pi^{-1}(X^\theta) = \sum_{j=1}^m \tilde{L}_j + \sum_{i=1}^n a_i C_i$ with*

$$(a_1, \cdots, a_n)^t = \mathcal{C}^{-1}(b_1, \cdots, b_n)^t. \tag{1.3.2}$$

We comment that X^θ is always a principal divisor in X except for two cases of θ in type A, in which we have variant methods to determine the multiplicities (c.f. Sections 4.3.2 and 4.3.4). The descriptions of the scheme-theoretic preimages $\pi^{-1}(X^\theta)$ are given in Propositions 4.3.2, 4.3.4, 4.3.7, 4.3.9, 4.4.5, 4.4.7, 4.4.9, 4.4.11, 4.5.4, 4.5.6, 4.6.3 and 4.7.2.

Objective (3)

Lastly, we analyze the restriction of anti-Poisson involutions to formal neighborhoods of the singular points in a filtered Poisson deformation X_λ. Let θ as classified in Proposition 1.3.1. We say that λ and θ are *compatible* if there exists a unique anti-Poisson involution $\theta_\lambda \colon X_\lambda \to X_\lambda$ such that $\operatorname{gr}\theta_\lambda = \theta$.

Let $\mathcal{X}$ be the universal graded Poisson deformation of X and consider the structure morphism $\varphi \colon \mathcal{X} \to \mathfrak{h}/W$ with the identification $X \simeq \varphi^{-1}(0)$. Note that $\mathbb{C}[\mathfrak{h}/W]$ coincides with the Poisson center of $\mathbb{C}[\mathcal{X}]$ (c.f. Proposition 5.1.12). Each filtered Poisson deformation X_λ can be obtain through the pullback of φ to a particular point $\lambda \in \mathfrak{h}/W$. Proposition 1.3.5 below tells us when λ and θ are compatible.

Proposition 1.3.5. *There is a unique anti-Poisson involution $\Theta \colon \mathcal{X} \to \mathcal{X}$ whose restriction to the zero fiber $X \simeq \varphi^{-1}(0)$ recovers θ. We call Θ a* UPD-lift *of θ. Moreover, λ and θ are compatible if and only if $\lambda \in (\mathfrak{h}/W)^\Theta$.*

The existence of the UPD-lift is dealt with in Proposition 5.1.14, and the explicit forms of Θ are given in Propositions 5.3.1, 5.3.3, 5.3.5, 5.3.7, 5.3.9 and 5.3.11. We examine when λ and θ are compatible in all types ADE case by case, and the results are provided in Propositions 5.3.2, 5.3.4, 5.3.6, 5.3.8, 5.3.10 and 5.3.12. The term *UPD-lift* is used to distinguish this notion from the lift $\tilde{\theta}$ to the minimal resolutions of Kleinian singularities, which is constructed in Theorem 1.3.3.

Consider the triple (X_λ, ι, p), where X_λ is a filtered deformation of a Kleinian singularity, ι is the Roman Numeral indicating the Type of an anti-Poisson involution θ of X as classified in Proposition 1.3.1, and p is a singular point of X_λ.

We call the triple (X_λ, ι, p) an *Involution-Singularity Triple* (abbreviated as IST) if ι and λ are compatible, and that p is fixed by θ_λ. We write $(X_\lambda, \iota, p) \sim (Q, \iota')$, where Q denotes the singularity type of $\widehat{X}_{\lambda,p}$ and ι' denotes the Type of the restriction of the anti-Poisson involution to the formal neighborhood $\widehat{\theta}_{\lambda,p} \colon \widehat{X}_{\lambda,p} \to \widehat{X}_{\lambda,p}$, and say that the IST (X_λ, ι, p) is of type (Q, ι').

Theorem 1.3.6. *Let (X_λ, ι, p) be an Involution-Singularity Triple (IST). The pair (Q, ι'), with $(X_\lambda, \iota, p) \sim (Q, \iota')$, can be determined explicitly as explained in Propositions 5.4.8 and 5.4.9 for type A, and in Proposition 5.4.13 for type D.*

1.3.2 Structure of the thesis

Chapter 2 provides the necessary preliminaries. We review basic facts about Kleinian singularities and construct the Poisson structures on them.

Chapter 3 addresses Objective (1). We begin by introducing anti-Poisson involutions of Kleinian singularities $X := \mathbb{C}^2/\Gamma$. In Section 3.2, we investigate the relations between anti-Poisson involutions and the normalizers of Γ in $\mathrm{GL}_2(\mathbb{C})$. In Section 3.3, we classify the anti-Poisson involutions on X up to conjugation by Poisson automorphisms and compute the fixed point loci X^θ case by case.

Chapter 4 addresses Objective (2). We first review the construction of Kleinian singularities as Nakajima quiver varieties in Section 4.1. In Section 4.2, we discuss general procedures for determining the scheme-theoretic preimage $\pi^{-1}(X^\theta)$. Sections 4.3 to 4.7 provide explicit realizations of Kleinian singularities as Nakajima quiver varieties and study the preimages $\pi^{-1}(X^\theta)$, following the recipe given in Section 4.2.

Chapter 5 addresses Objective (3). In Sections 5.1 and 5.2, we review key facts about conical symplectic singularities their Poisson deformations and construct the Poisson brackets on the universal graded Poisson deformations of Kleinian singularities. In Section 5.3, we construct the UPD-lifts of θ and investigate when θ and λ are compatible case by case. In Section 5.4, we study the restriction of anti-Poisson involutions to singular points in the filtered Poisson deformations, specifically determining the type of the Involution-Singularity Triple (IST) (X_λ, ι, p).

Finally, we would like to point out a parallel and independent work by Bhaduri et al. [Bha+24] that studied the set-theoretic preimage $\pi^{-1}(X^\theta)_{\mathrm{red}}$, which is part of Objective (2). They address some of the anti-Poisson involutions (half of the cases in types D_n, E_6, and all the cases in types E_7, E_8) using a different approach. Their motivation was to generalize the McKay correspondence to some complex reflection groups in $\mathrm{GL}_2(\mathbb{C})$.

1.3.3 Future directions

Motivated by the broader framework of classifying unipotent Harish-Chandra $(\mathfrak{g}, K)$-modules, as discussed in Section 1.2, we outline two future research directions.

Restriction of anti-Poisson involutions to formal neighborhoods in exceptional types

As noted in the statement of Theorem 1.3.6, Objective (3) has so far only been completed for types A and D. A natural next step is to extend the study to type E.

Problem 1.3.7. *Determine the type of an Involution-Singularity Triple (IST) (X_λ, ι, p) when X_λ is a filtered Poisson deformation in type E.*

At present, we do not have an explicit description of the singularity type $\widehat{X}_{\lambda,p}$ similar to that of Propositions 5.4.7 and 5.4.12 because the generators of the invariant functions $\mathbb{C}[\mathfrak{h}]^W$ are extremely complicated in exceptional types. However, there is a Lie-theoretic way to determine the singularity type, as explained in [Slo80, Section 6.5] and reviewed in Section 5.4.1. In particular, the singular points of X_λ are in bijection with the connected components of the Dynkin diagram of the semisimple part of the Levi subalgebra $Z_{\mathfrak{g}}(t)$, where t is a preimage of λ under the quotient map $\mathfrak{h} \to \mathfrak{h}/W$.

As explained in Section 1.2.1, Lie-algebra involutions can produce anti-Poisson involutions on Kleinian singularities. In fact, all anti-Poisson involutions arise in this way. The type A case is discussed in 3 of Remark 3.3.3. The cases of types D and E will appear in a revised version of [Hu25]; we thank the anonymous referee for providing the instructions.

It is therefore natural to reformulate the entire study of anti-Poisson involutions in the deformations and resolutions of Kleinian singularities through a purely Lie-theoretic perspective. This approach would complement the quiver variety approach in Chapter 4 and enable us to relate our results to the classical theory of real semisimple Lie groups in a more uniform, case-independent manner. It could provide a pathway to solving Problem 1.3.7, the restriction problem in type E.

Problem 1.3.8. *Perform the formal neighborhood analysis of anti-Poisson involutions in Objective (3) for all types ADE using a Lie-theoretic approach.*

In particular, we would like to examine the action of θ on the Levi subalgebra $Z_{\mathfrak{g}}(t)$. A key obstacle is that θ naturally acts on $\mathfrak{h}/W$, the deformation parameter space of the Kleinian singularities X, but does not directly acts on the Cartan subalgebra $\mathfrak{h}$, where t lies. It may be possible to construct an action of θ on $\mathfrak{h}$, but this action might no longer be an involution (i.e., of order 2).

Deformation of singular Lagrangians in the simultaneous resolutions of Kleinian singularities

We described the fixed point locus X^θ and and the preimage $\pi^{-1}(X^\theta)$ in Objectives (1) and (2). Let X_λ be a filtered Poisson deformation of X. Recall that for compatible θ and λ, there exists a unique anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$ such that $\operatorname{gr}\theta_\lambda = \theta$. Let $\pi_\lambda\colon Y_\lambda \to X_\lambda$ denote the minimal resolution. A natural question, as an extension of Objectives (1) and (2), is Problem 1.3.9 below.

Problem 1.3.9. *Describe the fixed point locus $X_\lambda^{\theta_\lambda}$ and the preimage $\pi_\lambda^{-1}(X_\lambda^{\theta_\lambda})$.*

The fixed point $X_\lambda^{\theta_\lambda}$ can be calculated out explicitly, as both X_λ and θ_λ have explicit forms (c.f. Proposition 1.3.5). For the preimage $\pi_\lambda^{-1}(X_\lambda^{\theta_\lambda})$, it is sufficient to work in formal neighborhoods. Restricting the minimal resolution map π_λ to the formal neighborhood of each singular point p gives rise to the minimal resolution of the completion $\widehat{X}_{\lambda,p}$, which is a Kleinain singularity:

$$\widehat{\pi}_{\lambda,p}\colon \widehat{Y}_{\lambda,\pi^{-1}(p)} \to \widehat{X}_{\lambda,p}. \tag{1.3.3}$$

Using these formal neighborhoods, describing the preimage $\pi^{-1}(X_\lambda^{\theta_\lambda})$ reduces to analyzing the restriction of the action of θ_λ to $\widehat{X}_{\lambda,p}$, for those p fixed by θ_λ, and combining it with the solution to Objective (2), i.e., the description of the preimage $\pi^{-1}(X^\theta)$ for Kleinian singularities. Determining the action of $\widehat{\theta}_{\lambda,p}$ of $\widehat{X}_{\lambda,p}$ is precisely the goal of Objective (3). Hence, Problem 1.3.9 is solved for types A and D using Theorem 1.3.6; for type E, it depends on the progress of the previous future project.

An upgraded question is to describe how $X_\lambda^{\theta_\lambda}$ and its preimage $\pi_\lambda^{-1}(X_\lambda^{\theta_\lambda})$ vary as λ deforms along curves in the parameter space $\mathfrak{h}/W$, as mentioned in Section 1.2.3. We formulate a more precise version of this question in Problem 1.3.10.

Let $\mathfrak{g}$ be a simple Lie algebra with ADE type labeling $X \simeq \mathbb{C}^2/\Gamma$. Recall the Slodowy slice to a subregular nilpotent element $S \subset \mathfrak{g}$, defined in eq. (1.2.1). In fact, $S \simeq \mathcal{X}$ is the universal graded Poisson deformation of the Kleinian singularity X. The restriction of the Grothendieck-Springer resolution to S and its preimage $\widetilde{S}$ under the morphism $G \times^B \mathfrak{b} \to \mathfrak{g}$ provides simultaneous resolutions of filtered Poisson deformations of Kleinian singularities:

$$\begin{tikzcd} \widetilde{S} \arrow[r, "\Phi"] \arrow[d] & S \arrow[d, "\varphi"] \\ \mathfrak{h} \arrow[r] & \mathfrak{h}/W \end{tikzcd} \tag{1.3.4}$$

Let $C \subset \mathfrak{h}$ be an affine smooth curve such that $\bar{C} \subset (\mathfrak{h}/W)^\Theta$, where $\bar{C}$ is the image of C under the quotient map $\mathfrak{h} \to \mathfrak{h}/W$, and Θ is the UPD-lift of θ, defined in Proposition 1.3.5.

Set $S_{\bar{C}} := \varphi^{-1}(\bar{C})$ and $\mathfrak{X}_C := \Phi^{-1}(S^{\Theta}_{\bar{C}})$. Consider the morphism

$$f\colon \mathfrak{X}_C \to C, \tag{1.3.5}$$

defined as the composition $\mathfrak{X}_C \hookrightarrow \widetilde{S} \to \mathfrak{h}$, whose image lies in C. The morphism f is flat with 1-dimensional fibers. Moreover, it is locally trivial outside a finite set of points $\{c_1, \cdots, c_k\}$. Set $\mathfrak{X}_{c_i} := f^{-1}(c_i),\ i \in \{1, \cdots, k\}$. Let $C^0 := C \setminus \{c_1, \cdots, c_k\}$ and $\mathfrak{X}_{C^0} := f^{-1}(C^0)$. Let $Z \subset \mathfrak{X}_{C^0}$ be an irreducible component and let $\bar{Z}$ denote its closure in $\mathfrak{X}_C$. Consider the scheme-theoretic intersection $\bar{Z} \cap \mathfrak{X}_{c_i}$, for some $i \in \{1, \cdots, k\}$.

Problem 1.3.10. *Determine the irreducible components and their multiplicities in the intersection $\bar{Z} \cap \mathfrak{X}_{c_i}$.*

Chapter 2

Preliminaries

2.1 Reminder on Kleinian singularities

We give a reminder on Kleinian singularities, following [CS98, Sections 1–2] and [Slo83, Section 5]. Let $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$ denote a finite subgroup. Up to conjugacy, there are five classes of such groups. We list the conjugacy classes below, where ϵ_n to denote a primitive n-th root of unity for $n = 5, 8, 10$.

(A_n) The cyclic group of order $n+1$, i.e. $\Big\{ \begin{pmatrix} \epsilon & 0 \\ 0 & \epsilon^{-1} \end{pmatrix} |\epsilon^{n+1} = 1,\ n \geqslant 1 \Big\}$.

(D_n) The binary dihedral group of order $4(n-2)$ i.e. $\Big\{ \begin{pmatrix} \epsilon & 0 \\ 0 & \epsilon^{-1} \end{pmatrix}, \begin{pmatrix} 0 & \epsilon \\ -\epsilon^{-1} & 0 \end{pmatrix} |\epsilon^{2(n-2)} = 1,\ n \geqslant 4 \Big\}$.

(E_6) The binary tetrahedral group of order 24.

It is generated by $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \dfrac{1}{1-i} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$

(E_7) The binary octahedral group of order 48.

It is generated by $\begin{pmatrix} \epsilon_8 & 0 \\ 0 & \epsilon_8^{-1} \end{pmatrix}, \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \dfrac{1}{1-i} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix},$

(E_8) The binary icosahedral group of order 120.

It is generated by $\begin{pmatrix} \epsilon_{10} & 0 \\ 0 & \epsilon_{10}^{-1} \end{pmatrix}, \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \dfrac{1}{\sqrt{5}} \begin{pmatrix} \epsilon_5 - \epsilon_5^4 & \epsilon_5^2 - \epsilon_5^3 \\ \epsilon_5^2 - \epsilon_5^3 & -\epsilon_5 + \epsilon_5^4. \end{pmatrix}.$

The *Kleinian singularity* attached to a finite subgroup Γ is the quotient singularity

$$X := \mathbb{C}^2/\Gamma = \operatorname{Spec} \mathbb{C}[u, v]^{\Gamma}.$$

Klein showed that X can be viewed as a hypersurface in $\mathbb{C}^3$

$$X \simeq \{(x, y, z) \in \mathbb{C}^3 | F(x, y, z) = 0\},$$

with an isolated singularity at 0 ([Kle93]). Here F denotes the relation among the three fundamental invariants x, y, z in $\mathbb{C}[u, v]^\Gamma$. We list the most commonly used form of these relations below.

(A_n) $x_1 y_1 + z_1^{n+1} = 0$ with $n \geqslant 1$.

(D_n) $x_1^{n-1} + x_1 y_1^2 + z_1^2 = 0$ with $n \geqslant 4$.

(E_6) $x_1^4 + y_1^3 + z_1^2 = 0$.

(E_7) $x_1^3 y_1 + y_1^3 + z_1^2 = 0$.

(E_8) $x_1^5 + y_1^3 + z_1^2 = 0$.

The reason that we use x_1, y_1, z_1 with subscripts here is because x, y, z are saved for different choices of generators in the later part of this thesis (c.f. Proposition 2.3.1). One can observe that the above list of finite subgroups of $\mathrm{SL}_2(\mathbb{C})$ is in bijection with ADE type Dynkin diagrams. There are two ways to attach a simply-laced Dynkin diagram to a Kleinian singularity X. McKay observed a representation-theoretic way, called the Mckay correspondence, in 1979 [McK80]. Let $\mathrm{Irr}(\Gamma) := \{V_0, \cdots, V_n\}$ denote the set of irreducible representations of Γ and assume that V_0 is the trivial representation. Set

$$a_{ij} := \dim \mathrm{Hom}_\Gamma(V_i \otimes \mathbb{C}^2, V_j),$$

where $\mathbb{C}^2$ is the tautological representation of $\mathrm{SL}_2(\mathbb{C})$. Then we have

$$\dim \bigoplus_{\substack{0 \leqslant j \leqslant n \\ a_{ij} \neq 0}} V_j = \dim(V_i \otimes \mathbb{C}^2) = 2 \dim V_i. \tag{2.1.1}$$

Moreover, it easy to check that $a_{ij} = a_{ji}$ because $\mathbb{C}^2 \simeq (\mathbb{C}^2)^*$ is self-dual as Γ-representations. Consider a graph with vertices $0, \cdots, n$ and the number of unoriented edges between i and j equal to a_{ij}. This is called the *McKay graph* of Γ, which is exactly the extended Dynkin diagram in the corresponding type with the trivial representation V_0 corresponding to the extended vertex 0.

The other identification is through an algebro-geometric way. Recall that by a resolution of X, one means a smooth variety $\widetilde{X}$ equipped with a projective birational morphism to X. Since we are in dimension 2, there is a *minimal* resolution $\pi\colon \widetilde{X} \to X$; the minimality condition means that any other resolution factors through $\widetilde{X}$. The minimal resolution $\pi\colon \widetilde{X} \to X$ had been essentially studied by Du Val [Pat34].

For a Kleinian singularity $X := \mathbb{C}^2/\Gamma$, the reduced fiber over the isolated singularity 0 under the minimal resolution $\pi\colon \widetilde{X} \to X$ is a connected union of projective lines

$$\pi^{-1}(0)_{\mathrm{red}} = C_1 \cup \cdots \cup C_n,\ C_i \simeq \mathbb{P}^1,$$

with pairwise transversal intersection according dually to the Dynkin diagram labeling Γ. Elaborating, if $C_i \cap C_j \neq \emptyset$, then it is a single point, and each C_i has self-intersection number −2. Let $\mathcal{C}$ denote the Cartan matrix of the corresponding type ADE. Then the intersection matrix $(C_i \cdot C_j)$ equals −$\mathcal{C}$. The intersection pairing will be defined rigorously in Section 2.2.

Remark 2.1.1. The scheme-theoretic exceptional fiber $\pi^{-1}(0)$ is not reduced except in type A. Let $\mathfrak{g}$ be a simple Lie algebra of types ADE, and $\{\alpha_1, \cdots, \alpha_n\}$ be the set of simple roots. There is a unique maximal root $\alpha = \sum_{i=1}^n \delta_i \alpha_i$ in the adjoint representation. Then $\pi^{-1}(0) = \sum_{i=1}^n \delta_i C_i$ as a divisor, [Art66, Theorem 4]. For example, in type A_n, we have $\delta_i = 1,\ \forall\ i$, so $\pi^{-1}(0)$ is generically reduced. (Actually, $\pi^{-1}(0)$ is reduced in type A_n, which can be seen from iterated blow-ups.) In type D_4, we have $\delta_i = 1, i \neq 2$ and $\delta_2 = 2$, so $\pi^{-1}(0)$ is not reduced.

2.2 Intersection pairing

We define an intersection pairing on $\widetilde{X}$ rigorously in this section. The main purpose of the discussion is to pave the road for the proof of Proposition 4.2.8. Note that the variety $\widetilde{X}$ is not projective, so it is not possible to define a pairing with good properties on the entire group of Cartier divisors on $\widetilde{X}$. Instead, we define the pairing on a smaller subset of divisors on $\widetilde{X}$ (c.f. Proposition 2.2.1). We consider a slightly more general situation, following the exposition in [Dol09, Section 2.1]. Let S be a smooth surface, and Z be an effective Cartier divisor on S with support projective over $\mathbb{C}$. We further assume that each irreducible component in the support of Z is smooth. Though we will only care about the situation $S = \widetilde{X}$ and $Z = \pi^{-1}(0)$, it is easier to write down the results in a more general setting.

Let $\mathrm{Div}(S)$ be the group of Cartier divisors on S, which coincides with the group of Weil divisors on S because S is smooth. Let $\mathrm{Div}_Z(S)$ denote the subgroup of $\mathrm{Div}(S)$ that consists of divisors with support contained in the support of Z. We identify effective divisors on S with closed (not necessary reduced) subschemes of S defined by the ideal sheaf $\mathcal{O}_S(-D)$. Let $\mathcal{F}$ be a coherent sheaf on the projective scheme Z. We can define the Euler characteristic of $\mathcal{F}$ by $\hat{\chi}(\mathcal{F}) := \sum_i (-1)^i \dim_{\mathbb{C}} H^i(Z, \mathcal{F})$ (c.f. [Har77, Exercise

III.5.1]). Let C be an effective divisor in $\mathrm{Div}_Z(S)$. For any invertible sheaf $\mathcal{L}$ on C, we set

$$\deg(\mathcal{L}) := \hat{\chi}(\mathcal{L}) - \hat{\chi}(\mathcal{O}_C). \tag{2.2.1}$$

For any divisor D on S, we set

$$C \cdot D := \deg \mathcal{O}_S(D) \otimes_{\mathcal{O}_S} \mathcal{O}_C. \tag{2.2.2}$$

Let $P = \sum_i n_i P_i$ be a Weil divisor on C. Define $\deg P := \sum_i n_i$ as the degree of the divisor P. Further assume that C is irreducible, then C is a smooth projective curve by our assumption. By the Riemann-Roch Theorem [Har77, Theorem IV.1.3], we have

$$\deg(\mathcal{O}_C(P)) = \deg P.$$

Let f be a rational function on S, we denote by $\mathrm{div}(f)$ its corresponding divisor. Any divisor which is equal to the divisor of a function is called a *principal divisor*.

Proposition 2.2.1. *[Ful98, Example 2.4.9] The pairing*

$$\mathrm{Div}_Z(S) \times \mathrm{Div}(S) \to \mathbb{Z},\ (C, D) \mapsto C \cdot D \tag{2.2.3}$$

satisfies the following properties

(1) If C and D are smooth curves that meet transversally, then $C \cdot D = \#(C \cap D)$.

(2) If $C, D \in \mathrm{Div}_Z(S)$, then $C \cdot D = D \cdot C$.

(3) $C \cdot (D_1 + D_2) = C \cdot D_1 + C \cdot D_2$.

(4) If D is a principal divisor, then $C \cdot D = 0$.

We apply Proposition 2.2.1 to the situation $S = \widetilde{X},\ Z = \pi^{-1}(0)$. The properties in Proposition 2.2.1 will be useful in the proof of Proposition 4.2.8.

2.3 Poisson brackets on Kleinian singularities

Consider the algebra $\mathbb{C}[u, v]$, with grading given by the degree of the polynomial function in variables u, v. There is a standard Poisson bracket of degree (-2) on $\mathbb{C}[u, v]$.

$$\{f_1, f_2\} = \frac{\partial f_1}{\partial u}\frac{\partial f_2}{\partial v} - \frac{\partial f_1}{\partial v}\frac{\partial f_2}{\partial u}. \tag{2.3.1}$$

Let Γ be a finite subgroup of $\mathrm{SL}_2(\mathbb{C})$ and consider the Kleinian singularity $X := \mathbb{C}^2/\Gamma$. The finite subgroup Γ preserves the Poisson bracket eq. (2.3.1), then the algebra of invariant functions $\mathbb{C}[X] = \mathbb{C}[u, v]^\Gamma$ is a graded Poisson subalgebra of $\mathbb{C}[u, v]$. We use $\deg f$ to denote the degree of a homogeneous element $f \in \mathbb{C}[X]$ and $\mathbb{C}[X]_d$ to denote the degree d-th component of $\mathbb{C}[X]$.

Next, we write down the fundamental invariants in $\mathbb{C}[X] = \mathbb{C}[u, v]^\Gamma$, and calculate the Poisson brackets among the generators. We take the fundamental invariants of $\mathbb{C}[u.v]^\Gamma$ to be of the following form, altered from [Dol09, Section 1.3], which align better with the fundamental invariants obtained through Nakajima quiver varieties in Section 4.1.2.

Proposition 2.3.1. *Let $\Gamma \subset \mathrm{SL}_2(C)$ be a finite subgroup. The subalgebra $\mathbb{C}[u, v]^\Gamma$ is generated by the following polynomials.*

$(A_n,\ n \geqslant 1)$ $x = u^{n+1}$,

$y = v^{n+1}$,

$z = uv$,

with relation $xy = z^{n+1}$.

$(D_n,\ n \geqslant 4 \text{ even})$ $x = u^2v^2$,

$y = -\frac{1}{4}(u^{2n-4} - 2u^{n-2}v^{n-2} + v^{2n-4})$,

$z = \frac{1}{4}uv(u^{2n-4} - v^{2n-4})$,

with relation $xy(y - x^{\frac{n-2}{2}}) = z^2$.

$(D_n,\ n \geqslant 5 \text{ odd})$ $x = u^2v^2$,

$y = \frac{1}{4}(u^{2n-4} - v^{2n-4})$,

$z = -\frac{1}{4}uv(u^{2n-4} - 2u^{n-2}v^{n-2} + v^{2n-4})$,

with relation $xy^2 = z(z - x^{\frac{n-1}{2}})$.

(E_6) $x = 108\sqrt{3}(1 + i)uv(u^4 - v^4)$,

$y = -108(u^8 + 14u^4v^4 + v^8)$,

$z=648(\sqrt{3}(u^{12}-33u^8v^4-33u^4v^8+v^{12})-12iu^2v^2(u^4-v^4)^2)$,

with relation $z^2+zx^2+y^3=0$.

(E_7) $x=-2^23(u^8+14u^4v^4+v^8)$,

$y=2^43^3(u^5v-uv^5)^2$,

$z=2^53^3uv(u^8-v^8)(u^8-34u^4v^4-v^8)$,

with relation $x^3y+y^3+z^2=0$.

(E_8) $x=-3^3uv(u^{10}+11u^5v^5-v^{10})$,

$y=-\frac{3^4}{2^2}(u^{20}+v^{20}-228(u^{15}v^5-u^5v^{15})+494u^{10}v^{10})$,

$z=\frac{3^6}{2^3}(u^{30}+v^{30}+522(u^{25}v^5-u^5v^{25})-10005(u^{20}v^{10}+u^{10}v^{20}))$,

with relation $x^5+y^3+z^2=0$.

It is easy to see that the relations among the fundamental invariants x,y,z in Proposition 2.3.1 are equivalent to those relations among the variables x_1,y_1,z_1 listed in Section 2.1 up to a change of coordinates. For example, in type D_n, with n even, a change of coordinate can be given by $x_1=x,\ y_1=-2iy+ix^{\frac{n-2}{2}},\ z_1=2z$. In type D_n, with n odd, a change of coordinate can be given by $x_1=x,\ y_1=2y,\ z_1=-2iz+ix^{\frac{n-1}{2}}$. In type E_6, a change of coordinate can be given by $x_1=\frac{\epsilon_8 x}{\sqrt{2}},\ y_1=y,\ z_1=z+\frac{x^2}{2}$. The Poisson brackets among the fundamental invariants can be calculated either by hand or by software such as Mathematica. We summarize the results in the following proposition.

Proposition 2.3.2. *The Poisson brackets among the fundamental invariants in Proposition 2.3.1 are given by*

$$
\begin{aligned}
(A_n,\ n\geqslant 1)\qquad & \{x,y\}=(n+1)^2z^n,\\
& \{x,z\}=(n+1)x,\\
& \{y,z\}=-(n+1)y.\\
(D_n,\ n\geqslant 4\ \text{even})\qquad & \{x,y\}=(4n-8)z,\\
& \{x,z\}=(2n-4)x(2y-x^{\frac{n-2}{2}}),\\
& \{y,z\}=(n-2)y(nx^{\frac{n-2}{2}}-2y).\\
(D_n,\ n\geqslant 5\ \text{odd})\qquad & \{x,y\}=(2n-4)(2z-x^{\frac{n-1}{2}}),\\
& \{x,z\}=(4n-8)xy,\\
& \{y,z\}=-(n-2)((n-1)x^{\frac{n-3}{2}}z+2y^2).\\
(E_6)\qquad & \{x,y\}=24\sqrt{3}(1+i)(2z+x^2),
\end{aligned}
$$

$$\{x, z\} = -72\sqrt{3}(1+i)y^2,$$

$$\{y, z\} = 48\sqrt{3}(1+i)xz.$$

$$(E_7) \qquad \{x, y\} = -96z,$$

$$\{x, z\} = 48(x^3 + 3y^2),$$

$$\{y, z\} = -144x^2y.$$

$$(E_8) \qquad \{x, y\} = -120z,$$

$$\{x, z\} = 180y^2,$$

$$\{y, z\} = -300x^4.$$

Chapter 3

Anti-Poisson Involutions of Kleinian Singularities and Their Fixed Point Loci

We classify the conjugacy classes of anti-Poisson involution of a Kleinian singularity and compute their fixed point loci in this section. We use $X = \mathbb{C}^2/\Gamma$ to denote a Kleinian singularity throughout. Recall from Section 2.3, $\mathbb{C}[X] = \mathbb{C}[u, v]^\Gamma$ is a graded Poisson algebra with grading given by the degree of the invariant polynomials in variable u, v. The fundamental invariants of $\mathbb{C}[X]$ are listed in Proposition 2.3.1, and the Poisson brackets are given in Proposition 2.3.2.

3.1 Anti-Poisson involutions

Definition 3.1.1. By an *anti-Poisson involution* on X, we mean a graded algebra involution $\theta\colon \mathbb{C}[X] \to \mathbb{C}[X]$ such that

$$\theta(\{f_1, f_2\}) = -\{\theta(f_1), \theta(f_2)\},\ \forall\, f_1, f_2 \in \mathbb{C}[X]. \tag{3.1.1}$$

Example 3.1.2. Consider the Kleinian singularity of type A_1: $X = \mathbb{C}^2/\Gamma$ with $\Gamma = \mathbb{Z}/\mathbb{Z}_2 \simeq \{\pm I_2\}$. We have $\mathbb{C}[X] = \mathbb{C}[u, v]^\Gamma = \mathbb{C}[x = u^2, y = v^2, z = uv] \simeq \mathbb{C}[x, y, z]/(xy - z^2)$. The brackets among the generators are

$$\{x, y\} = 4z,\ \{x, z\} = 2x,\ \{y, z\} = -2y.$$

The graded automorphism $x \mapsto y,\ y \mapsto x,\ z \mapsto z$ is an anti-Poisson involution.

Similar to the anti-Poisson involution on Kleinian singularities (or any Poisson algebra), we can define the *anti-symplectic involution* on a smooth symplectic variety (N, ω), which is an involution $\eta\colon N \to N$ such that $\eta^*\omega = -\omega$. The following proposition will be very useful in Section 4.2.

Proposition 3.1.3. *Let (N, ω) be a smooth symplectic variety with an anti-symplectic involution η. Then the scheme-theoretic fixed point locus N^η is either empty or a smooth Lagrangian subvariety.*

Proof. The smoothness of the scheme-theoretic fixed point locus N^η is a corollary of Luna's slice theorem (c.f. [Dré04, Proposition 5.8]). The fact that N^η is a Lagrangian subvariety is an easy exercise in linear algebra. We leave it to the reader as an exercise. □

Definition 3.1.4. Let $X = \mathbb{C}^2/\Gamma$ be a Kleinian singularity with anti-Poisson involution θ. The *scheme-theoretic fixed point locus* of X under θ is the scheme

$$X^\theta = \operatorname{Spec} \mathbb{C}[X]/J, \tag{3.1.2}$$

where $J = (\theta(f) - f | f \in \mathbb{C}[X])$.

Remark 3.1.5. The scheme-theoretic fixed point locus X^θ turns out to be reduced, this follows from a case by case calculation in Corollaries 3.3.4, 3.3.7, 3.3.10, 3.3.13 and 3.3.19.

We have $0 \in X^\theta$ because θ is a graded automorphism. Similar to Proposition 3.1.3, one can show that the fixed point locus X^θ is either a singular Lagrangian subvariety (1-dimensional) or only contains the singular point (0-dimensional). The latter case only appears in the Type III anti-Poisson involutions for the Type A_n Kleinian singularity, with n odd (c.f. Corollary 3.3.4).

If two anti-Poisson involutions $\theta_i,\ i = 1, 2$, are conjugate to each other by a graded Poisson automorphism σ, then σ induces an isomorphism between the fixed point loci $X^{\theta_1} \xrightarrow{\sim} X^{\theta_2}$. We want to classify anti-Poisson involutions on $\mathbb{C}^2/\Gamma$ up to conjugation by graded Poisson automorphisms. It turns out that the conjugacy classes form a finite set, this will be proved in Corollary 3.2.3.

3.2 Normalizers and graded automorphisms

In this section, we discuss the action of the normalizer $N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)$ on the Kleinian singularity $\mathbb{C}^2/\Gamma$. Recall that $\mathbb{C}[X] = \mathbb{C}[u, v]^\Gamma$ is a graded subalgebra of $\mathbb{C}[u, v]$. The grading induces $\mathbb{C}^*$-actions on both $\mathbb{C}^2$ and $X = \mathbb{C}^2/\Gamma$, and the quotient morphism $q\colon \mathbb{C}^2 \to X$ is $\mathbb{C}^*$-equivariant. Let $X^0 := X \setminus \{0\}$ denote the smooth locus of X and $q^0\colon \mathbb{C}^2 \setminus \{0\} \to X^0$ denote the restriction of the quotient morphism. Then q^0 is an (analytic) universal covering map. Let us denote the group of graded automorphisms of the Kleinian singularity by

$\mathrm{Aut}_{\mathrm{gr}}(\mathbb{C}^2/\Gamma)$. Given an element $g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)$, we can produce an element in $\mathrm{Aut}_{\mathrm{gr}}(\mathbb{C}^2/\Gamma)$ by

$$\begin{aligned}\theta_g \colon \mathbb{C}[u,v]^\Gamma &\to \mathbb{C}[u,v]^\Gamma,\\ f(-) &\mapsto g\cdot f := f(g^{-1}-).\end{aligned}$$

Clearly, the elements in Γ induce the identity automorphism of $\mathbb{C}[u,v]^\Gamma$. Moreover, Γ is a normal subgroup of $N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)$. Therefore, we get a well-defined group homomorphism

$$N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)/\Gamma \to \mathrm{Aut}_{\mathrm{gr}}(\mathbb{C}^2/\Gamma). \tag{3.2.1}$$

Proposition 3.2.1. *eq.* (3.2.1) *is a bijection.*

Proof. We first show that the map eq. (3.2.1) is injective. Let $g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)$ such that g acts trivially on $X = \mathbb{C}^2/\Gamma$. Then g preserves the Γ-orbits of $\mathbb{C}^2$. Consider the linear subspaces

$$V_\gamma := \{v \in \mathbb{C}^2 | gv = \gamma v\},\ \gamma \in \Gamma.$$

For each particular $v \in \mathbb{C}^2$, there exists a $\gamma \in \Gamma$ such that $gv = \gamma v$. Therefore, we have

$$\bigcup_{\gamma\in\Gamma} V_\gamma = \mathbb{C}^2. \tag{3.2.2}$$

The left-hand side of eq. (3.2.2) is a finite union of linear subspaces of $\mathbb{C}^2$, therefore one of them, say V_{γ_0}, has to be the entire $\mathbb{C}^2$. It follows that $g = \gamma_0 \in \Gamma$.

Next, we show that the map eq. (3.2.1) is surjective. Let θ be a graded automorphism of X. Consider the restriction $\theta\colon X^0 \to X^0$, and lift it to the universal covering space $\hat{\theta}\colon \mathbb{C}^2 \setminus \{0\} \to \mathbb{C}^2 \setminus \{0\}$, i.e., we have $\theta \circ q^0 = q^0 \circ \hat{\theta}$. The lift $\hat{\theta}$ is still holomorphic (c.f. [For91, Theorem 4.9]). We claim that $\hat{\theta}$ commutes with the $\mathbb{C}^*$-action on $\mathbb{C}^2 \setminus \{0\}$ induced from that of $\mathbb{C}^2$. It suffices to show that $\lambda\hat{\theta} = \hat{\theta}\lambda$, for any $\lambda \in \mathbb{C}^*$, and we prove this pointwise. First. we have that

$$q(\lambda^{-1}\hat{\theta}(\lambda v)) = \theta(q(v)) = q(\hat{\theta}(v)),\ \forall\, v \in \mathbb{C}^2 \setminus \{0\},$$

because θ and q are both $\mathbb{C}^*$-equivariant. Hence, $\lambda^{-1}\hat{\theta}(\lambda v)$ and $\hat{\theta}(v)$ lie in the same fiber $q^{-1}(q(\hat{\theta}(v)))$. Next, let $\lambda(t),\ 0 \leqslant t \leqslant 1$ be a continuous path connecting $\lambda(0) = 1$ and $\lambda(1) = \lambda$ in $\mathbb{C}^*$. Then consider the following continuous map

$$\begin{aligned}\phi\colon [0,1] &\to q^{-1}(q(\hat{\theta}(v)))\\ t &\mapsto \lambda(t)^{-1}\hat{\theta}(\lambda(t)v).\end{aligned}$$

The range is a discrete space, so ϕ has to be constant. Thus, we obtain that $\lambda^{-1}\hat{\theta}(\lambda v) = \phi(1) = \phi(0) = \hat{\theta}(v)$. The final step is to extend the lifting $\hat{\theta}$ to a holomorphic map from $\mathbb{C}^2$ to $\mathbb{C}^2$ by Hartogs's Extension Theorem (c.f. [Hor73, Theorem 2.3.2]). The only possible choice is to send 0 to 0 because $\hat{\theta}\colon \mathbb{C}^2 \setminus \{0\} \to \mathbb{C}^2 \setminus \{0\}$ commutes with the $\mathbb{C}^*$-action. Now we have a holomorphic automorphism $\hat{\theta}$ on $\mathbb{C}^2$ that commutes with the $\mathbb{C}^*$-action, which has to be linear and invertible from looking at its Taylor expansion. Thus, $\hat{\theta}$ must come from an element $g \in \mathrm{GL}_2(\mathbb{C})$. Note that the lift $\hat{\theta}$ preserves each Γ-orbit, then so does g. It follows that $g \in N_{\mathrm{GL}_2}(\Gamma)$. □

Remark 3.2.2. Let $g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)$. By eq. (2.3.1), we obtain that

$$\{g \cdot f_1, g \cdot f_2\} = \det(g) g \cdot \{f_1, f_2\}. \tag{3.2.3}$$

There are two kinds of graded automorphisms in $\mathrm{Aut}_{\mathrm{gr}}(\mathbb{C}^2/\Gamma)$ that are worth special attention. The first is graded Poisson automorphisms, which come from elements in $N_{\mathrm{SL}_2}(\Gamma)$. The second is graded anti-Poisson automorphisms, which come from elements in $N_{\mathrm{GL}_2}(\Gamma)$ with determinant −1, which we denote by $N_{\mathrm{GL}_2}(\Gamma)^-$.

Let us denote the set of anti-Poisson involutions on $X = \mathbb{C}^2/\Gamma$ by $\mathrm{API}(\mathbb{C}^2/\Gamma)$ and the conjugacy classes of anti-Poisson involutions by $\mathrm{API}(\mathbb{C}^2/\Gamma)/\sim$. An element $g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)$ gives rise to anti-Poisson involution if and only if $\det(g) = -1$ and $g^2 \in \Gamma$. Proposition 3.2.1 tells us that there is a bijection between sets

$$\{g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^- | g^2 \in \Gamma\}/\Gamma \xrightarrow{\sim} \mathrm{API}(\mathbb{C}^2/\Gamma). \tag{3.2.4}$$

Consider the conjugation action of $N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)$ on the left-hand side of eq. (3.2.4), i.e., the set of Γ-cosets, then we further obtain the following bijection between sets

$$\big(\{g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^- | g^2 \in \Gamma\}/\Gamma\big)\big/N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma) \xrightarrow{\sim} \mathrm{API}(\mathbb{C}^2/\Gamma)/\sim. \tag{3.2.5}$$

Corollary 3.2.3. *There are finitely many anti-Poisson involutions on $\mathbb{C}^2/\Gamma$ up to conjugation by Poisson* *automorphisms.*

Proof. We prove the type A case and the non-type A cases separately. In type A_n, with $n \geqslant 1$, we have

$$\Gamma = \left\{ \begin{pmatrix} \epsilon & 0 \\ 0 & \epsilon^{-1} \end{pmatrix} | \epsilon^{n+1} = 1 \right\}$$

is the cyclic group. We can calculate that

$$N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma) = \left\{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} | ad \neq 0 \right\} \bigsqcup \left\{ \begin{pmatrix} 0 & c \\ b & 0 \end{pmatrix} | bc \neq 0 \right\}.$$

Following the bijection in eq. (3.2.4), we have

$$\left\{ \begin{pmatrix} a & 0 \\ 0 & -a^{-1} \end{pmatrix}, \begin{pmatrix} 0 & c \\ c^{-1} & 0 \end{pmatrix} \middle| \, a^{2n+2} = 1 \right\} \Big/ \Gamma \xrightarrow{\sim} \mathrm{API}(\mathbb{C}^2/\Gamma).$$

Note that

$$\left\{ \begin{pmatrix} 0 & c \\ c^{-1} & 0 \end{pmatrix} \middle| \, c \neq 0 \right\}$$

forms a single orbit under the conjugation action of $N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)$. Taking

$$h = \begin{pmatrix} a' & 0 \\ 0 & a'^{-1} \end{pmatrix} \in N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)$$

such that $a'^2 c = 1$, we can calculate that

$$h \begin{pmatrix} 0 & c \\ c^{-1} & 0 \end{pmatrix} h^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Therefore, $\mathrm{API}(\mathbb{C}^2/\Gamma)/\sim$ is a finite set.

For the non-type A cases, we first prove that $N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)$ is a finite group. Note that $N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)$ is an algebraic group. It suffices to show that the connected component $N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)^\circ$ is a singleton. This will follow from the following two observations.

1. The action of the connected group $N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)^\circ$ on the finite group Γ has to be trivial, so $N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)^\circ$

is contained in the centralizer $Z_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)$.

2. The tautological representation $\mathbb{C}^2$ is irreducible as a Γ-representation. According to Schur's Lemma, $Z_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)$ consists of scalar matrices of determinant 1.

Therefore, $N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)^\circ$ only consists of the identity matrix because it is connected. Next, observe that there is a bijection $N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^- \xrightarrow{\sim} N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)$ by sending

$$g \mapsto \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \cdot g.$$

Thus $N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$ is also finite. It follows that the set of anti-Poisson automorphisms on $\mathbb{C}^2/\Gamma$ forms a finite subset, so does $\mathrm{API}(\mathbb{C}^2/\Gamma)$, the set of anti-Poisson involutions. As a corollary, the set of conjugacy classes $\mathrm{API}(\mathbb{C}^2/\Gamma)/\sim$ is also finite. $\square$

Moreover, there exists a representative in each conjugacy class of anti-Poisson involutions that has a simple form. It can be written out explicitly in terms of generators of $\mathbb{C}[X] = \mathbb{C}[u,v]^\Gamma$. We list all the conjugacy classes of anti-Poisson involutions for Kleinian singularities in Propositions 3.3.1, 3.3.2, 3.3.5, 3.3.8, 3.3.11, 3.3.14 and 3.3.17. Moreover, we also explain which element in $N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$ gives the corresponding anti-Poisson involution in the subsequent remark after each of the proposition.

We would also like to know how an anti-Poisson involution/an element in the normalizer acts on the McKay graph of Γ. Recall that $\mathrm{Irr}(\Gamma) = \{V_0, \cdots, V_n\}$ denotes the set of irreducible representations of Γ. Let $g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$ and $V_i \in \mathrm{Irr}(\Gamma)$. We define V_i^g, the twist of the representation V_i under g, as follows. As a vector space $V_i^g = V_i$, but the action of Γ is different.

$$\gamma \cdot v := g\gamma g^{-1}(v),\ \gamma \in \Gamma,\ v \in V_i^g. \tag{3.2.6}$$

Then V_i^g is also an irreducible representation of Γ. Moreover, if $V_i \otimes \mathbb{C}^2 = \oplus_j V_j$, then we have $V_i^g \otimes \mathbb{C}^2 = \oplus_j V_j^g$ because $(\mathbb{C}^2)^g \simeq \mathbb{C}^2$ as Γ-representations, $\forall g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)$. Therefore, the twist eq. (3.2.6) induces an automorphism of the McKay graph of Γ (an extended Dynkin diagram). The twist of the trivial representation V_0 is still the trivial representation, no matter what g is. Thus, we obtain an automorphism of the finite Dynkin diagram labeling Γ (removing the vertex 0). We will discuss what kind of involutions of the Dynkin diagrams we can get in Remarks 3.3.3, 3.3.6, 3.3.9, 3.3.12, 3.3.15 and 3.3.18 when we take $g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$ that induces an anti-Poisson involution on $\mathbb{C}^2/\Gamma$.

3.3 Classification of anti-Poisson involutions

3.3.1 Type A_n

The Poisson algebra of interest is $\mathbb{C}[x,y,z]/(xy - z^{n+1})$. Recall from Propositions 2.3.1 and 2.3.2 that the degrees of the generators are $\deg x = \deg y = n+1,\ \deg z = 2$, and the Poisson brackets among them are given by

$$\{x,y\} = (n+1)^2 z^n,\ \{x,z\} = (n+1)x,\ \{y,z\} = -(n+1)y.$$

Proposition 3.3.1. *The Kleinian singularity of Type A_n, with n odd, has three conjugacy classes of anti-Poisson involutions.*

I. $\theta(x) = y,\ \theta(y) = x,\ \theta(z) = z.$

II. $\theta(x) = x,\ \theta(y) = y,\ \theta(z) = -z.$

III. $\theta(x) = -x,\ \theta(y) = -y,\ \theta(z) = -z.$

Note that when $n = 1$, the Poisson automorphism $x \mapsto \frac{1}{2}(x+y) - z,\ y \mapsto \frac{1}{2}(x+y) + z,\ z \mapsto \frac{1}{2}(x-y)$ conjugates Type II *anti-Poisson involution to Type* I.

Proof. When $n = 1$, we have $\Gamma = \{\pm I_2\}$ and $N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma) = \mathrm{GL}_2(\mathbb{C})$. An element $g \in \mathrm{GL}_2(\mathbb{C})$ gives rise to an anti-Poisson involution if and only if $\det(g) = -1$ and $g^2 = \pm I_2$ by Proposition 3.2.1 and Remark 3.2.2. It follows that g is diagonalizable, and its eigenvalues lie in $\{1, -1, i, -i\}$ with the requirement that the product of the two eigenvalues is -1. Then up to multiplication by Γ and conjugation by $N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma) = \mathrm{SL}_2(\mathbb{C})$, the element g is either $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ or $\begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}$, which correspond to Type I and Type III anti-Poisson involutions respectively. It is obvious that they are not conjugate to each other.

When $n > 1$, it is easy to verify that the involutions I $-$ III are anti-Poisson using the Poisson brackets given in Proposition 2.3.2. Conversely, we show that any anti-Poisson involution can be conjugated to one of Types I $-$ III. First, notice that we have $\theta(z) = \pm z$ because θ is a graded involution, and we have $\mathbb{C}[X]_2 = \mathbb{C}z$. Consider the operator defined by $\{z, \cdot\}$ on $\mathbb{C}[X]$. It is diagonalizable with integral eigenvalues, therefore produces a new grading on $\mathbb{C}[X]$, which we denote by $\deg_z := \{z, \cdot\}$. Under this new grading, we have $\deg_z(x) = -(n+1),\ \deg_z(y) = n+1,\ \deg_z(z) = 0$. When $\theta(z) = z$, it follows from

$$\{z, \theta(f)\} = \{\theta(z), \theta(f)\} = -\theta(\{z,f\}),\ \forall f \in \mathbb{C}[X], \tag{3.3.1}$$

that θ reverses the new grading $\deg_z$. Thus we must have

$$\theta(x) = ay,\ \theta(y) = bx,\ a, b \in \mathbb{C}.$$

Moreover, using that $\theta^2(x) = x,\ \theta^2(y) = y$, we get $ab = 1$. It is easy to verify that $\theta : x \mapsto ay,\ y \mapsto a^{-1}x,\ z \mapsto z$ is an anti-Poisson involution on $\mathbb{C}[X]$ and its conjugate $\sigma\theta\sigma^{-1}$ is the Type I anti-involution, where σ is the Poisson automorphism $x \mapsto a'x,\ y \mapsto a'^{-1}y,\ z \mapsto z$ such that $a'^2 a = 1$.

When $\theta(z) = -z$, similar to eq. (3.3.1), we can show that θ preserves the new grading $\deg_z$. Thus we must have

$$\theta(x) = ax,\ \theta(y) = by,\ a, b \in \mathbb{C}. \tag{3.3.2}$$

Moreover, using that $\theta^2(x) = x,\ \theta(y) = y$, we get $a^2 = b^2 = 1$. Plugging eq. (3.3.2) into

$$\theta(\{x, y\}) = -\{\theta(x), \theta(y)\},$$

we get $ab = 1$. The two solutions $a = b = 1$ and $a = b = -1$ correspond to Type II and Type III anti-Poisson involutions respectively.

Finally, we show that Types I $-$ III anti-Poisson involutions are not conjugate to each other. Any graded Poisson automorphism has to send z to a scalar multiple of z because $\mathbb{C}[X]_2 = \mathbb{C}z$, so it can not conjugate an anti-Poisson involution sending z to $-z$ (Types II, III) to those sending z to z (Type I). Furthermore, the fixed point locus of Type III anti-Poisson involution is the singular point, while the fixed point locus of Type II anti-Poisson clearly contains more than one point. Hence, Types II and III anti-Poisson involutions are not conjugate to each other, either. $\square$

Proposition 3.3.2. *The Kleinian singularity of Type A_n, with n even, has two conjugacy classes of anti-Poisson involutions.*

I. $\theta(x) = y,\ \theta(y) = x, \theta(z) = z.$

II. $\theta(x) = -x,\ \theta(y) = y, \theta(z) = -z.$

Proof. The proof is pretty similar to Proposition 3.3.1. We leave it to the reader. $\square$

Remark 3.3.3. 1. According to Remark 3.2.2, every anti-Poisson involution on X comes from an element in $N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)$ with determinant -1. We list here the elements in the normalizer that give rise to the anti-Poisson involutions for type A_n. When n is odd, the matrices $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} \epsilon & 0 \\ 0 & -\epsilon^{-1} \end{pmatrix}$ give rise to anti-Poisson involutions of Types $\mathrm{I}, \mathrm{II}, \mathrm{III}$ in Proposition 3.3.1 respectively, where ϵ is a $(2n+2)$-th primitive root of unity. When n is even, the matrices $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ give rise an anti-Poisson involutions of Types I, II in Proposition 3.3.2 respectively.

2. It is easy to see that the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ induces the automorphism of Dynkin diagram A_n by swapping the vertices $i, n+1-i$ (in both n even and odd cases), via twisting the irreducible representations of Γ as defined in eq. (3.2.6). The matrices $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} \epsilon & 0 \\ 0 & -\epsilon^{-1} \end{pmatrix}$ act trivially on the set $\mathrm{Irr}(\Gamma)$ because they commute with Γ, hence induce the trivial automorphisms on the Dynkin diagram A_n.

3. Consider the simple Lie algebra of type A_n, i.e. the Lie algebra $\mathfrak{g} = \mathfrak{sl}_{n+1}(\mathbb{C})$. Let $\mathcal{N}$ denote the nilpotent cone of $\mathfrak{g}$ that coincides with the closure of the regular nilpotent orbit, and S the Slodowy slice to a subregular nilpotent element e. Then we have $S \cap \mathcal{N} \simeq X$, the type A_n Kleinian singularity (c.f. [Bri71]). Let $\sigma\colon \mathfrak{g} \to \mathfrak{g}$ be a Lie algebra involution with the decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. As we discussed in Section 1.2.1, if we further assume that $e \in \mathfrak{p}$ and that the $\mathfrak{sl}_2$-triple associated with S is normal, then $\theta := -\sigma$ restricts to an anti-Poisson involution on $S \cap \mathcal{N} \simeq X$, and we have that $S \cap \mathcal{N} \cap \mathfrak{p} \simeq X^{\theta}$.

   We list below which Lie algebra involution σ an anti-Poisson involution in Propositions 3.3.1 and 3.3.2 corresponds to. In Table 3.1, we use M to denote a matrix in $\mathfrak{sl}_{n+1}(\mathbb{C})$, and M^t to denote its transpose. The matrix $I_{k,l}$ refers to a diagonal matrix with k-entries of 1 and l-entries of –1. The correspondence in Table 3.1 is not trivial, but not very hard. We leave it as an exercise to the reader.

	Type I	Type II	Type III
n odd	$\sigma : M \mapsto -M^t$	$\sigma : M \mapsto \mathrm{Ad}\, I_{\frac{n+1}{2},\frac{n+1}{2}}(M)$	$\sigma : M \mapsto \mathrm{Ad}\, I_{\frac{n+3}{2},\frac{n-1}{2}}(M)$
n even	$\sigma : M \mapsto -M^t$	$\sigma : M \mapsto \mathrm{Ad}\, I_{\frac{n+2}{2},\frac{n}{2}}(M)$	N/A

Table 3.1: Involutions of $\mathfrak{sl}_{n+1}$

   We make some comments to the case $n = 1$. All involutions of $\mathfrak{sl}_2(\mathbb{C})$ are inner because the Dynkin diagram A_1 has no nontrivial automorphism. The Type I, II anti-Poisson involutions conjugate to each other as mentioned in Proposition 3.3.1, which agrees with the fact that the corresponding Lie algebra involutions are conjugate. The Type III anti-Poisson involution comes from the trivial Lie algebra involution $\sigma = \operatorname{Ad} I_{2,0} = \mathrm{id}$.

A direct calculation gives the following description of the fixed point loci in Type A_n.

Corollary 3.3.4. *The fixed point loci of anti-Poisson involutions in Type A_n are of the following forms.*

When n is odd,

I. $J = (x - y)$. $X^\theta = \operatorname{Spec} \mathbb{C}[x, z]/(x^2 - z^{n+1})$*, the union of two copies of $\mathbb{A}^1$'s.*

II. $J = (z)$. $X^\theta = \operatorname{Spec} \mathbb{C}[x, y]/(xy)$*, the union of two copies of $\mathbb{A}^1$'s.*

III. $J = (x, y, z)$. $X^\theta = \operatorname{Spec} \mathbb{C}$*, a point.*

When n is even,

I. $J = (x - y)$. $X^\theta = \operatorname{Spec} \mathbb{C}[x, z]/(x^2 - z^{n+1})$*, a cusp.*

II. $J = (x, z)$. $X^\theta = \operatorname{Spec} \mathbb{C}[y] \simeq \mathbb{A}^1$.

Moreover, all the fixed point loci are reduced as schemes.

3.3.2 Type D_n, n even

The Poisson algebra of interest is $\mathbb{C}[X] = \mathbb{C}[x, y, z]/(xy(y - x^{\frac{n-2}{2}}) - z^2)$. Recall from Proposition 2.3.1 and Proposition 2.3.2 that the degrees of the generators are $\deg x = 4,\ \deg y = 2n - 4,\ \deg z = 2n - 2$, and the Poisson brackets among them are given by

$$\{x, y\} = (4n - 8)z,\ \{x, z\} = (2n - 4)x(2y - x^{\frac{n-2}{2}}),\ \{y, z\} = (n - 2)y(nx^{\frac{n-2}{2}} - 2y).$$

Proposition 3.3.5. *The Kleinian singularity of Type D_n, with n even, has two conjugacy classes of anti-Poisson involutions.*

I. $\theta(x) = x,\ \theta(y) = y,\ \theta(z) = -z.$

II. $\theta(x) = x,\ \theta(y) = x^{\frac{n-2}{2}} - y,\ \theta(z) = z.$

Proof. When $n = 4$, we have $\Gamma = \Big\{ \pm I_2,\ \pm \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix},\ \pm \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},\ \pm \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \Big\}$. An element $g \in N_{\mathrm{GL}_2 \mathbb{C}}(\Gamma)$ gives rise to an anti-Poisson involution if and only if $\det(g) = -1$ and $g^2 \in \Gamma$ by Proposition 3.2.1 and Remark 3.2.2. Then up to multiplication by Γ and conjugation by $N_{\mathrm{SL}_2(\mathbb{C})}(\Gamma)$, the element

g is either $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\begin{pmatrix} \epsilon_8 & 0 \\ 0 & \epsilon_8^3 \end{pmatrix}$, where ϵ_8 is an 8-th primitive root of unity. Using the formulas of the generators x, y, z given in Proposition 2.3.1, we see that the two matrices correspond to the Type I and the Type II anti-Poisson involutions respectively. It is obvious that they are not conjugate to each other.

When $n > 4$, it is easy to verify that the involutions I, II are anti-Poisson using the Poisson brackets given in Proposition 2.3.2. Conversely, let θ be an anti-Poisson involution of $\mathbb{C}[X]$. Note that $\deg z$ is not an integral linear combination of $\deg x$ and $\deg y$. Thus we may assume

$$\theta(x) = ax, \quad \theta(y) = by + b'x^{\frac{n-2}{2}}, \quad \theta(z) = cz,$$

as θ is graded. We have $\theta^2(x) = x,\ \theta^2(y) = y,\ \theta^2(z) = z$ as θ is an involution, which implies that $a^2 = b^2 = c^2 = 1$. From

$$0 = \theta(xy(y - x^{\frac{n-2}{2}}) - z^2) = axy^2 + ab(2b' - a^{\frac{n-2}{2}})yx^{\frac{n}{2}} + ab'(b' - a^{\frac{n-2}{2}})x^{n-1} - z^2,$$

we get $a = 1,\ ab(2b' - a^{\frac{n-2}{2}}) = -1,\ ab'(b' - a^{\frac{n-2}{2}}) = 0$ because $xy(y + 4x^{\frac{n-2}{2}}) - z^2 = 0$ is the only relation of degree $4n - 8$ in $\mathbb{C}[X]$ up to scaling. By plugging $a = 1$ into the latter two equations, we get $b(2b' - 1) = -1,\ b'(b' - 1) = 0$. There are only two solutions. The first is $b' = 0,\ b = 1$. Using

$$\{\theta(x), \theta(y)\} = -\theta(\{x, y\}) \tag{3.3.3}$$

as θ is anti-Poisson, we can further deduce that $c = -1$. This corresponds to Type I anti-Poisson involution. The second solution is $b' = 1,\ b = -1$. With a similar procedure as above, we can obtain that $c = 1$. This corresponds to the Type II anti-Poisson involution.

To see that Types I, II anti-Poisson involutions are non-conjugate, note that any graded automorphism has to send z to a scalar multiple of z because $\mathbb{C}[X]_{2n-2} = \mathbb{C}z$. Thus, an anti-involution sending z to $-z$ (Type I) can not conjugate to that sending z to z (Type II). □

Remark 3.3.6. (1) According to Proposition 3.2.1, every anti-Poisson involution on $\mathbb{C}^2/\Gamma$ should come from an element in $N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$. We list here the elements in the normalizer that give rise to the anti-Poisson involutions for type D_n, with n even. The matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ gives rise to the Type I anti-Poisson involution in Proposition 3.3.5, and the matrix $\begin{pmatrix} \epsilon & 0 \\ 0 & -\epsilon^{-1} \end{pmatrix}$ gives rise to the Type II anti-Poisson involution in Proposition 3.3.5, where ϵ is a $(4n-8)$-th primitive root of unity.

(2) To determine what kinds of automorphisms the matrices in (1) give rise to, we only need to look at their twists on the four 1-dimensional representations of Γ (including one trivial representation). Then it is not hard to discover that the matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ induces the trivial automorphism of the Dynkin diagram D_n, and the matrix $\begin{pmatrix} \epsilon & 0 \\ 0 & -\epsilon^{-1} \end{pmatrix}$ swaps two short arms of the Dynkin diagram D_n (when n is even). We can further determine which two vertices/1-dimensional representations are being swapped in type D_4, which is left as an exercise to the reader.

Corollary 3.3.7. *The fixed point loci of anti-Poisson involutions in Type D_n, with n even, are of the following forms.*

I. $J = (z)$. The fixed point locus $X^\theta = \operatorname{Spec}\mathbb{C}[x,y]/(xy(y - x^{\frac{n-2}{2}}))$ is the union of three copies of $\mathbb{A}^1$'s intersecting at the singular point.

II. $J = (2y - x^{\frac{n-2}{2}})$. The fixed point locus $X^\theta = \operatorname{Spec}\mathbb{C}[x,z]/(x^{n-1} + 4z^2)$ is a cusp.

Moreover, both fixed point loci are reduced as schemes.

3.3.3 Type $D_n,\ n$ odd

The Poisson algebra of interest is $\mathbb{C}[X] = \mathbb{C}[x,y,z]/(xy^2 - z(z - x^{\frac{n-1}{2}}))$. Recall from Proposition 2.3.1 and Proposition 2.3.2 that the degrees of the generators are $\deg x = 4,\ \deg y = 2n-4,\ \deg z = 2n-2$, and the Poisson brackets among them are given by

$$\{x,y\} = (2n-4)(2z - x^{\frac{n-1}{2}}),\ \{x,z\} = (4n-8)xy,\ \{y,z\} = -(n-2)((n-1)x^{\frac{n-3}{2}}z + 2y^2).$$

Proposition 3.3.8. *The Kleinian singularity of Type D_n, with n odd, has two conjugacy classes of anti-Poisson involutions.*

I. $\theta(x) = x,\ \theta(y) = -y,\ \theta(z) = z.$

II. $\theta(x) = x,\ \theta(y) = y,\ \theta(z) = x^{\frac{n-1}{2}} - z.$

Proof. The proof is pretty similar to Proposition 3.3.5. We leave it to the reader. $\square$

Remark 3.3.9. 1. According to Remark 3.2.2, every anti-Poisson involution on $\mathbb{C}^2/\Gamma$ comes from an element in $N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$. We list here the elements in the normalizer that give rise to the anti-Poisson involutions for type D_n, with n odd. The matrix $\begin{pmatrix} \epsilon & 0 \\ 0 & -\epsilon^{-1} \end{pmatrix}$ gives rise to the Type I anti-Poisson involution in Proposition 3.3.8, where ϵ is a $(4n-8)$-th primitive root of unity. The matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ gives rise to the Type II anti-Poisson involution in Proposition 3.3.8.

2. Similar to Remark 3.3.6, one can show that the matrix $\begin{pmatrix} \epsilon & 0 \\ 0 & -\epsilon^{-1} \end{pmatrix}$ induces the trivial automorphism of the Dynkin diagram D_n, and the matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ swaps the two short arms of the Dynkin diagram D_n (when n is odd).

Corollary 3.3.10. *The fixed point loci of anti-Poisson involutions in Type D_n, with n odd, are of the following forms.*

I. $J = (y)$. *The fixed point locus $X^\theta = \operatorname{Spec} \mathbb{C}[x, z]/(z(z - x^{\frac{n-1}{2}}))$ is the union of two copies of $\mathbb{A}^1$'s intersecting at the singular point.*

II. $J = (2z - x^{\frac{n-1}{2}})$. *The fixed point locus $X^\theta = \operatorname{Spec} \mathbb{C}[x, y]/(x(4y^2 + x^{n-2}))$ is the union of $\mathbb{A}^1$ and a cusp, intersecting at the singular point.*

Moreover, both fixed point loci are reduced as schemes.

We have finished the classification of conjugacy classes of anti-Poisson involutions in the classical types. Next, we consider the three exceptional types.

3.3.4 Type E_6

The Poisson algebra of interest is $\mathbb{C}[X] = \mathbb{C}[x, y, z]/(z^2 + zx^2 + y^3)$. Recall from Proposition 2.3.1 and Proposition 2.3.2 that the degrees of the generators are $\deg x = 6,\ \deg y = 8,\ \deg z = 12$, and the Poisson brackets among them are given by

$$\{x, y\} = 24\sqrt{3}(1 + i)(2z + x^2),\ \{x, z\} = -72\sqrt{3}(1 + i)y^2,\ \{y, z\} = 48\sqrt{3}(1 + i)xz.$$

Proposition 3.3.11. *The Kleinian singularity of Type E_6 has two conjugacy classes of anti-Poisson involutions.*

I. $\theta(x) = -x,\ \theta(y) = y,\ \theta(z) = z.$

II. $\theta(x) = x,\ \theta(y) = y,\ \theta(z) = -z - x^2.$

Proof. It is easy to verify the involutions I, II anti-Poisson using the Poisson brackets given in Proposition 2.3.2. Conversely, let θ be an anti-Poisson involution of $\mathbb{C}[X]$. We may assume

$$\theta(x) = ax,\ \theta(y) = by,\ \theta(z) = cz + c'x^2.$$

as θ is graded. We have $\theta^2(x) = x,\ \theta^2(y) = y,\ \theta^2(z) = z$, which implies that $a^2 = b^2 = c^2 = 1$. From

$$0 = \theta(z^2 + zx^2 + y^3) = c^2z^2 + c(2c' + a^2)zx^2 + b^3y^3 + c'(c' + a^2)x^4,$$

we get $b^3 = 1,\ c(2c' + a^2) = 1$ using the relation $c^2 = 1$ and the fact that $z^2 + zx^2 + y^3 = 0$ is the only relation of degree 24 in $\mathbb{C}[X]$ up to scaling. Together with $b^2 = 1$, we can deduce that $b = 1$. Using

$$\{\theta(x), \theta(z)\} = -\theta(\{x, z\})$$

as θ is anti-Poisson, we can further determine that $ac = -1$. Setting $a = -1,\ c = 1$, we get $c' = 0$ from $c(2c' + a^2) = 1$, which corresponds to Type I anti-Poisson involution. Setting $a = 1,\ c = -1$, we get $c' = -1$ from $c(2c' + a^2) = 1$, which corresponds to Type II anti-Poisson involution.

To see that Types I, II anti-Poisson involutions are non-conjugate, note that any graded automorphism has to send x to a scalar multiple of x because $\mathbb{C}[X]_6 = \mathbb{C}x$. Thus, an anti-involution sending x to $-x$ (Type I) can not conjugate to that sending x to x (Type II). $\square$

Remark 3.3.12. (1) According to Remark 3.2.2, every anti-Poisson involution on $\mathbb{C}^2/\Gamma$ comes from an element in $N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$. We list here the elements in the normalizer that give rise to the anti-Poisson involutions for the Kleinian singularity of type E_6. The matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ gives rise to Type I anti-Poisson involution in Proposition 3.3.11, and the matrix $\begin{pmatrix} \epsilon_8 & 0 \\ 0 & -\epsilon_8^{-1} \end{pmatrix}$ gives rise to Type II anti-Poisson involution in Proposition 3.3.11, where ϵ_8 is a 8-th primitive root of unity. It is not quite obvious to see that both $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} \epsilon_8 & 0 \\ 0 & -\epsilon_8^{-1} \end{pmatrix}$ are in $N^-_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)$. One way is to verify that they preserve $\mathbb{C}[X] = \mathbb{C}[u,v]^\Gamma$ by using the formulas of the fundamental invariants x, y, z given in Proposition 2.3.1 and deduce that they lie in $N^-_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)$ by Proposition 3.2.1, which is faster than computing that they normalize Γ directly.

(2) Similar to Remark 3.3.6, we can obtain the actions of the matrices in (1) on the Dynkin diagram E_6 by studying their twists on the three 1-dimension representations of the finite subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$ of type E_6, or through looking at the character table of Γ given in [Dol09, Example 5.2.3]. The matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ induces the trivial action on the Dynkin diagram E_6, and the matrix $\begin{pmatrix} \epsilon_8 & 0 \\ 0 & -\epsilon_8^{-1} \end{pmatrix}$ acts on the finite Dynkin diagram E_6 by swapping the two longer arms.

Corollary 3.3.13. *The fixed point loci of anti-Poisson involutions in Type E_6 are of the following forms.*

I. $J = (x)$. *The fixed point locus $X^\theta = \operatorname{Spec} \mathbb{C}[y,z]/(z^2 + y^3)$ is a cusp.*

II. $J = (2z + x^2)$. *The fixed point locus $X^\theta = \operatorname{Spec} \mathbb{C}[x,y]/(x^4 - 4y^3)$ is a cusp.*

Moreover, both fixed point loci are reduced as schemes.

3.3.5 Type E_7

The Poisson algebra of interest is $\mathbb{C}[X] = \mathbb{C}[x,y,z]/(x^3y + y^3 + z^2)$. Recall from Proposition 2.3.1 and Proposition 2.3.2 that the degrees of the generators are $\deg x = 8,\ \deg y = 12,\ \deg z = 18$, and the Poisson brackets among them are given by

$$\{x,y\} = -96z,\ \{x,z\} = 48(x^3 + 3y^2),\ \{y,z\} = -144x^2y.$$

Proposition 3.3.14. *The Kleinian singularity of Type E_7 has a unique anti-Poisson involution (up to conjugation).*

$$\theta(x) = x,\ \theta(y) = y,\ \theta(z) = -z$$

Proof. It is easy to verify the above automorphism is indeed an anti-Poisson involution of $\mathbb{C}[X]$. Conversely, let θ be an anti-Poisson involution of $\mathbb{C}[X]$. We may assume

$$\theta(x) = ax,\ \theta(y) = by,\ \theta(z) = cz.$$

as θ is graded. $\theta^2 = \mathrm{id}$ gives us that $a^2 = b^2 = c^2 = 1$. From

$$0 = \theta(x^3y + y^3 + z^2) = abx^3y + b^3y^3 + c^2z^2 = abx^3y + by^3 + z^2$$

we get $ab = b = 1$, hence $a = b = 1$, because $x^3y + y^3 + z^2 = 0$ is the only relation of degree 36 in $\mathbb{C}[X]$ up to scaling. Using the condition that θ is anti-Poisson, we have

$$\{\theta(x), \theta(y)\} = -\theta(\{x, y\}),$$

from which we can determine that $c = -1$. □

Remark 3.3.15. The matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$ gives rise to the unique anti-Poisson involution of the Kleinian singularity of type E_7. There is no nontrivial antomorphism on the Dynkin diagram E_7, thus the matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ induces the trivial action on the Dynkin diagram E_7.

Corollary 3.3.16. *The vanishing ideal of the anti-Poisson involution in Type E_7 is $J = (z)$. The corresponding fixed point locus is $X^\theta = \operatorname{Spec} \mathbb{C}[x, y]/(y(x^3 + y^2))$, which is the union of $\mathbb{A}^1$ and a cusp, intersecting at the singular point. Moreover, the fixed point locus is reduced as scheme.*

3.3.6 Type E_8

In Type E_8, the Poisson algebra of interest is $\mathbb{C}[X] = \mathbb{C}[x, y, z]/(x^5+y^3+z^2)$. Recall from Proposition 2.3.1 and Proposition 2.3.2 that the degrees of the generators are $\deg x = 12,\ \deg y = 20,\ \deg z = 30$, and the Poisson brackets among them are given by

$$\{x, y\} = -120z,\ \{x, z\} = 180y^2,\ \{y, z\} = -300x^4.$$

Proposition 3.3.17. *The Kleinian singularity of Type E_8 has a unique anti-Poisson involution (up to conjugation).*

$$\theta(x) = x,\ \theta(y) = y,\ \theta(z) = -z$$

Proof. The proof repeats that of type E_7 in Proposition 3.3.14. □

Remark 3.3.18. The matrix $\begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$ gives rise to the unique anti-Poisson involution of the Kleinian singularity of type E_8. There is no nontrivial antomorphism on the Dynkin diagram E_8, thus the matrix $\begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}$ induces the trivial action on the Dynkin diagram E_8.

Corollary 3.3.19. *The vanishing ideal of the anti-Poisson involution in Type E_8 is $J = (z)$, and the corresponding fixed point locus $X^\theta = \operatorname{Spec} \mathbb{C}[x, y]/(x^5 + y^3)$ is a cusp. Moreover, the fixed point locus is reduced as a scheme.*

Chapter 4

Preimages of Fixed Point Loci: Quiver Variety Approach

4.1 Nakajima quiver varieties

We will describe the preimages of the fixed point loci from Corollaries 3.3.4, 3.3.7, 3.3.10, 3.3.13, 3.3.16 and 3.3.19 through the realization of Kleinian singularities as Nakajima quiver varieties. Let us briefly review the construction of Nakajima quiver varieties, following [Nak94; Nak98].

4.1.1 Representations of quivers

By a quiver $Q = (I, \Omega)$ we mean an oriented graph with the vertex set $I = \{0, 1, \cdots, n\}$ and the edge set Ω. For an arrow $a\colon i \to j \in \Omega$, we denote by $t(a) = i \in I$ its initial vertex (tail) and by $h(a) = j \in I$ its terminal vertex (head). The quivers we care about are extended Dynkin quivers of types ADE. Let $V = (V_i)_{i\in I}$ be a collection of finite-dimensional vector spaces corresponding to the vertices. The dimension of V is a row vector

$$\dim V = (\dim V_0, \dim V_1, \cdots, \dim V_n) \in \mathbb{Z}^{n+1}_{\geqslant 0}.$$

For two collections of vector spaces V and W with $\delta = \dim V,\ w = \dim W$, we can consider the *framed representations* of the quiver Q.

$$R(Q, \delta, w) := \bigoplus_{a\in\Omega} \mathrm{Hom}_{\mathbb{C}}(V_{t(a)}, V_{h(a)}) \oplus \bigoplus_{i\in I} \mathrm{Hom}_{\mathbb{C}}(W_i, V_i).$$

Next, we consider the doubled quiver $\bar{Q} = (I, \bar{\Omega})$, where $\bar{\Omega} = \Omega \sqcup \Omega^*$, and $\Omega \xrightarrow{\sim} \Omega^*,\ a \mapsto a^*$, is the reversal of the orientation. The framed representations of the doubled quiver $\bar{Q}$ is

$$M(\delta, w) := \bigoplus_{a\in\Omega} \big(\mathrm{Hom}(V_{t(a)}, V_{h(a)}) \oplus \mathrm{Hom}(V_{h(a)}, V_{t(a)})\big) \oplus \bigoplus_{i\in I} \mathrm{Hom}(W_i, V_i) \oplus \bigoplus_{i\in I} \mathrm{Hom}(V_i, W_i). \tag{4.1.1}$$

An element in $M(\delta, w)$ can be considered as a quadruple (B, B^*, l, k) with

$$\begin{aligned} B = (B_a)_{a\in\Omega} &\in \bigoplus_{a\in\Omega} \mathrm{Hom}(V_{t(a)}, V_{h(a)}), \quad B^* = (B_{a^*})_{a\in\Omega} \in \bigoplus_{a\in\Omega} \mathrm{Hom}(V_{h(a)}, V_{t(a)}), \\ l = (l_i)_{i\in I} &\in \bigoplus_{i\in I} \mathrm{Hom}(W_i, V_i), \qquad k = (k_i)_{i\in I} \in \bigoplus_{i\in I} \mathrm{Hom}(V_i, W_i). \end{aligned} \tag{4.1.2}$$

We have $M(\delta, w) \simeq T^*R(Q, \delta, w)$ by identifying $\mathrm{Hom}_{\mathbb{C}}(V_i, V_j)^*$ with $\mathrm{Hom}_{\mathbb{C}}(V_j, V_i)$ via the trace form $(A, B) := \mathrm{tr}(AB)$. Then the cotangent bundle $M(\delta, w)$ is naturally endowed with a symplectic form

$$\omega((B_a, B_{a^*}, l, k), (B'_a, B'_{a^*}, l', k')) = \sum_{i\in I} \Big(\sum_{\substack{a\in\Omega \\ h(a)=i}} \mathrm{tr}(B_a B'_{a^*}) - \sum_{\substack{a\in\Omega \\ t(a)=i}} \mathrm{tr}(B_{a^*} B'_a) + \mathrm{tr}(l_i k'_i - k_i l'_i) \Big), \tag{4.1.3}$$

where $(B_a, B_{a^*}, l, k), (B'_a, B'_{a^*}, l', k')$ are regarded as tangent vectors.

Set $\mathrm{GL}(\delta) := \prod_{i\in I} \mathrm{GL}(V_i)$ and $\mathfrak{gl}(\delta) := \prod_{i\in I} \mathfrak{gl}(V_i)$. Given $g = (g_i)_{i\in I} \in \mathrm{GL}(\delta)$, it acts on the symplectic vector space $M(\delta, w)$ via

$$g \cdot B_a = g_{h(a)} B_a g_{t(a)}^{-1},\ g \cdot B_{a^*} = g_{t(a)} B_{a^*} g_{h(a)}^{-1},\ g \cdot l_i = g_i l_i,\ g \cdot k_i = k_i g_i^{-1}. \tag{4.1.4}$$

The $\mathrm{GL}(\delta)$-action on $M(\delta, w)$ preserves the symplectic form eq. (4.1.3), and is Hamiltonian. We can consider the corresponding moment map $\mu\colon M(\delta, w) \to \mathfrak{gl}(\delta)$, where $\mu = (\mu_i)_{i\in I}$, with

$$\mu_i(B, B^*, l, k) = \sum_{\substack{a\in\Omega, \\ h(a)=i}} B_a B_{a^*} - \sum_{\substack{a\in\Omega, \\ t(a)=i}} B_{a^*} B_a + l_i k_i. \tag{4.1.5}$$

For a reference to eq. (4.1.5), one may consult [Nak98, Section 3.i]. The equation $\mu = 0$ is also called the "ADHM equation", referring to Atiyah-Drinfeld-Hitchin-Manin. A *character* is a group homormophism $\chi\colon\ \mathrm{GL}(\delta) \to \mathbb{C}^*$. It is given by

$$\chi(g) = \prod_{i\in I} \det(g_i)^{\chi_i}, \text{ where } g = (g_i)_{i\in I},\ \chi = (\chi_i)_{i\in I} \in \mathbb{Z}^I$$

because the only nontrivial characters of $\mathrm{GL}_m(\mathbb{C})$ are integral powers of the determinant map and $\mathrm{GL}(\delta)$ is just a product of $\mathrm{GL}_m(\mathbb{C})$'s. Let $\lambda = (\lambda_i)_{i\in I} \in \mathfrak{gl}(\delta)^{\mathrm{GL}(\delta)}$. Then the fiber $\mu^{-1}(\lambda)$ is $\mathrm{GL}(\delta)$-invariant. Consider the *χ-semi-invariants* of weight m,

$$\mathbb{C}[\mu^{-1}(\lambda)]^{\mathrm{GL}(\delta),m\chi} := \{f \in \mathbb{C}[\mu^{-1}(\lambda)] | f(g^{-1}(B, B^*, l, k)) = \chi(g)^m f((B, B^*, l, k))\}. \tag{4.1.6}$$

The direct sum of $\mathbb{C}[\mu^{-1}(\lambda)]^{\mathrm{GL}(\delta),m\chi}$ with respect to $m \in \mathbb{Z}_{\geqslant 0}$ is a graded algebra. A *Nakajima quiver variety* is the GIT quotient

$$\mathcal{M}_\chi^\lambda(\delta, w) := \mu^{-1}(\lambda)//^\chi \mathrm{GL}(\delta) = \mathrm{Proj} \bigoplus_{m\geqslant 0} \mathbb{C}[\mu^{-1}(\lambda)]^{\mathrm{GL}(\delta),m\chi}. \tag{4.1.7}$$

When χ equals the trivial character $(0, \cdots, 0)$, we have

$$\mathcal{M}_0^\lambda(\delta, w) = \mu^{-1}(\lambda)// \mathrm{GL}(\delta) = \mathrm{Spec}\, \mathbb{C}[\mu^{-1}(\lambda)]^{\mathrm{GL}(\delta)}.$$

Moreover, there is a natural projective morphism

$$\pi \colon \mathcal{M}_\chi^\lambda(\delta, w) \to \mathcal{M}_0^\lambda(\delta, w). \tag{4.1.8}$$

In this paper, we mainly focus on the zero fiber of the moment map. We denote $\mathcal{M}_\chi(\delta, w) := \mathcal{M}_\chi^0(\delta, w)$ for simplicity. We use $[B, B^*, l, k]$ to denote the $\mathrm{GL}(\delta)$-orbit of a point (B, B^*, l, k).

Remark 4.1.1. If we take the framing vector w to be zero, then we get the usual representation space of quivers. The $\mathrm{GL}(\delta)$-action and the moment map μ pass along with $l = 0,\ k = 0$. We can follow our previous notation to denote it as $M(\delta, 0)$.

Next, we discuss the χ-semistable points. If f is a semi-invariant, then the set of non-vanishing locus $\mu^{-1}(0)_f$ is a $\mathrm{GL}(\delta)$-invariant open subset of $\mu^{-1}(0)$. We define the *χ-semistable locus*

$$\mu^{-1}(0)^{\chi-ss} := \{u \in \mu^{-1}(0) | \exists f \in \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta),m\chi} \text{ for } m > 0 \text{ s.t. } f(u) \neq 0\}. \tag{4.1.9}$$

We have $\mu^{-1}(\lambda)^{\chi-ss} = \bigcup_f \mu^{-1}(0)_f$, where f runs through all the nonzero semi-invariants. King first interprets χ-semistability of representations in terms of subrepresentations in [Kin94]. Nakajima gives the following description of the semistable locus for the character $\chi = (-1, \cdots, -1)$.

Proposition 4.1.2 ([Nak98], Lemma 3.8). *Let $\chi = (-1, \cdots, -1)$. Then a point (B, B^*, l, k) is χ-semistable if and only if $\mathrm{im}(l)$ generates the entire collection of vector spaces $V = (V_i)_{i\in I}$ under the action of (B, B^*).*

Corollary 4.1.3. *Let $\chi = (-1, \cdots, -1)$. Then we have*

1. $\mathrm{GL}(\delta)$ *acts on* $\mu^{-1}(0)^{\chi-ss}$ *freely, and each* $\mathrm{GL}(\delta)$*-orbit is closed.*
2. *We have* $\mathcal{M}_\chi(\delta, w) \simeq \mu^{-1}(0)^{\chi-ss} /\!/ \mathrm{GL}(\delta)$*, and it is smooth.*
3. *The symplectic form eq.* (4.1.3) *induces a symplectic form on* $\mathcal{M}_\chi(\delta, w)$.

4.1.2 Applications to Kleinian singularities

Take $Q = (I, \Omega)$ to be an extended Dynkin quiver of types ADE, i.e., the vertex set I is given by the affine simple roots and the edge set Ω is given by the extended Dynkin diagram with some orientation. Let $\alpha_0, \alpha_1, \cdots, \alpha_n$ denote the set of simple affine roots. Then there is a unique relation $\sum_{0\leqslant i\leqslant n} \delta_i\alpha_i = 0$ where δ_i are nonnegative integers with $\delta_0 = 1$. Following the notations in Section 4.1.1, we consider the space $M(\delta, w)$ with the dimension and framing vectors

$$\delta = (\delta_0, \delta_1, \cdots, \delta_n),\ w = (1, 0, \cdots, 0), \tag{4.1.10}$$

i.e. there is a single framing of dimension 1 attached to the extended vertex 0. We have $l = (l_0, 0, \cdots, 0),\ k = (k_0, 0, \cdots 0)$. Moreover, the moment map eq. (4.1.5) becomes

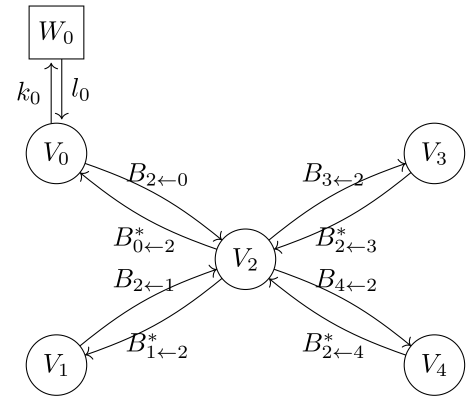
$$\mu_i(B_a, B_{a^*}, l, k) = \sum_{\substack{a\in\Omega,\\ h(a)=i}} B_aB_{a^*} - \sum_{\substack{a\in\Omega,\\ t(a)=i}} B_{a^*}B_a + \delta_{i,0}l_0k_0,\ i \in I, \tag{4.1.11}$$

where $\delta_{i,0}$ means the Kronecker delta. Setting eq. (4.1.11) equal to zero, we can further get the ADHM equation

$$\sum_{\substack{a\in\Omega,\\ h(a)=i}} B_aB_{a^*} - \sum_{\substack{h\in\Omega,\\ t(a)=i}} B_{a^*}B_a = l_0k_0 = 0,\ 0 \leqslant i \leqslant n. \tag{4.1.12}$$

by summing up all $\sum_{i=0}^n \mathrm{tr}(\mu_i) = 0$ and using the fact that V_0 is 1-dimensional. Moreover, the χ-semistability condition in Proposition 4.1.2 implies that if $(B, B^*, l_0, k_0) \in \mu^{-1}(0)^{\chi-ss}$, then $l_0 \neq 0,\ k_0 = 0$.

Example 4.1.4. For the extended Dynkin quiver $\widetilde{D}_4$, we have $\delta = (1, 1, 2, 1, 1),\ w = (1, 0, 0, 0, 0)$. The representation space $M(\delta, w)$ is illustrated in Picture 4.1.



Picture 4.1: Extended Dynkin quiver $\widetilde{D}_4$

Next, we relate Nakajima quiver varieties of extended Dynkin quivers to Kleinian singularities and their minimal resolutions.

Lemma 4.1.5. *Let $Q = (I, \Omega)$ be an extended Dynkin quiver. Then we have*

$$\mathcal{M}_0(\delta, w) \xrightarrow{\sim} \mathcal{M}_0(\delta, 0). \tag{4.1.13}$$

Proof. We denote the moment map for the space $M(\delta, w)$ (resp., $M(\delta, 0)$) by μ (resp., $\bar{\mu}$). From eq. (4.1.12), we see that

$$\mathbb{C}[\mu^{-1}(0)] = \mathbb{C}[\bar{\mu}^{-1}(0)][k_0^*, l_0^*]/(k_0^* l_0^*).$$

The scalar subgroup $\mathbb{C}^* \subset \mathrm{GL}(\delta)$ acts on $M(\delta, w)$ by $\lambda \cdot (B, B^*, l_0, k_0) = (B, B^*, \lambda l_0, \lambda^{-1} k_0)$. Thus, we have

$$\mathbb{C}[\mu^{-1}(0)]^{\mathbb{C}^*} = \mathbb{C}[\bar{\mu}^{-1}(0)],$$

which implies that

$$\mathbb{C}[\bar{\mu}^{-1}(0)]^{\mathrm{GL}(\delta)} \xrightarrow{\sim} \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}.$$

$\square$

Recall that a basis in the *path algebra* $\mathbb{C}\bar{Q}$ consists of the paths in the doubled quiver $\bar{Q}$, including a trivial path e_i for each vertex i. There is a natural grading on $\mathbb{C}\bar{Q}$ given by the number of arrows in a path. Consider the *preprojective algebra*

$$\Pi^0 := \mathbb{C}\bar{Q}/(\sum_{a\in\Omega}[a, a^*]).$$

Theorem 4.1.6 ([CBH98], Lemma 1.1, Corollary 3.5)**.** *Let $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$ be a finite subgroup and $Q = (I, \Omega)$ the corresponding extended Dynkin quiver. There is a graded algebra isomorphism*

$$e_0 \Pi^0 e_0 \simeq \mathbb{C}[u, v]^\Gamma.$$

There is also a natural grading on $\mathbb{C}[M(\delta, w)]$ given by the degree of the polynomial functions, which is equivalent to the following $\mathbb{C}^*$-action on $M(\delta, w)$:

$$t \cdot (B, B^*, l_0, k_0) := (t^{-1}B, t^{-1}B^*, t^{-1}l_0, t^{-1}k_0),\ t \in \mathbb{C}^*. \tag{4.1.14}$$

Note that this grading passes to $\mathbb{C}[\mathcal{M}_0(\delta, w)] = \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ because the ADHM equation eq. (4.1.12) is homogeneous.

Theorem 4.1.7 ([CBH98], Lemmas 8.1, 8.5, Theorem 8.10)**.** *There is a graded isomorphism of algebras*

$$e_0 \Pi^0 e_0 \xrightarrow{\sim} \mathbb{C}[\mathcal{M}_0(\delta, 0)]. \tag{4.1.15}$$

given by sending a loop $p = a_1 \cdots a_m$ to the trace function $\mathrm{tr}(B_{a_1} \cdots B_{a_m})$.

Combining Lemma 4.1.5, Theorem 4.1.6, and Theorem 4.1.7, we obtain that

Corollary 4.1.8. *There is an isomorphism of graded algebras*

$$C[\mathcal{M}_0(\delta, w)] \simeq \mathbb{C}[u, v]^\Gamma. \tag{4.1.16}$$

Moreover, $\mathbb{C}[\mathcal{M}_0(\delta, w)]$ is generated by trace functions of paths that start and end at the vertex 0.

We know that $\mathcal{M}_\chi(\delta, w) = \mu^{-1}(0)^{\chi-ss} /\!/ \mathrm{GL}(\delta)$ a smooth symplectic variety from Corollary 4.1.3. By the fact that symplectic resolutions are minimal in dimension 2, we obtain the following theorem.

Theorem 4.1.9 ([Nak98], Proposition 3.24)**.** *Let $\chi = (-1, \cdots, -1)$. The projective morphism*

$$\pi \colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_0(\delta, w)$$

in eq. (4.1.8) *gives the minimal resolution of Kleinian singularities.*

At the end of this section, we comment that $\dim V_0 = 1$, so we can identify $\mathrm{End}(V_0)$ with $\mathbb{C}$ by taking traces. The graded isomorphism in eq. (4.1.16) allows us to pick out the generators of $\mathbb{C}[X] \simeq$

$\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ in terms of trace functions (c.f. Propositions 4.4.1, 4.5.1, 4.6.4 and 4.7.3), which is essential to the construction of the liftings in Sections 4.3 to 4.7.

4.2 Lifting anti-Poisson involutions to the minimal resolutions

Recall that $\pi\colon \widetilde{X} \to X$ denotes the minimal resolution of a Kleinian singularity. We would like to describe the preimage $\pi^{-1}(X^\theta)$ in terms of the following aspects.

(i) What are the irreducible components?

(ii) How do the irreducible components intersect with each other?

(iii) Is $\pi^{-1}(X^\theta)$ (generically) reduced?

Our main approach is through lifting the anti-Poisson involution θ to the minimal resolution $\widetilde{X}$ and examining its action on the exceptinal fiber $\pi^{-1}(0)$. By a *lift* of θ, we mean an anti-symplectic involution $\tilde{\theta}\colon \tilde{X} \to \tilde{X}$ such that

$$\pi \circ \tilde{\theta} = \theta \circ \pi. \tag{4.2.1}$$

Note that the action of $\tilde{\theta}$ is determined by its action on the open dense subset $\pi^{-1}(X \setminus \{0\})$, and π is an isomorphism away from the exceptional fiber. Therefore, Therefore, any automorphism $\tilde{\theta}$ of $\widetilde{X}$ satisfying eq. (4.2.1) is automatically an anti-symplectic involution.

Note that the lift $\tilde{\theta}$ always exists and is unique due to the definition and uniqueness of the minimal resolution $\pi\colon \widetilde{X} \to X$. However, to analyze the action of $\tilde{\theta}$ on $\widetilde{X}$, especially the action on the exceptional fiber $\pi^{-1}(0) = \bigcup_{1\leqslant i\leqslant n} C_i$ with $C_i \simeq \mathbb{P}^1$, we need an explicit construction of the lift $\tilde{\theta}$. We use Proposition 4.2.1 as a guideline to construct the lifts in Sections 4.3 to 4.7. Recall the symplectic form $\omega(-,-)$ defined on $M(\delta, w)$ in eq. (4.1.3).

Proposition 4.2.1. *Let θ be an anti-Poisson involution of X. Let $\Theta\colon M(\delta,w) \to M(\delta,w),\ u \mapsto \Theta(u)$ be an anti-symplectic linear automorphism satisfying the following conditions.*

(1) There is an automorphism $g \mapsto g^\Theta$ of $\mathrm{GL}(\delta)$ such that $\Theta g \Theta^{-1}$ equals the action of g^Θ on $M(\delta,w)$.

(2) $\Theta(x) = \theta(x),\ \Theta(y) = \theta(y),\ \Theta(z) = \theta(z)$, where x, y, z are the generators of $\mathbb{C}[X]$.

Then Θ descends to an anti-symplectic involution $\tilde{\theta}\colon \widetilde{X} \to \widetilde{X},\ [u] \mapsto [\Theta(u)]$. Moreover, $\tilde{\theta}$ is a lift of θ.

Proof. We show that the condition (1) implies that the following three things.

(1a) Θ preserves $\mu^{-1}(0)$. The natural $\mathrm{GL}(\delta)$-action on $M(\delta, w)$ induces the $\mathfrak{gl}(\delta)$-action on $M(\delta, w)$ and the condition (1) implies that the conjugation $A^{\Theta} := \Theta A \Theta^{-1}$ is obtained by permuting the factors of $A = (A_i)_{i \in I} \in \mathfrak{gl}(\delta)$. One can compute that $\mu(\Theta(u)) = -\mu(u)^{\Theta}$, and the claim follows.

(1b) Θ preserves $\mu^{-1}(0)^{\chi-ss}$. Recall that $\chi(g) = \prod_{i \in I} \det(g_i)^{-1}$. One can show that $\chi((\cdot)^{\Theta}) = \chi(\cdot)$ by the condition (1), hence $\Theta(f) \in \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta), m\chi}$ iff $f \in \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta), m\chi}$. It follows that $\Theta(u)$ is χ-semistable iff u is (c.f. eq. (4.1.9)).

(1c) Θ normalizes the $\mathrm{GL}(\delta)$-action because the conjugation $g^{\Theta} = \Theta g \Theta^{-1}$ still lies in $\mathrm{GL}(\delta)$.

Together with the condition (2), we see that Θ descends to the anti-Poisson involution θ on $X \simeq \mu^{-1}(0) /\!/ \mathrm{GL}(\delta)$ and the anti-symplectic involution $\tilde{\theta}$ on $\widetilde{X} \simeq \mu^{-1}(0)^{ss} /\!/ \mathrm{GL}(\delta)$. We automatically have that $\pi \circ \tilde{\theta} = \theta \circ \pi$, hence $\tilde{\theta}$ is a lift of θ. $\square$

Consider the scheme-theoretic fixed point locus $\widetilde{X}^{\tilde{\theta}}$. It is a smooth Lagrangian subvariety of $\widetilde{X}$ (c.f. Proposition 3.1.3). It is easy to show that

$$\pi^{-1}(X^{\theta})_{\mathrm{red}} = \pi^{-1}(0)_{\mathrm{red}} \cup \widetilde{X}^{\tilde{\theta}}. \tag{4.2.2}$$

The advantage of the lift $\tilde{\theta}$ is that the smooth Lagrangian $\widetilde{X}^{\tilde{\theta}}$ is much easier to analyze than $\pi^{-1}(X^{\theta})_{\mathrm{red}}$. We study the variety $\pi^{-1}(X^{\theta})_{\mathrm{red}}$ in Section 4.2, and the scheme structure of $\pi^{-1}(X^{\theta})$ in Section 4.2.

Now we address the three questions asked at the beginning of Section 4.2 separately.

Question (i)

Recall that the natural grading on $\mathbb{C}[M(\delta, w)]$ given by the degree of polynomial functions induces the following $\mathbb{C}^*$-action on $M(\delta, w)$, as described in eq. (4.1.14):

$$t \cdot (B, B^*, l_0, k_0) := (t^{-1}B, t^{-1}B^*, t^{-1}l_0, t^{-1}k_0),\ t \in \mathbb{C}^*.$$

The symplectic form ω in eq. (4.1.3) is transformed into $t^{-2}\omega$ under the $\mathbb{C}^*$-action. The $\mathbb{C}^*$-action commutes with the $\mathrm{GL}(\delta)$-action, and the subsets $\mu^{-1}(0)$, $\mu^{-1}(0)^{\chi-ss}$ are $\mathbb{C}^*$-stable. It follows that the $\mathbb{C}^*$-action descends to $X \simeq \mu^{-1}(0) /\!/ \mathrm{GL}(\delta)$ and $\widetilde{X} \simeq \mu^{-1}(0)^{\chi-ss} /\!/ \mathrm{GL}(\delta)$. Moreover, the minimal resolution $\pi\colon \widetilde{X} \to X$ is $\mathbb{C}^*$-equivariant.

There is an easier way to observe this $\mathbb{C}^*$-action on X. Recall the grading on $\mathbb{C}[X] = \mathbb{C}[u, v]^{\Gamma}$ given by the degree of the polynomial functions on u, v. We know from Corollary 4.1.8 that $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)} \simeq$

$\mathbb{C}[u,v]^\Gamma$ is a graded algebra automorphism. Therefore, the $\mathbb{C}^*$-action on X induced from the grading of $\mathbb{C}[u,v]^\Gamma$ will be the same from the one descended from eq. (4.1.14). Let x, y, z denote the generators of $\mathbb{C}[X] = \mathbb{C}[u,v]^\Gamma$ given in Proposition 2.3.1. We have

$$t \cdot (x_0, y_0, z_0) = (t^{-\deg x} x_0, t^{-\deg y} y_0, t^{-\deg z} z_0),\ (x_0, y_0, z_0) \in X. \tag{4.2.3}$$

Let $L \subset X^\theta$ be an 1-dimensional irreducible component. Then from Corollaries 3.3.4, 3.3.7, 3.3.10, 3.3.13, 3.3.16 and 3.3.19, we know that L is either $\mathbb{A}^1$ or a cusp, and it is $\mathbb{C}^*$-stable. Moreover, one can deduce that the $\mathbb{C}^*$-action is transitive on the open part $L \setminus \{0\}$ by looking at the coordinate ring of L. Consider the variety

$$\tilde{L} := \overline{\pi^{-1}(L \setminus \{0\})} \subset \widetilde{X}^{\tilde{\theta}}, \tag{4.2.4}$$

which is smooth as being an irreducible component of the smooth Lagrangian $\widetilde{X}^{\tilde{\theta}}$. We will further show that $\tilde{L} \simeq \mathbb{A}^1$ in Proposition 4.2.2. Pick an arbitrary point $s \in L \setminus \{0\}$. It doesn't matter which s we choose because the $\mathbb{C}^*$-action on $L \setminus \{0\}$ is transitive. Set

$$s_0 := \lim_{t \to 0} t^{-1} \cdot \pi^{-1}(s) \in \tilde{L} \cap \pi^{-1}(0), \tag{4.2.5}$$

which exists because the composition $\tilde{L} \overset{i}{\hookrightarrow} \pi^{-1}(L) \xrightarrow{\pi} L$ is proper and the limit point $\lim_{t \to 0} t^{-1} \cdot s = 0$ exists.

Proposition 4.2.2. *We have an isomorphism of varieties*

$$\tilde{L} \xrightarrow{\sim} \mathbb{A}^1$$

with the limit point s_0 being mapped to the origin.

Proof. We first show that the morphism $f := \pi \circ i \colon \tilde{L} \to L$ is quasi-finite. It suffices to show that the fiber $f^{-1}(0)$ consists of finitely many points as f is an isomorphism outside of $f^{-1}(0)$. Note that the morphism f is of finite type, hence quasi-compact. Therefore, it is enough to show that $\dim f^{-1}(0) = 0$. If $\dim f^{-1}(0) > 0$, then $\dim f^{-1}(0) = 1$. It follows that $f^{-1}(0) = \tilde{L}$ as $\tilde{L}$ is irreducible, a contradiction.

Next, note that the morphism f is also proper, hence finite. It follows that $\tilde{L}$ is affine as an affine scheme over the affine scheme L. Moreover, $\tilde{L}$ is smooth as an irreducible component of the smooth Lagrangian $\widetilde{X}^{\tilde{\theta}}$, and rational because it contains $\pi^{-1}(L \setminus \{0\}) \simeq \mathbb{C}^*$ as an open dense subset. It follows that $\tilde{L}$ is isomorphic to $\mathbb{A}^1$ with finitely many points removed.

Let $U := \pi^{-1}(L \setminus \{0\}) \cup \{s_0\} \subset \tilde{L}$, then it admits an open embedding into $\mathbb{A}^1$. More importantly, U is equipped with a contracting $\mathbb{C}^*$-action to the point s_0 from eqs. (4.1.14) and (4.2.5). By a version of Luna's slice theorem [Dré04, Theorem 5.4], we can identify U with its tangent space, i.e., $U \simeq T_{s_0}U \simeq \mathbb{A}^1$. This implies that

$$U = \tilde{L} \simeq \mathbb{A}^1.$$

because an open proper subset of $\mathbb{A}^1$ cannot be isomorphic to $\mathbb{A}^1$, which can be seen from comparing the invertible functions on U and $\mathbb{A}^1$. □

Then it is easy to see that the irreducible components of $\pi^{-1}(X^\theta)$ are $\mathbb{P}^1$ that are irreducible components of $\pi^{-1}(0)$ and $\mathbb{A}^1$'s that are normalizations of the irreducible components of X^θ. Denote the $\mathbb{P}^1$'s by $C_1, \cdots, C_n$ and the $\mathbb{A}^1$'s by $\tilde{L}_1, \cdots, \tilde{L}_m$, where m is the number of 1-dimensional irreducible components of X^θ. We would like to point out that the variety $\widetilde{X}^{\tilde{\theta}}$ may contain not only all of the $\tilde{L}_j$'s, but also some of the C_i's, which is the situation in most of the cases.

Question (ii)

We study how the irreducible components of $\pi^{-1}(X^\theta)$ intersect with each other. The $\mathbb{A}^1$'s do not intersect each other because they are irreducible components of the smooth variety $\widetilde{X}^{\tilde{\theta}}$. The $\mathbb{P}^1$'s intersect with each according dually to the Dynkin diagram labeling Γ. The remaining thing is to determine how the $\mathbb{A}^1$'s intersect with the $\mathbb{P}^1$'s. A detailed answer will be given in Proposition 4.2.5.

The lift $\tilde{\theta}$ preserves $\pi^{-1}(0)$ because θ fixes the singular point 0. Note that $\tilde{\theta}$ is of order 2, so $\tilde{\theta}$ either preserves a component C_i, or swaps it with another component C_j. Moreover, the intersection points of the $\mathbb{A}^1$'s and the $\mathbb{P}^1$'s must lie in $\pi^{-1}(0)^{\tilde{\theta}}$ (c.f. eq. (4.2.5) and Proposition 4.2.2). Lemma 4.2.3 will be useful in determining $\pi^{-1}(0)^{\tilde{\theta}}$.

Lemma 4.2.3. *(1) If $\tilde{\theta}$ swaps two components C_i and C_j, then $C_i \cap C_j$ is either empty, or a single point contained in $\pi^{-1}(0)^{\tilde{\theta}}$.*

(2) If $\tilde{\theta}$ preserves a component C_i, then $\tilde{\theta}$ either acts trivially on C_i, or has exactly two fixed points.

(3) If $\tilde{\theta}$ preserves two components C_i and C_j such that $C_i \cap C_j \neq \emptyset$, then $\tilde{\theta}$ acts trivially on one of the components, and has exactly two fixed points on the other component.

Proof. (1) Note that if $C_i \cap C_j$ is non-empty, then it is a single point. The rest follows.

(2) Any automorphism of $C_i \simeq \mathbb{P}^1$ is given by a Mӧbius transformation (c.f. [Har77, Exercise I. 6.6]). It follows that an involution of $\mathbb{P}^1$ either acts trivially or has exactly two fixed points.

(3) We denote by p the unique point in $C_i \cap C_j$. The tangent map $\mathrm{d}_p\tilde{\theta} \colon T_p\widetilde{X} \to T_p\widetilde{X}$ is an anti-symplectic linear involution of the 2-dimensional symplectic vector space $T_p\widetilde{X}$ with two distinct eigenvalues $1, -1$. By assumption, T_pC_i, T_pC_j are eigenspaces of the linear map $\mathrm{d}_p\tilde{\theta}$. It follows that $\mathrm{d}_p\tilde{\theta}$ acts on one of T_pC_i, T_pC_j by eigenvalue 1, and acts on the other by eigenvalue -1. The rest follows from (2).

□

We emphasize that when we say that $\tilde{\theta}$ preserves a component, we mean that $\tilde{\theta}(C_i) \subset C_i$; when we say that $\tilde{\theta}$ fixes a component, we mean that $\tilde{\theta}$ fixes C_i pointwise, i.e. acts trivially. A direct application of (3) in Lemma 4.2.3 gives Corollary 4.2.4.

Corollary 4.2.4. *Consider a subset $\{C_{i'}\}_{i' \in I'} \subset \{C_i\}_{i \in I}$. Assume that*

(1) The lift $\tilde{\theta}$ preserves each component $C_{i'},\ i' \in I'$.

(2) The $C_{i'}$'s have pairwise transversal intersection according dually to a Dynkin diagram of type A.

Then the $\tilde{\theta}$-fixed and non-fixed components appear alternatingly.

Proposition 4.2.5 follows from Proposition 4.2.2, Lemma 4.2.3, and the fact that $\widetilde{X}^{\tilde{\theta}}$ is a smooth Lagrangian (c.f. Proposition 3.1.3).

Proposition 4.2.5. *(1) $\pi^{-1}(0)^{\tilde{\theta}}$ is a disjoint union of $\mathbb{P}^1$'s and finitely many points. The number of isolated points in $\pi^{-1}(0)^{\tilde{\theta}}$ equals the number of irreducible components of X^{θ}.*

(2) For each of the isolated point $p \in \pi^{-1}(0)^{\tilde{\theta}}$, there is a unique $\tilde{L}_j$, such that $\tilde{L}_j \cap \pi^{-1}(0)^{\tilde{\theta}} = \{p\}$.

(3) If $\tilde{L}_j \cap C_i \neq \emptyset$, then $\tilde{L}_j \cap C_i = \tilde{L}_j \cap \pi^{-1}(0)^{\tilde{\theta}} = \{p\}$. Moreover, $\tilde{L}_j$ and C_i meet transversally at p.

To figure out which C_i the component $\tilde{L}_j$ intersects with exactly, we need a description of C_i in terms of quiver data. Recall $B_a \colon V_{t(a)} \to V_{h(a)}$. Set

$$\epsilon B_a := \begin{cases} B_a,\ a \in \Omega \\ -B_a,\ a \in \Omega^*. \end{cases}$$

Let $(B_a)_{\substack{a\in\bar{\Omega}\\ t(a)=i}}$ denote the collection of the entries of (B, B^*) with source V_i. Fix the character $\chi = (-1, \cdots, -1)$. Pick a point $(B, B^*, l_0, k_0) \in \mu^{-1}(0)^{\chi-ss}$, and consider the following chain of linear maps for each $i = 1, \cdots, n$.

$$V_i \xrightarrow{(B_a)_{a\in\bar{\Omega},\ t(a)=i}} \bigoplus_{\substack{a\in\bar{\Omega}\\ t(a)=i}} V_{h(a)} \xrightarrow{\epsilon(B_{a^*})_{a\in\bar{\Omega},\ t(a)=i}} V_i. \tag{4.2.6}$$

The composition of the maps in eq. (4.2.6) is 0 by the ADHM equation (c.f. eq. (4.1.12)). Moreover, the second map in eq. (4.2.6) is surjective (c.f.Proposition 4.1.2). Recall from eq. (2.1.1), we have that

$$\dim \bigoplus_{\substack{a\in\bar{\Omega}\\ t(a)=i}} V_{h(a)} = 2\dim V_i.$$

Therefore, eq. (4.2.6) is an exact sequence if and only if the first map in eq. (4.2.6) is injective, i.e.

$$\operatorname{rank}(B_a)_{\substack{a\in\bar{\Omega}\\ t(a)=i}} = \dim V_i,$$

where $\operatorname{rank}(B_a)_{\substack{a\in\bar{\Omega}\\ t(a)=i}}$ denotes the rank of the linear map $(B_a)_{\substack{a\in\bar{\Omega}\\ t(a)=i}} : V_i \to \bigoplus_{\substack{a\in\bar{\Omega}\\ t(a)=i}} V_{h(a)}$.

Proposition 4.2.6. *Define*

$$C_i = \left\{[B, B^*, l_0, 0] \in \mathcal{M}_\chi(\delta, w) \,\middle|\, \operatorname{rank}(B_a)_{\substack{a\in\bar{\Omega}\\ t(a)=i}} < \dim V_i\right\},\ 1 \leqslant i \leqslant n.$$

Then $\pi^{-1}(0) = \bigcup_{1\leqslant i\leqslant n} C_i$*, and* $C_i \simeq \mathbb{P}^1$ *corresponds to the vertex* i *in the Dynkin diagram labeling* Γ.

To prove Proposition 4.2.6, we also need to consider quiver varieties without framing. Let $M(\delta, 0)$ be the representation space of an extended Dynkin quiver of types ADE without framing. Consider the character $\chi' = (n, -1 \cdots, -1)$. Then $\pi_{\chi'} \colon \mathcal{M}_{\chi'}(\delta, 0) \to \mathcal{M}_0(\delta, 0) \simeq \mathbb{C}^2/\Gamma$ gives the minimal resolution of a Kleinian singularity ([Kro89, Corollary 3.2 and Proposition 3.10]). Note that we use $\pi_{\chi'}$ to distinguish it from the minimal resolution $\pi\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_0(\delta, w)$ given in Theorem 4.1.9. Next, define the dual representation space $M(\delta^*, 0)$ by replacing each V_i in the definition eq. (4.1.1) by V_i^*. Consider the character $-\chi' = (-n, 1 \cdots, 1)$. Similarly to the case of $\pi_{\chi'}$, we have that $\pi_{-\chi'} \colon \mathcal{M}_{-\chi'}(\delta^*, 0) \to \mathcal{M}_0(\delta^*, 0) \simeq \mathbb{C}^2/\Gamma$ is the minimal resolution of a Kleinian singularity. For a linear map $B_a \colon V_{t(a)} \to V_{h(a)}$, we denote by $B'_{a^*} \colon V^*_{h(a)} \to V^*_{t(a)}$ its dual. There is an obvious linear isomorphism of vector spaces $M_{\chi'}(\delta, 0) \xrightarrow{\sim}$

$M_{-\chi'}(\delta^*, 0),\ (B, B^*, 0, 0) \mapsto ((B^*)', B', 0, 0)$. It naturally induces the following commutative diagram.

$$\begin{array}{ccc} \mathcal{M}_{\chi'}(\delta, 0) & \xrightarrow{\phi_1} & \mathcal{M}_{-\chi'}(\delta^*, 0) \\ {\scriptstyle \pi_{\chi'}}\downarrow & & \downarrow{\scriptstyle \pi_{-\chi'}} \\ \mathcal{M}_0(\delta, 0) & \xrightarrow{\phi_0} & \mathcal{M}_0(\delta^*, 0) \end{array} \tag{4.2.7}$$

Moreover, ϕ_1 and ϕ_0 are both isomorphisms of varieties (c.f. [Kir16, Lemma 10.29]). It follows that ϕ_1 restricts to an isomorphism on the exceptional fibers $\pi_{\chi'}^{-1}(0) \xrightarrow{\sim} \pi_{-\chi'}^{-1}(0)$. Theorem 4.2.7 below, due to Nakajima, is essential to the proof of Proposition 4.2.6.

Theorem 4.2.7. *[Nak96, Theorem 5.10] Define*

$$C_i' := \left\{ [B', B'^*, 0, 0] \in \mathcal{M}_{-\chi'}(\delta^*, 0) \big| \operatorname{rank}(B'_{a^*})_{\substack{a \in \bar{\Omega} \\ t(a) = i}} < \dim V_i^* \right\},\ 1 \leqslant i \leqslant n.$$

Then $\pi_{-\chi'}^{-1}(0) = \bigcup_{1 \leqslant i \leqslant n} C_i'$, *and* $C_i' \simeq \mathbb{P}^1$ *corresponds to the vertex* i *in the Dynkin diagram labeling* Γ.

Proof of Proposition 4.2.6. Consider the projection map

$$M_{\chi}(\delta, w) \to \mathcal{M}_{\chi'}(\delta, 0)$$
$$(B, B^*, l_0, k_0) \mapsto (B, B^*, 0, 0).$$

We show that it induces an isomorphism of varieties

$$\begin{aligned} \varphi_1 \colon \mathcal{M}_{\chi}(\delta, w) &\to \mathcal{M}_{\chi'}(\delta, 0) \\ [B, B^*, l_0, 0] &\mapsto [B, B^*, 0, 0]. \end{aligned} \tag{4.2.8}$$

We denote the moment map for the space $M(\delta, w)$ (resp., $M(\delta, 0)$) by μ (resp., $\bar{\mu}$). According to eq. (4.1.12), $(B, B^*, l_0, k_0) \in \mu^{-1}(0)$ if and only if $(B, B^*, 0, 0) \in \bar{\mu}^{-1}(0)$ and $l_0 k_0 = 0$. Moreover, if $(B, B^*, l_0, k_0) \in \mu^{-1}(0)^{\chi-ss}$, then $l_0 \neq 0,\ k_0 = 0$ (c.f. Proposition 4.1.2). To prove that eq. (4.2.8) is an isomorphism, it suffices to show that $(B, B^*, l_0, 0) \in \mu^{-1}(0)^{\chi-ss}$ if and only if $(B, B^*, 0, 0) \in \bar{\mu}^{-1}(0)^{\chi'-ss}$. We have that

(1) $(B, B^*, l_0, 0)$ is χ-semistable if and only if $\operatorname{im}(l_0) = V_0$ and it generates the entire collection of vector spaces $V = (V_i)_{1 \leqslant i \leqslant n}$ under the action of (B, B^*), according to Proposition 4.1.2.

(2) $(B, B^*, 0, 0)$ is χ'-semistable if and only if any (B, B^*)-stable subspace $S = (S_i)_{i \in I}$ satisfies

$$\chi'(\dim S) := \sum_{i=0}^{n} \chi_i' \dim S_i \leqslant 0,$$

according to [Nak96, Lemma 2.4].

Note that $\dim V_i = \delta_i$ and especially $\dim V_0 = \delta_0 = 1$. For a subspace $S = (S_i)_{i\in I} \subset (V_i)_{i\in I}$, if $S_0 = \{0\}$, then we automatically have that $\chi'(\dim S) \leqslant 0$. If $S_0 = V_0$, then $\chi'(\dim S) \leqslant 0$ implies that $S_i = V_i$. Therefore, we can restate (2) into the following.

(2)′ $(B, B^*, 0, 0)$ is χ'-semistable if and only if V_0 generates the entire V under the action of (B, B^*).

From the semistability conditions (1) and (2)′, it is obvious that $(B, B^*, l_0, 0) \in \mu^{-1}(0)^{\chi-ss}$ if and only if $(B, B^*, 0, 0) \in \bar{\mu}^{-1}(0)^{\chi'-ss}$. The isomorphism φ_1 in eq. (4.2.8) then follows. Similar to showing that ϕ_1 restricts to an isomorphism $\pi_{\chi'}^{-1}(0) \xrightarrow{\sim} \pi_{-\chi'}^{-1}(0)$ (c.f. the commutative diagram in eq. (4.2.7)), one can show that φ_1 also restricts to an isomorphism $\pi^{-1}(0) \xrightarrow{\sim} \pi_{\chi'}^{-1}(0)$. The rest follows from Theorem 4.2.7 by computing that $(\phi_1 \circ \varphi_1)^{-1}(C_i') = C_i$. □

Question (iii)

We can write the corresponding divisor of $\pi^{-1}(X^\theta)$ as a linear combination of the irreducible components $\tilde{L}_j,\ 1 \leqslant j \leqslant m,\ C_i,\ 1 \leqslant i \leqslant n$. The coefficient of $\tilde{L}_j$ has to be 1 because we know that X^θ is reduced from Corollaries 3.3.4, 3.3.7, 3.3.10, 3.3.13, 3.3.16 and 3.3.19, and that π is an isomorphism away from the exceptional fiber. Therefore, we have

$$\pi^{-1}(X^\theta) = \sum_{j=1}^{m} \tilde{L}_j + \sum_{i=1}^{n} a_i C_i,\ a_i \geqslant 1$$

as a divisor. Our goal is to determine the positive integers a_i. Recall the intersection pairing defined in Proposition 2.2.1.

$$\mathrm{Div}_{\pi^{-1}(0)}(\widetilde{X}) \times \mathrm{Div}(\widetilde{X}) \mapsto \mathbb{Z},\ (C, D) \mapsto C \cdot D$$

Recall that $\mathcal{C}$ denotes the Cartan matrix of the simply-laced Dynkin diagram labeling Γ, and we know that the intersection matrix $(C_i \cdot C_j)_{1\leqslant i,j\leqslant n}$ equals $-\mathcal{C}$. Define

$$b_i := \#\{1 \leqslant j \leqslant m \mid \tilde{L}_j \cap C_i \neq \emptyset\},\ \ 1 \leqslant i \leqslant n. \tag{4.2.9}$$

Note that if $\tilde{L}_j \cap C_i \neq \emptyset$ is nonempty, then the two curves meet transversally at a single point by (3) of Proposition 4.2.5. Applying the property (1) of the intersection pairing in Proposition 2.2.1, we can deduce that

$$b_i = \sum_{j=1}^{m} C_i \cdot \tilde{L}_j. \tag{4.2.10}$$

Proposition 4.2.8. *If $X^\theta = \mathrm{div}(f) \subset X$ is a principal divisor, then*

$$(a_1, \cdots, a_n)^t = \mathcal{C}^{-1}(b_1, \cdots, b_n)^t \tag{4.2.11}$$

Proof. If $X^\theta = \mathrm{div}(f)$, then $\pi^{-1}(X^\theta) = \mathrm{div}(\pi^* f) = \sum_{j=1}^m \tilde{L}_j + \sum_{i=1}^n a_i C_i$ is also a principal divisor. Applying properties (3) and (4) in Proposition 2.2.1, we have that

$$\sum_{j=1}^{m} C_i \cdot \tilde{L}_j + \sum_{k=1}^{n} a_k C_i \cdot C_k = C_i \cdot \mathrm{div}(\pi^* f) = 0,\ 1 \leqslant i \leqslant n. \tag{4.2.12}$$

Rearranging the terms, by eq. (4.2.10) and the fact that $(C_i \cdot C_j)_{1 \leqslant i,j \leqslant n} = -\mathcal{C}$, we get

$$\mathcal{C}(a_1, \cdots, a_n)^t = (\sum_{j=1}^{m} C_1 \cdot \tilde{L}_j, \cdots, \sum_{j=1}^{m} C_n \cdot \tilde{L}_j)^t = (b_1, \cdots, b_n)^t.$$

Then eq. (4.2.11) follows from the fact that the Cartan matrix $\mathcal{C}$ is invertible. □

Remark 4.2.9. (1) We comment that $a_i = 1,\ 1 \leqslant i \leqslant n$ does not imply that $\pi^{-1}(X^\theta)$ is reduced, but rather generically reduced. However, for a principal divisor, being generically reduced is equivalent to being reduced.

(2) Assume X^θ is a principal divisor. If C_i intersects with more than 2 irreducible components in $\{\tilde{L}_j | 1 \leqslant j \leqslant m\} \cup \{C_k | 1 \leqslant k \leqslant n\}$, then $a_i > 1$. This follows from eq. (4.2.12).

(3) Note that $\pi^{-1}(X^\theta)$ is definitely not reduced in types D, E because $\pi^{-1}(0)$ is not reduced in these types (c.f. Remark 2.1.1). However, $\pi^{-1}(X^\theta)$ can be non-reduced in type A (c.f. Propositions 4.3.2 and 4.3.7), though $\pi^{-1}(0)$ is reduced in type A.

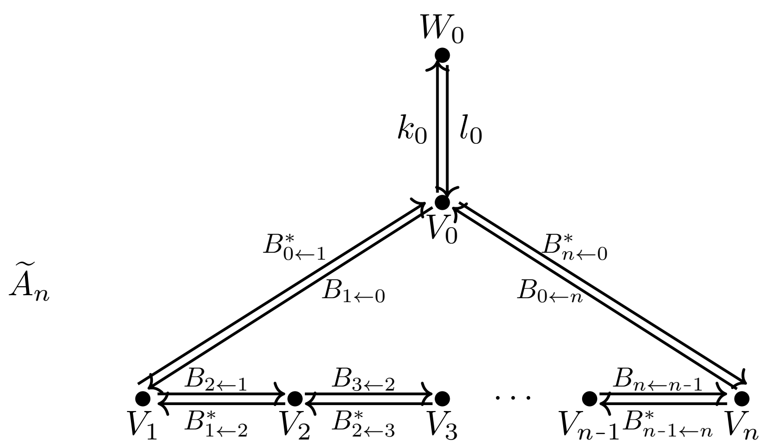
(4) Let $\mathrm{div}(f) \subset X$ be a reduced principal divisor. We write $\mathrm{div}(\pi^* f) = D + \sum_{i=1}^n a_i C_i$ such that D does not involve any C_i. Further assume that each irreducible component of D intersects with at most one C_i, and the intersection is transversal. Define b_i similarly as in eq. (4.2.9). Then eq. (4.2.11) still holds.

From Corollaries 3.3.4, 3.3.7, 3.3.10, 3.3.13, 3.3.16 and 3.3.19, we see that the corresponding divisor of X^θ is principal except for the following two cases.

1. Type A_n, with n odd, case III, where $X^\theta = \{0\}$. We have $\pi^{-1}(X^\theta) = \pi^{-1}(0) = \sum_{i=1}^n C_i$ is reduced, according to Remark 2.1.1.

2. Type A_n, with n even, case II, where $X^\theta = \operatorname{Spec} \mathbb{C}[X]/(x, z)$. The preimage $\pi^{-1}(X^\theta)$ is generically reduced (c.f. (3) of Proposition 4.3.9).

4.3 Preimages of fixed point loci for type A_n Kleinian singularity



Picture 4.2: Extended Dynkin quiver $\widetilde{A}_n$

We realize the Kleinian singularity of type $A_n : X = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - z^{n+1})$ as a Nakajima quiver variety explicitly. Recall that $\delta = (1, \cdots, 1)$. Using the notations in Section 4.1.2, we have $\dim W_0 = \dim V_i = 1,\ 1 \leqslant i \leqslant n$, and $B_{i+1\leftarrow i} \in \operatorname{Hom}(V_i, V_{i+1}),\ B^*_{i\leftarrow i+1} \in \operatorname{Hom}(V_i, V_{i+1})^* \simeq \operatorname{Hom}(V_{i+1}, V_i)$. The representation space is

$$M(\delta, w) = \bigoplus_{i=0}^{n} \operatorname{Hom}(V_i, V_{i+1}) \oplus \bigoplus_{i=0}^{n} \operatorname{Hom}(V_{i+1}, V_i) \oplus \operatorname{Hom}(W_0, V_0) \oplus \operatorname{Hom}(V_0, W_0).$$

The group $\operatorname{GL}(\delta) = \prod_{i=0}^n \operatorname{GL}(V_i)$ acts on $M(\delta, w)$ naturally with the moment map

$$\mu = (\mu_i)_{0\leqslant i\leqslant n} \colon M(\delta, w) \to \mathfrak{gl}(\delta),$$

where

$$\begin{aligned} \mu_0 &= B_{0\leftarrow n} B^*_{n\leftarrow 0} - B^*_{0\leftarrow 1} B_{1\leftarrow 0} + l_0 k_0, \\ \mu_i &= B_{i\leftarrow i-1} B^*_{i-1\leftarrow i} - B^*_{i\leftarrow i+1} B_{i+1\leftarrow i},\ 1 \leqslant i \leqslant n. \end{aligned} \tag{4.3.1}$$

According to Corollary 4.1.8 and Proposition 2.3.1, the algebra of invariant functions $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ is generated by the trace functions of loops that start and end at the vertex 0 of degrees 2 or $n+1$, which are

$$\begin{aligned} x := B_{0\leftarrow n}\cdots B_{2\leftarrow 1}B_{1\leftarrow 0},\quad y := B^*_{0\leftarrow 1}\cdots B^*_{n-1\leftarrow n}B^*_{n\leftarrow 0},\\ z := B^*_{0\leftarrow 1}B_{1\leftarrow 0} \overset{(4.3.1)}{=} B^*_{i\leftarrow i+1}B_{i+1\leftarrow i},\ 1\leqslant i\leqslant n \text{ (assume } V_{n+1}=V_0). \end{aligned} \tag{4.3.2}$$

It follows that $xy = z^{n+1}$. Thus we can construct a surjective homomorphism

$$\mathbb{C}[x,y,z]/(xy - z^{n+1}) \twoheadrightarrow \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}, \tag{4.3.3}$$

which turns out to be an isomorphism because the left-hand side is a domain of dimension 2 and the right hand side is also of dimension 2 from Corollary 4.1.8. The explicit isomorphism eq. (4.3.3) has been known before. For example, one can find a treatment in [Nak96, eq. (5.3)]. By Proposition 4.2.6, we can write down the equation of each irreducible component of $\pi^{-1}(0)$.

$$C_i = \big\{[B, B^*, l_0, 0] \in \mathcal{M}_\chi(\delta, w) \big| B_{i+1\leftarrow i} = B^*_{i-1\leftarrow i} = 0\big\},\ 1\leqslant i\leqslant n. \tag{4.3.4}$$

Consider the following automorphism τ of the extended Dynkin diagram of type $\widetilde{A}_n$.

$$\tau(0) = 0;\ \tau(i) = n+1-i,\ 1\leqslant i\leqslant n. \tag{4.3.5}$$

It induces an automorphism of the extended Dynkin quiver $\widetilde{A}_n$ in Picture 4.2 by sending an arrow $a : i \to j$ to the arrow $\tau(a) : \tau(i) \to \tau(j)$. The automorphism τ will be used to construct lifts of the anti-Poisson involutions in Sections 4.3.1 and 4.3.4. Note that τ coincides with the nontrivial automorphism of the McKay graph of type $\widetilde{A}_n$ induced by an element $g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$ (c.f. Remark 3.3.3). Sections 4.3.1 to 4.3.3 study the preimages of fixed point loci of anti-Poissons for Kleinian singularities of type A_n, with n odd. Sections 4.3.4 and 4.3.5 deal with the cases with n even.

4.3.1 Type A_n, n odd, case I

In this section, we consider the anti-Poisson involution

$$\theta(x) = y,\ \theta(y) = x,\ \theta(z) = z.$$

The fixed point locus $X^\theta = \operatorname{Spec}\mathbb{C}[X]/(x-y) = \operatorname{Spec}\mathbb{C}[x,z]/(x^2-z^{n+1})$ has two irreducible components $L_j,\ j=1,2$. Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0)\cup\tilde{L}_1\cup\tilde{L}_2$.

Proposition 4.3.1. *The involution $\tilde{\theta}\colon \mathcal{M}_\chi(\delta,w)\to\mathcal{M}_\chi(\delta,w)$ defined by*

$$\tilde{\theta}([B_{i+1\leftarrow i}, B^*_{j-1\leftarrow j}, l_0, k_0]) = ([B^*_{\tau(i+1)\leftarrow\tau(i)}, B_{\tau(j-1)\leftarrow\tau(j)}, l_0, -k_0]) \tag{4.3.6}$$

is anti-symplectic. Moreover, it is a lift of $\theta\colon X\to X$.

Proof. We consider the following linear anti-symplectic involution of the representation space.

$$\Theta\colon M(\delta,w)\to M(\delta,w),\ \ (B_{i+1\leftarrow i}, B^*_{j-1\leftarrow j}, l_0, k_0)\mapsto(B^*_{\tau(i+1)\leftarrow\tau(i)}, B_{\tau(j-1)\leftarrow\tau(j)}, l_0, -k_0).$$

It satisfies the conditions listed in Proposition 4.2.1.

1. $(g_i)^\Theta_{i\in I} = (g_{\tau(i)})_{i\in I}$.

2. $\Theta(x)=y,\ \Theta(y)=x,\ \Theta(z)=z$.

The involution $\tilde{\theta}$ defined in eq. (4.3.6) is the descent of Θ to $\mathcal{M}_\chi(\delta,w)$. The rest follows from Proposition 4.2.1. $\square$

Consider the action of $\tilde{\theta}$ on the exceptional fiber $\pi^{-1}(0)$, we have the following results.

Proposition 4.3.2. *(1) The involution $\tilde{\theta}$ preserves only the component $C_{\frac{n+1}{2}}$, and swaps the component C_i with C_{n+1-i} for $i\neq\frac{n+1}{2}$. We have $\pi^{-1}(0)^{\tilde{\theta}} = \{p_1,p_2\}\subset C_{\frac{n+1}{2}}\setminus(C_{\frac{n-1}{2}}\cup C_{\frac{n+3}{2}})$.*

(2) With a suitable choice of labeling, we have $\tilde{L}_i\cap\pi^{-1}(0) = \tilde{L}_i\cap C_{\frac{n+1}{2}} = \{p_i\},\ i=1,2$.

(3) The scheme $\pi^{-1}(X^\theta)$ is not reduced (except in type A_1). As a divisor, we have $\pi^{-1}(X^\theta) = \tilde{L}_1 + \tilde{L}_2 + \sum_{i=1}^n a_iC_i$, where $a_i = \min\{i, n+1-i\}$.

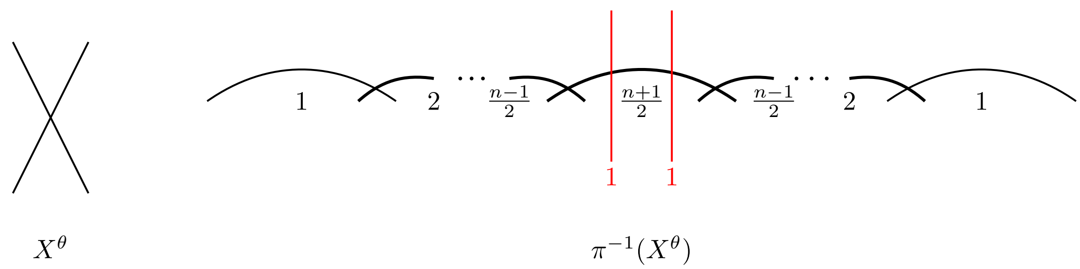
Proof. (1) It is easy to see that $\tilde{\theta}(C_i) = C_{\tau(i)} = C_{n+1-i}$ thanks to eq. (4.3.4). It follows that $\pi^{-1}(0)^{\tilde{\theta}}\subset C_{\frac{n+1}{2}}$. The intersection point $C_{\frac{n-1}{2}}\cap C_{\frac{n+1}{2}}$ is not $\tilde{\theta}$-stable because $\tilde{\theta}(C_{\frac{n-1}{2}}) = C_{\frac{n+3}{2}}$. Similarly, the intersection point $C_{\frac{n+1}{2}}\cap C_{\frac{n+3}{2}}$ is not $\tilde{\theta}$-stable, either.

(2) This follows from Proposition 4.2.5.

(3) Recall that we defined $b_i := \#\{1 \leqslant j \leqslant m | \tilde{L}_j \cap C_i \neq \emptyset\}$ in eq. (4.2.9). We know that $b_{\frac{n+1}{2}} = 2$ and $b_i = 0,\ i \neq \frac{n+1}{2}$ from (2). The divisor $X^\theta = \operatorname{div}(x-y)$ is principal. Then applying Proposition 4.2.8, we can determine that $a_i = \min\{i, n+1-i\}$.

□

We depict the fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ in Picture 4.3. The curly lines are $\mathbb{P}^1$'s and the straight lines are $\mathbb{A}^1$'s. The number below each component indicates its multiplicity. Thickened lines indicate that the multiplicities are larger than 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.



Picture 4.3: $A_n,\ n$ odd, Type I

4.3.2 Type $A_n,\ n$ odd, case II

In this section, we consider the anti-Poisson involution

$$\theta(x) = x,\ \theta(y) = y,\ \theta(z) = -z.$$

The fixed point locus $X^\theta = \operatorname{Spec}\mathbb{C}[X]/(z) = \operatorname{Spec}\mathbb{C}[x,y]/(xy)$ has two irreducible components $L_j,\ j = 1, 2$. Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}_1 \cup \tilde{L}_2$.

Proposition 4.3.3. *The involution* $\tilde{\theta}\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w)$ *defined by*

$$\tilde{\theta}([B, B^*, l_0, k_0]) = [-B, B^*, l_0, -k_0] \tag{4.3.7}$$

is anti-symplectic. Moreover, it is a lift of $\theta\colon X \to X$.

Proof. We consider the following linear anti-symplectic involution on the representation space.

$$\Theta\colon M(\delta, w) \to M(\delta, w),\ \ (B, B^*, l_0, k_0) \mapsto (-B, B^*, l_0, -k_0).$$

It satisfies the conditions listed in Proposition 4.2.1.

1. $g^\Theta = g$.

2. $\Theta(x) = (-1)^{n+1}x = -x,\ \Theta(y) = y,\ \Theta(z) = -z$.

The involution $\tilde{\theta}$ defined in eq. (4.3.7) is the descent of Θ to $\mathcal{M}_\chi(\delta, w)$. The rest follows from Proposition 4.2.1. ☐

Proposition 4.3.4. *(1) The involution $\tilde{\theta}$ preserves all the components $C_i,\ 1 \leqslant i \leqslant n$. We have $\pi^{-1}(0)^{\tilde{\theta}} = \bigcup_{i \text{ even}} C_i \cup \{p_1, p_2\}$, where $p_1 \in C_1 \setminus C_2$ and $p_2 \in C_n \setminus C_{n-1}$.*

(2) With a suitable choice of labeling, $\tilde{L}_1 \cap \pi^{-1}(0) = \tilde{L}_1 \cap C_1 = \{p_1\},\ \tilde{L}_2 \cap \pi^{-1}(0) = \tilde{L}_2 \cap C_n = \{p_2\}$.

(3) The scheme $\pi^{-1}(X^\theta)$ is reduced.

Proof. (1) First, the lift $\tilde{\theta}$ preserves all the components C_i thanks to eq. (4.3.4). It follows from Corollary 4.2.4 that the $\tilde{\theta}$-fixed components and non-fixed components appear alternatingly. If $\tilde{\theta}$ fixes all the odd components, then there are no isolated points in $\pi^{-1}(0)^{\tilde{\theta}}$. However, (1) of Proposition 4.2.5 tells us that there are two isolated points in $\pi^{-1}(0)^{\tilde{\theta}}$. Therefore, $\tilde{\theta}$ fixes each C_i with i even. Then it follows from (3) of Lemma 4.2.3 that $\tilde{\theta}$ has exactly two fixed points on each C_j with j odd. The intersection points $C_j \cap C_{j-1}$ and $C_j \cap C_{j+1}$ exhaust both $\tilde{\theta}$-fixed points of C_j when $j \neq 1, n,\ j$ odd. However, when $j = 1$ (resp. $j = n$), the intersection point $C_! \cap C_2$ (resp. $C_n \cap C_{n-1}$) only accounts for one of the $\tilde{\theta}$-fixed points of C_1 (resp. C_n), resulting in an extra $\tilde{\theta}$-fixed point on C_1 (resp. C_n) that does not lie in C_2 (resp. C_{n-1}).

(2) This follows from Proposition 4.2.5.

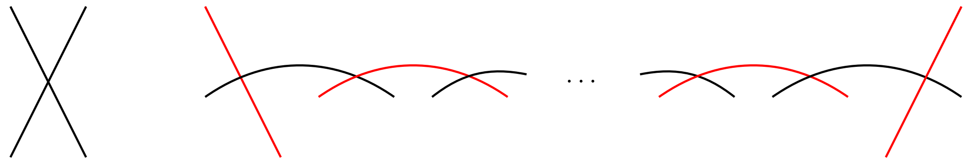
(3) Recall the definition of b_i in eq. (4.2.9). We have $(b_1, b_2, \cdots, b_{n-1}, b_n) = (1, 0, \cdots, 0, 1)$ by (2). The divisor $X^\theta = \operatorname{div}(z)$ is principal. Then applying Proposition 4.2.8, we can determine that

$$(a_1, \cdots, a_n) = (1, \cdots, 1).$$

Therefore, $\pi^{-1}(X^\theta)$ is generically reduced. Note that $\pi^{-1}(X^\theta) = \operatorname{div}(\pi^* z)$ is a principal divisor, so being generically reduced is equivalent to being reduced (c.f. (1) of Remark 4.2.9).

☐

We depict the fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ in Picture 4.4. All the components have multiplicity 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.



X^θ $\qquad$ $\pi^{-1}(X^\theta)$

Picture 4.4: $A_n,\ n$ odd, Type II

4.3.3 Type $A_n,\ n$ odd, case III

In this section, we consider the anti-Poisson involution

$$\theta(x) = -x,\ \theta(y) = -y,\ \theta(z) = -z.$$

The fixed point locus X^θ is singular point 0. The preimage $\pi^{-1}(X^\theta)$ is the exceptional fiber $\pi^{-1}(0)$ whose irreducible components are n-$\mathbb{P}^1$'s with pairwise transversal intersection according dually to the Dynkin diagram A_n. Moreover, $\pi^{-1}(0)$ is reduced (according to Remark 2.1.1). All questions for this case have been solved without constructing a lift.

Remark 4.3.5. However, a lift of θ still exists, and it is given by

$$\tilde{\theta}\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w),\ \ [B, B^*, l_0, k_0] \mapsto [-\epsilon B, \epsilon^{-1}B^*, l_0, -k_0], \tag{4.3.8}$$

where ϵ is a primitive $(2n+2)$-th root of unity. Moreover, $\tilde{\theta}$ fixes each C_i with i odd. The proofs are similar to those of Propositions 4.3.3 and 4.3.4.

4.3.4 Type $A_n,\ n$ even, case I

In this section, we consider the anti-Poisson involution

$$\theta(x) = y,\ \theta(y) = x,\ \theta(z) = -z.$$

The fixed point locus $X^\theta = \operatorname{Spec}\mathbb{C}[X]/(x-y) = \operatorname{Spec}\mathbb{C}[x,y,z]/(x^2-z^{n+1})$ is a cusp (a single irreducible component L). Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}$. This section is very similar to Section 4.3.1 with some minor changes due to the parity. We state the results, but omit the proofs. Recall the automorphism τ of the extended Dynkin diagram of type $\widetilde{A}_n$ defined in eq. (4.3.5).

Proposition 4.3.6. *The involution $\tilde{\theta}\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w)$ defined by*

$$\tilde{\theta}([B_{i+1\leftarrow i}, B^*_{j-1\leftarrow j}, l_0, k_0]) = [B^*_{\tau(i+1)\leftarrow\tau(i)}, B_{\tau(j-1)\leftarrow\tau(j)}, l_0, -k_0] \tag{4.3.9}$$

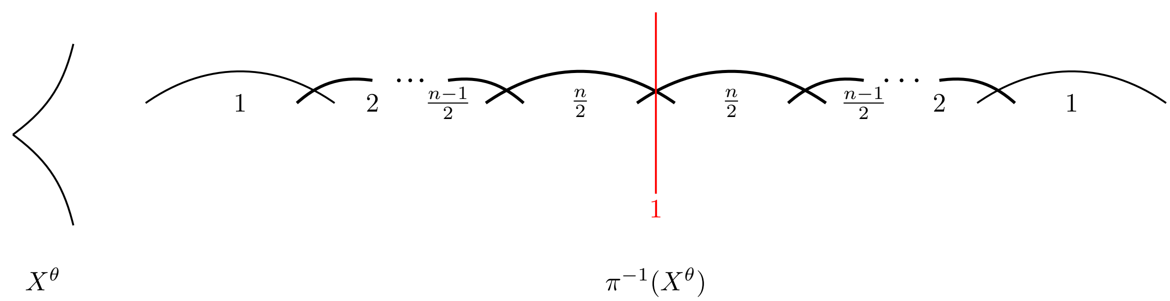
is anti-symplectic. Moreover, it is a lift of $\theta\colon X \to X$.

Proposition 4.3.7. *1. The involution $\tilde{\theta}$ swaps the component C_i with C_{n+1-i}, $\forall i$. We have $\pi^{-1}(0)^{\tilde{\theta}} = C_{\frac{n-1}{2}} \cap C_{\frac{n+1}{2}} = \{p\}$.*

2. $\tilde{L} \cap \pi^{-1}(0) = \tilde{L} \cap C_{\frac{n-1}{2}} \cap C_{\frac{n+1}{2}} = \{p\}$.

3. The scheme $\pi^{-1}(X^\theta)$ is not reduced. As a divisor, we have $\pi^{-1}(X^\theta) = \tilde{L}_1 + \tilde{L}_2 + \sum_{i=1}^n a_i C_i$, where $a_i = \min\{i, n+1-i\}$.

The fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ are depicted in Picture 4.5. The number below each component indicates its multiplicity. Thickened lines indicate that the multiplicities are larger than 1. The unique $\tilde{\theta}$-fixed component of $\pi^{-1}(X^\theta)$ is $\tilde{L}$, and it is colored in red.



Picture 4.5: $A_n,\ n$ even, Type I

4.3.5 Type $A_n,\ n$ even, case II

In this section, we consider the anti-Poisson involution

$$\theta(x) = -x,\ \theta(y) = y,\ \theta(z) = -z.$$

The fixed point locus $X^\theta = \operatorname{Spec} \mathbb{C}[X]/(x, z) \simeq \mathbb{A}^1$ has a single irreducible component L. Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}$. Similar to Proposition 4.3.3, one can construct the following lift of θ.

Proposition 4.3.8. *The involution $\tilde{\theta}\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w)$ defined by*

$$\tilde{\theta}([B, B^*, l_0, k_0]) = [-B, B^*, l_0, -k_0] \tag{4.3.10}$$

is anti-symplectic. Moreover, it is a lift of $\theta\colon X \to X$.

Proposition 4.3.9. *(1) $\pi^{-1}(0)^{\tilde{\theta}} = \bigcup_{i \text{ even}} C_i \cup \{p\}$ where $p \in C_1 \setminus C_2$.*

(2) We have $\tilde{L} \cap \pi^{-1}(0) = \tilde{L} \cap C_1 = \{p\}$.

(3) The scheme $\pi^{-1}(X^\theta)$ is generically reduced.

Proof. The proofs to (1) and (2) are almost identical to those of Proposition 4.3.4, so we leave it to the reader. The proof to (3) will be a bit different from that of Proposition 4.3.4 because X^θ is not a principal divisor. Set

$$Y := \operatorname{Spec} \mathbb{C}[X]/(z) = \operatorname{Spec} \mathbb{C}[x, y, z]/(z, xy) \supset X^\theta. \tag{4.3.11}$$

The closed embedding $\pi^{-1}(X^\theta) \hookrightarrow \pi^{-1}(Y)$ implies that $\pi^{-1}(X^\theta) \leqslant \pi^{-1}(Y)$ as divisors. We prove that $\pi^{-1}(Y)$ is reduced in Lemma 4.3.10. It follows that $\pi^{-1}(X^\theta)$ is generically reduced as well. $\square$

Lemma 4.3.10. *The scheme $\pi^{-1}(Y)$ is reduced.*

Proof. Note that $\pi^{-1}(Y) = \operatorname{div}(\pi^* z)$ is a principal divisor. We want to apply Proposition 4.2.8 to show that $\pi^{-1}(Y)$ is reduced (c.f. (1) and (4) of Remark 4.2.9). To figure out what are the b_i's as defined in eq. (4.2.9), we need to determine the irreducible components of $\pi^{-1}(Y)$ and study how different components intersect with each other. The scheme $Y = \operatorname{Spec} \mathbb{C}[X]/(z) = \operatorname{Spec}[x, y]/(xy)$ has two irreducible components $L = \operatorname{Spec} \mathbb{C}[y]$ (coinciding with X^θ) and $L' := \operatorname{Spec} \mathbb{C}[x]$. Similar to $\tilde{L}$, one can define $\tilde{L}' := \overline{\pi^{-1}(L' \setminus \{0\})}$. Then the irreducible components of $\pi^{-1}(Y)$ are $C_1, \cdots, C_n, \tilde{L}, \tilde{L}'$.

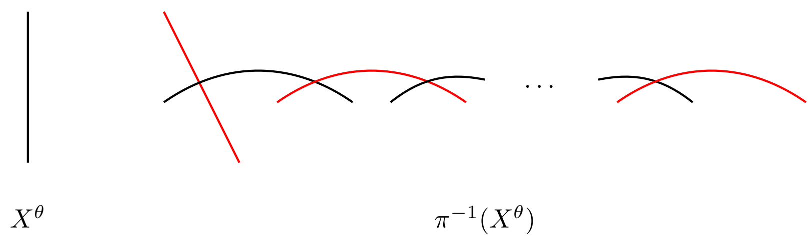
There is a recipe in Section 4.2 that allows us to understand how different components of $\pi^{-1}(X^\eta)$ intersect with each other when η is an anti-Poisson involution of X. However, Y is not the fixed point locus of an anti-Poison involution when n is even (the setup of Section 4.3.5). Nevertheless, we can write Y as the union of the fixed point loci of two different anti-Poisson involutions θ and θ' of X, where

$$\theta'\colon \mathbb{C}[X] \to \mathbb{C}[X],\ \ x \mapsto x,\ y \mapsto -y,\ z \mapsto z.$$

We have $X^{\theta'} = L'$. It is easy to see that θ' is conjugate to θ from the classification in Proposition 3.3.2. Recall that we have $\tilde{L} \cap \pi^{-1}(0) = \tilde{L} \cap C_1 = \{p\} \subset C_1 \setminus C_2$ from (2) of Proposition 4.3.9. Similarly, one can show that $\tilde{L}' \cap \pi^{-1}(0) = \tilde{L}' \cap C_n = \{p'\} \subset C_n \setminus C_{n-1}$. Therefore, Y and $\pi^{-1}(Y)_{\mathrm{red}}$ look like in Picture 4.4 (forgetting about the red coloring).

Lastly, we compute the multiplicities of the components of $\pi^{-1}(Y)$. We write $\pi^{-1}(Y) = \tilde{L} + \tilde{L}' + \sum_{i=1}^{n} a_i C_i$ as a divisor of $\widetilde{X}$. Recall the definition of b_i in eq. (4.2.9). We have $(b_1, b_2, \cdots, b_{n-1}, b_n) = (1, 0, \cdots, 0, 1)$ from the previous paragraph. We can determine that $a_i = 1$ by eq. (4.2.11). Therefore, $\pi^{-1}(Y)$ is generically reduced. $\square$

We depict the fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ in Picture 4.6. All the components have multiplicity 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.



Picture 4.6: $A_n,\ n$ even, Type II

Remark 4.3.11. There is also a computational way to study the scheme-theoretic preimage $\pi^{-1}(X^\theta)$. We explain the method below, but omit some details.

Recall that we have $X \simeq \mu^{-1}(0)/\!/\operatorname{GL}(\delta) \simeq \operatorname{Spec}\mathbb{C}[x, y, z]/(xy - z^{n+1})$, where the generators x, y, z are defined in eq. (4.3.2). Moreover, $\widetilde{X} = \mu^{-1}(0)/\!/^{\chi}\operatorname{GL}(\delta)$, and the natural projective morphism $\pi \colon \widetilde{X} \to X$ gives the minimal resolution (c.f. Theorem 4.1.9). Set $B_{j\leftarrow 0} := B_{j\leftarrow j-1} \cdots B_{1\leftarrow 0}$, $B^*_{j\leftarrow 0} := B^*_{j\leftarrow j+1} \cdots B^*_{n\leftarrow 0}$. We identify $B_{0\leftarrow 0} = B^*_{n+1\leftarrow 0} = \mathrm{id}_{V_0}$ and $B_{n+1\leftarrow 0} = x$, $B^*_{0\leftarrow 0} = y$. Consider $f_i := (\prod_{j=0}^{i} B_{j\leftarrow 0} l_0)(\prod_{j=i+1}^{n} B^*_{j\leftarrow 0} l_0) \in \mathbb{C}[\mu^{-1}(0)]^{\operatorname{GL}(\delta),\chi}$. The principal affine open subset $U_i := \mu^{-1}(0)_{f_i} \subset \mu^{-1}(0)^{\chi-ss}$ is $\operatorname{GL}(\delta)$-invariant, and one can show that $\{U_i\}_{i\in I}$ covers $\mu^{-1}(0)^{\chi-ss}$ by Proposition 4.1.2. It follows that $\{U_i/\!/\operatorname{GL}(\delta)\}_{i\in I}$ forms an affine open cover of $\widetilde{X}$. Moreover, for each $i \in I$, set $u_i := \frac{B^*_{i\leftarrow 0}}{B_{i\leftarrow 0}}$, $v_i := \frac{B_{i+1\leftarrow 0}}{B^*_{i+1\leftarrow 0}}$. One can prove that

$$U_i/\!/\operatorname{GL}(\delta) = \operatorname{Spec}\mathbb{C}[U_i]^{\operatorname{GL}(\delta)} = \operatorname{Spec}\mathbb{C}[u_i, v_i] \simeq \mathbb{C}^2, \tag{4.3.12}$$

using a version of the Zariski's Main Theorem [Gro67, Corollary 18.12.13]. Moreover, two affine open subsets $U_i/\!/\operatorname{GL}(\delta)$, $U_j/\!/\operatorname{GL}(\delta)$ with $i < j$ can be glued via $v_i u_j z^{j-i-1} = 1$. The composition $\pi_i \colon U_i/\!/\operatorname{GL}(\delta) \hookrightarrow$

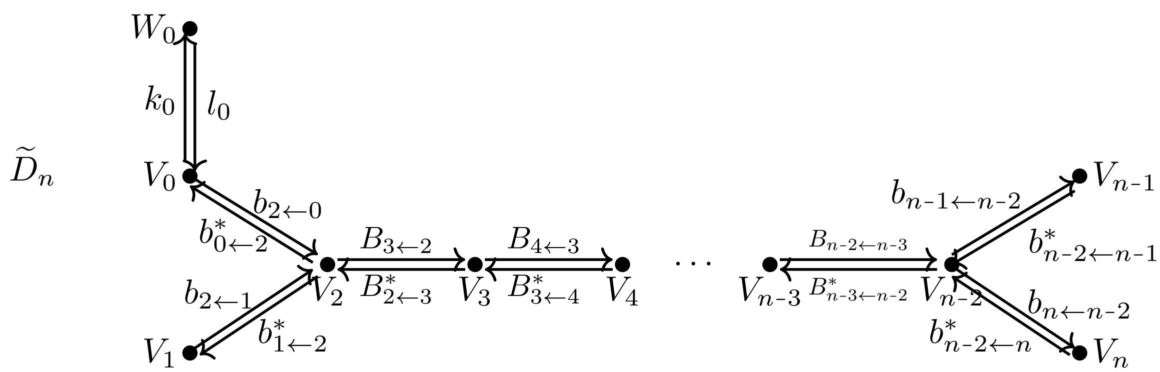
$\widetilde{X} \xrightarrow{\pi} X$, in terms of coordinates, is given by

$$(u_i, v_i) \mapsto (x = u_i^{n-i} v_i^{n-i+1},\ y = u_i^{i+1} v_i^{i},\ z = u_i v_i). \tag{4.3.13}$$

We can determine the preimage $\pi^{-1}(Y)$ of any subscheme $Y \subset X$ by computing the preimages $\pi_i^{-1}(Y) = \pi^{-1}(Y) \cap (U_i /\!/ \mathrm{GL}(\delta))$ using eq. (4.3.13), and then glue them together.

The idea of the computational method can be carried beyond type A as well, but the computations are not manageable.

4.4 Preimages of fixed point loci for type D_n Kleinian singularity



Picture 4.7: Extended Dynkin quiver $\widetilde{D}_n$

We realize the Kleinian singularity of type D_n as a Nakajima quiver variety explicitly. Recall that $\delta = (1, 1, 2, \cdots, 2, 1, 1)$ Using notations in Section 4.1.2, we have that $W_0, V_0, V_1, V_{n-1}, V_n$ are 1-dimensional vector spaces, and $V_i,\ 2 \leqslant i \leqslant n-2$ are 2-dimensional vector spaces. We use $b_{j\leftarrow i},\ b^*_{i\leftarrow j}$ to denote the linear maps between a 1-dimensional vector space and a 2-dimensional vector space; and $B_{i+1\leftarrow i},\ B^*_{i\leftarrow i+1}$ to denote the linear maps between 2-dimensional vector spaces. The representation space is

$$\begin{aligned} M(\delta, w) = T^*\big(&\operatorname{Hom}(V_0, V_2) \oplus \operatorname{Hom}(V_1, V_2) \oplus \bigoplus_{i=2}^{n\text{-}3} \operatorname{Hom}(V_i, V_{i+1}) \\ &\oplus \operatorname{Hom}(V_{n\text{-}2}, V_{n\text{-}1}) \oplus \operatorname{Hom}(V_{n\text{-}2}, V_n) \oplus \operatorname{Hom}(W_0, V_0)\big). \end{aligned}$$

The group $\mathrm{GL}(\delta) = \prod_{i=0}^{n} \mathrm{GL}(V_i)$ acts on $M(\delta, w)$ naturally with the moment map

$$\mu = (\mu_i)_{0 \leqslant i \leqslant n} \colon M(\delta, w) \to \mathfrak{gl}(\delta),$$

where, for $n \geqslant 5$, we have

$$
\begin{aligned}
\mu_0 &= -b^*_{0\leftarrow 2}b_{2\leftarrow 0} + l_0k_0,\\
\mu_1 &= -b^*_{1\leftarrow 2}b_{2\leftarrow 1},\\
\mu_2 &= b_{2\leftarrow 0}b^*_{0\leftarrow 2} + b_{2\leftarrow 1}b^*_{1\leftarrow 2} - B^*_{2\leftarrow 3}B_{3\leftarrow 2},\\
\mu_i &= B_{i\leftarrow i-1}B^*_{i-1\leftarrow i} - B^*_{i\leftarrow i+1}B_{i+1\leftarrow i},\ 3\leqslant i\leqslant n-3,\\
\mu_{n-2} &= B_{n-2\leftarrow n-3}B^*_{n-3\leftarrow n-2} - b^*_{n-2\leftarrow n-1}b_{n-1\leftarrow n-2} - b^*_{n-2\leftarrow n}b_{n\leftarrow n-2},\\
\mu_{n-1} &= b_{n-1\leftarrow n-2}b^*_{n-2\leftarrow n-1},\\
\mu_n &= b_{n\leftarrow n-2}b^*_{n-2\leftarrow n}.
\end{aligned}
\tag{4.4.1}
$$

When $n = 4$, we have

$$\mu_2 = \mu_{n-2} = b_{2\leftarrow 0}b^*_{0\leftarrow 2} + b_{2\leftarrow 1}b^*_{1\leftarrow 2} - b^*_{2\leftarrow 3}b_{3\leftarrow 2} - b^*_{2\leftarrow 4}b_{4\leftarrow 2},$$

and $\mu_0,\ \mu_1,\ \mu_3 = \mu_{n-1},\ \mu_4 = \mu_n$ are determined by the same equations as in eq. (4.4.1). Picture 4.1 does a better job in illustrating the extended Dynkin quiver $\widetilde{D}_4$. Recall from Proposition 4.2.6, we have that

$$C_i = \left\{[B, B^*, l_0, 0] \in \mathcal{M}_\chi(\delta, w) \middle|\, \operatorname{rank}(B_a)_{\substack{a\in\bar{\Omega}\\ t(a)=i}} < \dim V_i\right\},\ 1\leqslant i\leqslant n.$$

To simplify the writing, we introduce the following notations.

$$
\begin{aligned}
B_{j\leftarrow i} &:= B_{j\leftarrow j-1}\cdots B_{i+1\leftarrow i} \colon V_i \to V_j,\ 2\leqslant i\leqslant j\leqslant n-2,\\
B^*_{i\leftarrow j} &:= B^*_{i\leftarrow i+1}\cdots B^*_{j-1\leftarrow j} \colon V_j \to V_i,\ 2\leqslant i\leqslant j\leqslant n-2.
\end{aligned}
$$

According to Corollary 4.1.8 and Proposition 2.3.1, the algebra of invariant functions $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ is generated by trace functions of loops that start and end at the vertex 0 of degrees $4,\ 2n-4$ and $2n-2$. The explicit generators are given in Proposition 4.4.1.

Proposition 4.4.1. *The trace functions*

$$
\begin{aligned}
x :=& b^*_{0\leftarrow 2}b_{2\leftarrow 1}b^*_{1\leftarrow 2}b_{2\leftarrow 0}, \quad y := b^*_{0\leftarrow 2}B^*_{2\leftarrow n-2}b^*_{n-2\leftarrow n}b_{n\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0},\\
z :=& b^*_{0\leftarrow 2}B^*_{2\leftarrow n-2}b^*_{n-2\leftarrow n}b_{n\leftarrow n-2}b^*_{n-2\leftarrow n-1}b_{n-1\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0}.
\end{aligned}
\tag{4.4.2}
$$

generate $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$. *Moreover, we have*

$$\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)} \simeq \begin{cases} \mathbb{C}[x,y,z]/(xy(y-x^{\frac{n-2}{2}})-z^2),\ n \text{ even}, \\ \mathbb{C}[x,y,z]/(xy^2-z(z-x^{\frac{n-1}{2}})),\ n \text{ odd}. \end{cases} \tag{4.4.3}$$

Lemmas 4.4.2 and 4.4.3 will be needed for the proof of Proposition 4.4.1.

Lemma 4.4.2. *Consider the extended Dynkin quiver* $\widetilde{D}_n,\ n \geqslant 5$. *The following relations hold in* $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$.

$$\begin{aligned} &b^*_{1\leftarrow 2}(B^*_{2\leftarrow 3}B_{3\leftarrow 2})^r b_{2\leftarrow 1} = b_{n-1\leftarrow n-2}(B_{n-2\leftarrow n-3}B^*_{n-3\leftarrow n-2})^r b^*_{n-2\leftarrow n-1} \\ =&b^*_{0\leftarrow 2}(B^*_{2\leftarrow 3}B_{3\leftarrow 2})^r b_{2\leftarrow 0} = b_{n\leftarrow n-2}(B_{n-2\leftarrow n-3}B^*_{n-3\leftarrow n-2})^r b^*_{n-2\leftarrow n} = \begin{cases} x^{\frac{r+1}{2}},\ r \text{ odd}, \\ 0,\ r \text{ even}. \end{cases} \end{aligned} \tag{4.4.4}$$

Proof. We only prove the equalities in the second line of eq. (4.4.4). The remaining equalities can be shown similarly. We begin with $r=1$, i.e., we show that

$$b^*_{0\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}b_{2\leftarrow 0} = b_{n\leftarrow n-2}B_{n-2\leftarrow n-3}B^*_{n-3\leftarrow n-2}b^*_{n-2\leftarrow n} = x. \tag{4.4.5}$$

Note that by eq. (4.1.12), we have $l_0k_0 = 0$. Then together with $\mu_0 = 0$, where μ_0 is given in eq. (4.4.1), we obtain that

$$-b^*_{0\leftarrow 2}b_{2\leftarrow 0} = 0. \tag{4.4.6}$$

Consider the first term in eq. (4.4.5), we have

$$b^*_{0\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}b_{2\leftarrow 0} \xlongequal{\mu_2} b^*_{0\leftarrow 2}(b_{2\leftarrow 1}b^*_{1\leftarrow 2}+b_{2\leftarrow 0}b^*_{0\leftarrow 2})b_{2\leftarrow 0} \xlongequal{eq.\ (4.4.6)} b^*_{0\leftarrow 2}b_{2\leftarrow 1}b^*_{1\leftarrow 2}b_{2\leftarrow 0} \xlongequal{eq.\ (4.4.2)} x.$$

For the second term in eq. (4.4.5), using $\mu_{n-2} = \mu_n = 0$, we obtain that

$$b_{n\leftarrow n-2}B_{n-2\leftarrow n-3}B^*_{n-3\leftarrow n-2}b^*_{n-2\leftarrow n} = b_{n\leftarrow n-2}b^*_{n-2\leftarrow n-1}b_{n-1\leftarrow n-2}b^*_{n-2\leftarrow n}.$$

Therefore, to prove eq. (4.4.5), it remains to prove eq. (4.4.7) below.

$$b_{n\leftarrow n-2}b^*_{n-2\leftarrow n-1}b^*_{n-1\leftarrow n-2}b^*_{n-2\leftarrow n} = b^*_{0\leftarrow 2}b_{2\leftarrow 1}b^*_{1\leftarrow 2}b_{2\leftarrow 0} = x, \tag{4.4.7}$$

which follows from the following computation.

$$
\begin{aligned}
2b_{n\leftarrow n-2}b^*_{n-2\leftarrow n-1}b_{n-1\leftarrow n-2}b^*_{n-2\leftarrow n} &\xlongequal{\mu_{n\text{-}1},\ \mu_n} \operatorname{tr}((b^*_{n-2\leftarrow n-1}b_{n-1\leftarrow n-2}+b^*_{n-2\leftarrow n}b_{n\leftarrow n-2})^2)\\
&\xlongequal{\mu_{n\text{-}2}} \operatorname{tr}((B_{n-2\leftarrow n-3}B^*_{n-3\leftarrow n-2})^2) \xlongequal{\mu_i} \operatorname{tr}((B^*_{2\leftarrow 3}B_{3\leftarrow 2})^2)\\
&\xlongequal{\mu_2} \operatorname{tr}((b_{2\leftarrow 0}b^*_{0\leftarrow 2}+b_{2\leftarrow 1}b^*_{1\leftarrow 2})^2) = 2b^*_{0\leftarrow 2}b_{2\leftarrow 1}b^*_{1\leftarrow 2}b_{2\leftarrow 0} = 2x.
\end{aligned}
\tag{4.4.8}
$$

Next, we prove the equalities in the second line of eq. (4.4.4) for any r.

$$
\begin{aligned}
b^*_{0\leftarrow 2}(B^*_{2\leftarrow 3}B_{3\leftarrow 2})^r b_{2\leftarrow 0} &\xlongequal{\mu_2} \begin{cases} \underline{b^*_{0\leftarrow 2}b_{2\leftarrow 1}b^*_{1\leftarrow 2}b_{2\leftarrow 0}}b^*_{0\leftarrow 2}\cdots b_{2\leftarrow 0}b^*_{0\leftarrow 2}b_{2\leftarrow 1}b^*_{1\leftarrow 2}b_{2\leftarrow 0},\ r \text{ odd},\\ \underline{b^*_{0\leftarrow 2}b_{2\leftarrow 0}}b^*_{0\leftarrow 2}b_{2\leftarrow 1}b^*_{1\leftarrow 2}\cdots b_{2\leftarrow 0}b^*_{0\leftarrow 2}b_{2\leftarrow 1}b^*_{1\leftarrow 2}b_{2\leftarrow 0},\ r \text{ even}.\end{cases}\\
&= \begin{cases} (b^*_{0\leftarrow 2}b_{2\leftarrow 1}b^*_{1\leftarrow 2}b_{2\leftarrow 0})^{\frac{r+1}{2}} = x^{\frac{r+1}{2}},\ r \text{ odd},\\ 0,\ r \text{ even}.\end{cases}
\end{aligned}
\tag{4.4.9}
$$

The computation to show that

$$
b_{n\leftarrow n-2}(B_{n-2\leftarrow n-3}B^*_{n-3\leftarrow n-2})^r b^*_{n-2\leftarrow n} = \begin{cases} x^{\frac{r+1}{2}},\ r \text{ odd},\\ 0,\ r \text{ even},\end{cases}
\tag{4.4.10}
$$

is similar to that of eq. (4.4.9) by using the $r=1$ case:

$$
b_{n\leftarrow n-2}B_{n-2\leftarrow n-3}B^*_{n-3\leftarrow n-2}b^*_{n-2\leftarrow n} = b_{n\leftarrow n-2}b^*_{n-2\leftarrow n-1}b_{n-1\leftarrow n-2}b^*_{n-2\leftarrow n} = x,
$$

and we leave it to the reader. □

Lemma 4.4.3. *Let q be a loop on the extended Dynkin quiver $\widetilde{D}_n,\ n \geqslant 5$ that starts and ends at the vertex 0. Assume that q does not contain the vertices $n-1, n$. Then the following holds in $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$.*

$$
\operatorname{tr}(B_q) \begin{cases} \in \mathbb{C}x^{\frac{\deg q}{4}},\ 4|\deg q,\\ = 0,\ 4\nmid \deg q.\end{cases}
\tag{4.4.11}
$$

Proof. We may assume that the vertex 0 appears only once in q. Otherwise, we can decompose that $q = q_1q_2$ where q_1, q_2 are loops that both start and end at 0 but with lower degrees. We have $\operatorname{tr}(B_q) = \operatorname{tr}(B_{q_1})\operatorname{tr}(B_{q_2})$. Proving eq. (4.4.11) breaks down to showing the corresponding statements for $\operatorname{tr}(B_{q_1})$ and $\operatorname{tr}(B_{q_2})$.

Note that q does not contain the vertices $n-1, n$ by assumption. We may further assume that q does not contain the vertex 1. Otherwise, we can replace the appearance of $b_{2\leftarrow 1}b^*_{1\leftarrow 2}$ in $\mathrm{tr}(B_q)$ with $B^*_{2\leftarrow 3}B_{3\leftarrow 2} - b_{2\leftarrow 0}b^*_{0\leftarrow 2}$ using the moment map $\mu_2 = 0$.

It remains to prove eq. (4.4.11) when q is a loop that starts and ends at the vertex 0 on the subquiver $0 \rightleftarrows 2 \rightleftarrows 3 \rightleftarrows \cdots \rightleftarrows n-2$ of type A. Using the moment maps $\mu_i = 0,\ 3 \leqslant i \leqslant n-3$ in eq. (4.4.1) iteratively, we can deduce that $\mathrm{tr}(B_q) = b^*_{0\leftarrow 2}(B^*_{2\leftarrow 3}B_{3\leftarrow 2})^{\frac{\deg q-2}{2}}b_{2\leftarrow 0}$. Then eq. (4.4.11) follows from Lemma 4.4.2. $\square$

Proof of Proposition 4.4.1. We only prove the statement for $n \geqslant 5$. The case $n = 4$ is similar, but easier.

We first show that x, y, z generate $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$. It suffices to show that the degree d-th component $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_d$ with $d = 2, 2n-4, 2n-2$, is contained in the subalgebra generated by x, y, z.

The only loops of degree 4 that start and end at the vertex 0 are $0 \to 2 \to 0 \to 2 \to 0,\ 0 \to 2 \to 1 \to 2 \to 0,\ 0 \to 2 \to 3 \to 2 \to 0$. The corresponding trace functions are

$$
\begin{aligned}
b^*_{0\leftarrow 2}b_{2\leftarrow 0}b^*_{0\leftarrow 2}b_{2\leftarrow 0} &\overset{\mu_0}{=} 0,\\
b^*_{0\leftarrow 2}b_{2\leftarrow 1}b^*_{1\leftarrow 2}b_{2\leftarrow 0} &\overset{\mathrm{def}}{=} x,\\
b^*_{0\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}b_{2\leftarrow 0} &\overset{\mu_2}{=} b^*_{0\leftarrow 2}(b_{2\leftarrow 0}b^*_{0\leftarrow 2} + b_{2\leftarrow 1}b^*_{1\leftarrow 2})b_{2\leftarrow 0} = 0 + x = x.
\end{aligned}
$$

Thus, $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_4 = \mathbb{C}x$.

A loop of degree $2n-4$ that starts and ends at 0 can contain at most one of $n-1, n$. If it contains neither $n-1$ nor n, Lemma 4.4.3 tells us that the corresponding trace function is a scalar multiple of $x^{\frac{n-2}{2}}$ (resp. 0) when n is even (resp. odd). If the loop contains exactly one of $n-1, n$, then it must be of the form $0 \to 2 \to \cdots \to n-2 \to n \to n-2 \to \cdots \to 2 \to 0$ or $0 \to 2 \to \cdots \to n-2 \to n-1 \to n-2 \to \cdots \to 2 \to 0$, and the corresponding trace functions are

$$
\begin{aligned}
y &\overset{\mathrm{def}}{=} b^*_{0\leftarrow 2}B^*_{2\to n-2}b^*_{n-2\to n}b_{n\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0},\\
y' &:= b^*_{0\leftarrow 2}B^*_{2\to n-2}b^*_{n-2\to n-1}b_{n-1\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0}.
\end{aligned}
\tag{4.4.12}
$$

There is a relation between y and y'.

$$\begin{aligned}&\overset{\mu_{n-2}}{=} b^*_{0\leftarrow 2}B^*_{2\to n-2}B_{n-2\leftarrow n-3}B^*_{n-3\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0}{=}b^*_{0\leftarrow 2}(B^*_{2\to 3}B_{3\leftarrow 2})^{n-3}b_{2\leftarrow 0}\\ &\underline{\underline{\ \textit{Lemma 4.4.2}\ }}\begin{cases}x^{\frac{n-2}{2}}, n \text{ even},\\ 0,\ n \text{ odd}.\end{cases}\end{aligned}\tag{4.4.13}$$

It follows that $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_{2n-4} = \mathbb{C}y + \mathbb{C}x^{\frac{n-2}{2}}$ when n is even, and $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_{2n-4} = \mathbb{C}y$ when n is odd.

Next, we consider the following two loops of degree $2n-2$ that start and end at 0.

$$\begin{aligned}z &= b^*_{0\leftarrow 2}B^*_{2\leftarrow n-2}b^*_{n-2\leftarrow n}b_{n\leftarrow n-2}b^*_{n-2\leftarrow n-1}b_{n-1\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0},\\ z' &:= b^*_{0\leftarrow 2}B^*_{2\leftarrow n-2}b^*_{n-2\leftarrow n-1}b_{n-1\leftarrow n-2}b^*_{n-2\leftarrow n}b_{n\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0}.\end{aligned}\tag{4.4.14}$$

One can compute that

$$z+z' = \begin{cases}0,\ n \text{ even},\\ x^{\frac{n-1}{2}},\ n \text{ odd}.\end{cases}\tag{4.4.15}$$

Similarly to the case of degree $2n-4$, one can show that that $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_{2n-2} = \mathbb{C}z$ when n is even, and $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_{2n-2} = \mathbb{C}z + \mathbb{C}x^{\frac{n-1}{2}}$ when n is odd. Therefore, $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ is generated by the functions x, y, z defined in eq. (4.4.2) for both n even and odd. Lastly, we compute that

$$\begin{aligned}zz' &= b^*_{0\leftarrow 2}B^*_{2\leftarrow n-2}b^*_{n-2\leftarrow n}b_{n\leftarrow n-2}b^*_{n-2\leftarrow n-1}\underline{b_{n-1\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0}}\\ &\quad \underline{\cdot b^*_{0\leftarrow 2}B^*_{2\leftarrow n-2}b^*_{n-2\leftarrow n-1}}b_{n-1\leftarrow n-2}b^*_{n-2\leftarrow n}b_{n\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0}\\ &= y'b^*_{0\leftarrow 2}B^*_{2\leftarrow n-2}b^*_{n-2\leftarrow n}\underline{b_{n\leftarrow n-2}b^*_{n-2\leftarrow n-1}b_{n-1\leftarrow n-2}b^*_{n-2\leftarrow n}}b_{n\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0}\\ &= xy'b^*_{0\leftarrow 2}B^*_{2\leftarrow n-2}b^*_{n-2\leftarrow n}b_{n\leftarrow n-2}B_{n-2\leftarrow 2}b_{2\leftarrow 0} = xyy',\end{aligned}\tag{4.4.16}$$

where the first and the last equalities only use the definitions of y, y' and z, z' in eqs. (4.4.12) and (4.4.14), and the second equality follows from Lemma 4.4.2 (taking $r=1$). Then combining eqs. (4.4.13), (4.4.15) and (4.4.16), we obtain that

$$0 = xyy' - zz' = \begin{cases}xy(y - x^{\frac{n-2}{2}}) - z^2,\ n \text{ even},\\ xy^2 - z(z - x^{\frac{n-1}{2}}),\ n \text{ odd}.\end{cases}\tag{4.4.17}$$

The isomorphism in eq. (4.4.3) follows from the same argument as we show that eq. (4.3.3) is an isomorphism in type A_n. $\square$

Consider the following automorphism defined on the extended Dynkin diagram of type $\widetilde{D}_n$.

$$\tau(i) = i,\ 0 \leqslant i \leqslant n-2;\ \tau(n-1) = n;\ \tau(n) = n-1. \tag{4.4.18}$$

It induces an automorphism of the extended Dynkin quiver $\widetilde{D}_n$ in Picture 4.7 by sending an arrow $a : i \to j$ to the arrow $\tau(a) : \tau(i) \to \tau(j)$, which will be used to construct the lifts of the anti-Poisson involutions in Sections 4.4.2 and 4.4.4. Note that τ coincides with the the order 2 automorphism of the McKay graph of type $\widetilde{D}_n$ induced by an element $g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$ (c.f. Remarks 3.3.6 and 3.3.9). Sections 4.4.1 and 4.4.2 study the preimages of fixed point loci of anti-Poissons for Kleinian singularities of type D_n, with n even. We have $X = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy(y - x^{\frac{n-2}{2}}) - z^2)$.

4.4.1 Type $D_n,\ n$ even, case I

In this section, we consider the anti-Poisson involution

$$\theta(x) = x,\ \theta(y) = y,\ \theta(z) = -z.$$

The fixed point locus $X^\theta = \operatorname{Spec} \mathbb{C}[X]/(z) = \operatorname{Spec} \mathbb{C}[x, y]/(xy(y - x^{\frac{n-2}{2}}))$ has three irreducible components $L_j,\ j = 1, 2, 3$, and each of them is an $\mathbb{A}^1$. Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}_1 \cup \tilde{L}_2 \cup \tilde{L}_3$.

Proposition 4.4.4. *The involution $\tilde{\theta}\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w)$ defined by*

$$\tilde{\theta}([B, B^*, l_0, k_0]) = [-B, B^*, l_0, -k_0] \tag{4.4.19}$$

is anti-symplectic. Moreover, it is a lift of $\theta\colon X \to X$.

Proof. We consider the following linear anti-symplectic involution on the representation space.

$$\Theta\colon M(\delta, w) \to M(\delta, w),\ \ (B, B^*, l_0, k_0) \mapsto (-B, B^*, l_0, -k_0).$$

It satisfies the conditions listed in Proposition 4.2.1.

1. $g^\Theta = g$.

2. $\Theta(x) = (-1)^2 x = x,\ \Theta(y) = (-1)^{n-2} y = y,\ \Theta(z) = (-1)^{n-1} z = -z.$

The involution $\tilde{\theta}$ defined in eq. (4.4.19) is the descent of Θ to $\mathcal{M}_\chi(\delta, w)$. The rest follows from Proposition 4.2.1. $\square$

Consider the action of $\tilde{\theta}$ on the reduced exceptional fiber $\pi^{-1}(0)$, we have the following results.

Proposition 4.4.5. *(1) The involution $\tilde{\theta}$ preserves all the components $C_i,\ 1 \leqslant i \leqslant n$. We have $\pi^{-1}(0)^{\tilde{\theta}} = \bigcup_{\substack{i\ even \\ 2 \leqslant i \leqslant n-2}} C_i \cup \{p_1, p_2, p_3\}$ where $p_1 \in C_1 \setminus C_2,\ p_2 \in C_{n-1} \setminus C_{n-2}$, and $p_3 \in C_n \setminus C_{n-2}$.*

(2) With a suitable choice of labeling, we have $\tilde{L}_1 \cap \pi^{-1}(0) = \tilde{L}_1 \cap C_1 = \{p_1\},\ \tilde{L}_2 \cap \pi^{-1}(0) = \tilde{L}_2 \cap C_{n-1} = \{p_2\}$, and $\tilde{L}_3 \cap \pi^{-1}(0) = \tilde{L}_3 \cap C_n = \{p_3\}$.

(3) The scheme $\pi^{-1}(X^\theta)$ is not reduced. As a divisor, we have $\pi^{-1}(X^\theta) = \tilde{L}_1 + \tilde{L}_2 + \tilde{L}_3 + \sum_{i=1}^n a_i C_i$, where $a_i = i + 1,\ 1 \leqslant i \leqslant n-2$ and $a_{n-1} = a_n = \frac{n}{2}$.

Proof. (1) It is easy to see that $\tilde{\theta}(C_i) = C_i$ thanks to Proposition 4.2.6. The component C_{n-2} intersects with three other components $C_{n-3},\ C_{n-1},\ C_n$. By (1) of Lemma 4.2.3, there are at least three $\tilde{\theta}$-fixed points on C_{n-2} (the intersection points with the adjacent components). It follows from (2) of Lemma 4.2.3 that $\tilde{\theta}$ acts trivially on C_{n-2}. By (3) of Lemma 4.2.3, $\tilde{\theta}$ has exactly two fixed points on each of C_{n-3}, C_{n-1}, C_n, resulting in three isolated points in $\pi^{-1}(0)^{\tilde{\theta}}$. The fact that $\tilde{\theta}$ fixes each $C_i,\ 2 \leqslant i \leqslant n-2$ with i even, follows from Corollary 4.2.4.

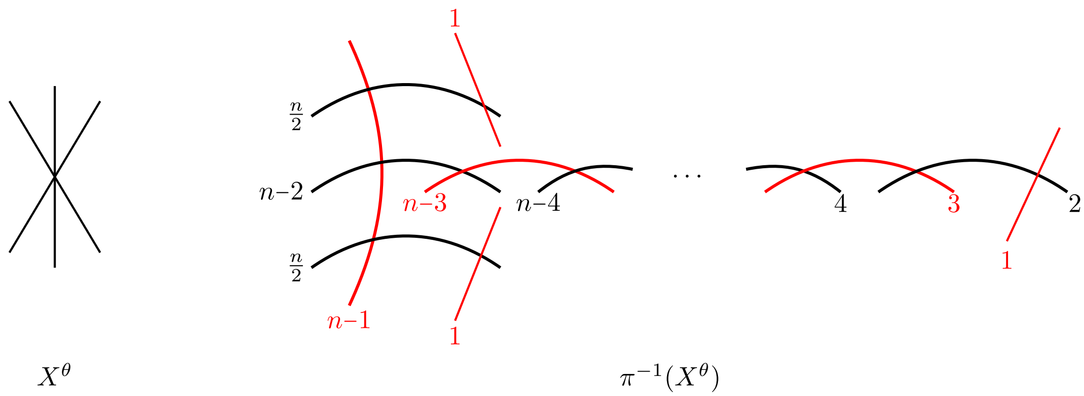
(2) This follows from Proposition 4.2.5.

(3) Recall the definition of b_i in eq. (4.2.9). We know that $(b_1, b_2, \cdots, b_{n-2}, b_{n-1}, b_n) = (1, 0, \cdots, 0, 1, 1)$ from (2). The divisor $X^\theta = \mathrm{div}(z)$ is principal. Then applying Proposition 4.2.8, we can determine that

$$(a_1, \cdots, a_i, \cdots, a_{n-2}, a_{n-1}, a_n) = (2, \cdots, i+1, \cdots, n-1, \frac{n}{2}, \frac{n}{2}).$$

$\square$

The fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ are depicted in Picture 4.8. The number next to each component indicates its multiplicity. Thickened lines indicate that the multiplicities are larger than 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.



Picture 4.8: $D_n,\ n$ even, Type I

4.4.2 Type $D_n,\ n$ even, case II

In this section, we consider the anti-Poisson involution

$$\theta(x) = x,\ \theta(y) = x^{\frac{n-2}{2}} - y,\ \theta(z) = z.$$

The fixed point locus $X^\theta = \operatorname{Spec}\mathbb{C}[X]/(2y - x^{\frac{n-2}{2}}) = \operatorname{Spec}\mathbb{C}[x,z]/(x^{n-1} + z^2)$ is a cusp (a single irreducible components L). Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}$.

Proposition 4.4.6. *The involution $\tilde{\theta}\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w)$ defined by*

$$\tilde{\theta}([B_a, B_a^*, l_0, k_0]) = [-B_{\tau(a)}, B^*_{\tau(a)}, l_0, -k_0] \tag{4.4.20}$$

is anti-symplectic. Moreover, it is a lift of $\theta\colon X \to X$.

Proof. We consider the following linear anti-symplectic involution on the representation space.

$$\Theta\colon M(\delta, w) \to M(\delta, w),\ \ (B_a, B_a^*, l_0, k_0) \mapsto (-B_{\tau(a)}, B^*_{\tau(a)}, l_0, k_0).$$

It satisfies the conditions listed in Proposition 4.2.1.

1. $(g_i)^\Theta_{i\in I} = (g_{\tau(i)})_{i\in I}$.
2. $\Theta(x) = (-1)^2 x = x,\ \Theta(y) = (-1)^{n-2}y' \xlongequal{eq.\ (4.4.13)} x^{\frac{n-2}{2}} - y,\ \Theta(z) = (-1)^{n-1}z' \xlongequal{eq.\ (4.4.15)} z.$

The involution $\tilde{\theta}$ defined in eq. (4.4.20) is the descent of Θ to $\mathcal{M}_\chi(\delta, w)$. The rest follows from Proposition 4.2.1. $\square$

Consider the action of $\tilde{\theta}$ on the exceptional fiber $\pi^{-1}(0)$, we have the following results.

Proposition 4.4.7. *(1) The involution $\tilde{\theta}$ preserves the components $C_i,\ 1 \leqslant i \leqslant n-2$, and swaps the two components C_{n-1}, C_n. We have $\pi^{-1}(0)^{\tilde{\theta}} = \bigcup_{\substack{i \text{ odd} \\ 1 \leqslant i \leqslant n-3}} C_i \cup \{p\}$ where $p \in C_{n-2} \setminus (C_{n-3} \cup C_{n-1} \cup C_n)$.*

(2) We have $\tilde{L} \cap \pi^{-1}(0) = \tilde{L} \cap C_{n-2} = \{p\}$.

(3) The scheme $\pi^{-1}(X^\theta)$ is not reduced. As a divisor, we have $\pi^{-1}(X^\theta) = \tilde{L} + \sum_{i=1}^{n} a_i C_i$, where $a_i = i,\ 1 \leqslant i \leqslant n-2$ and $a_{n-1} = a_n = \frac{n-2}{2}$.

Proof. (1) It is easy to see that $\tilde{\theta}(C_i) = C_{\tau(i)}$ thanks to Proposition 4.2.6. The intersection points $C_{n-1} \cap C_{n-2},\ C_n \cap C_{n-2}$ are permuted by $\tilde{\theta}$. It follows from (2) of Lemma 4.2.3 that there are exactly two $\tilde{\theta}$-fixed points on C_{n-2}. The intersection point $C_{n-3} \cap C_{n-2}$ is one of the $\tilde{\theta}$-fixed points on C_{n-2}, and the other one is an isolated point in $\pi^{-1}(0)^{\tilde{\theta}}$. It follows from Corollary 4.2.4 that the $\tilde{\theta}$-fixed components in $\{C_i \mid 1 \leqslant i \leqslant n-2\}$ appear alternatingly.

(2) This follows from Proposition 4.2.5.

(3) Recall the definition of b_i in eq. (4.2.9). We have $(b_1, \cdots, b_{n-3}, b_{n-2}, b_{n-1}, b_n) = (0, \cdots, 0, 1, 0, 0)$ by (2). Moreover, $X^\theta = \operatorname{div}(2y - x^{\frac{n-2}{2}})$ is a principal divisor. Then applying Proposition 4.2.8, we can determine that

$$(a_1, \cdots, a_i, \cdots, a_{n-2}, a_{n-1}, a_n) = (1, \cdots, i, \cdots, n-2, \frac{n-2}{2}, \frac{n-2}{2}).$$

$\square$

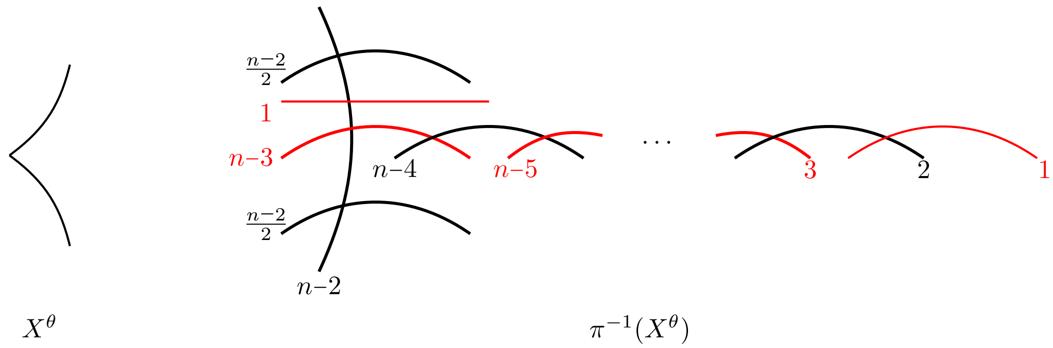
The fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ are depicted in Picture 4.9. The number next to each component indicates its multiplicity. Thickened lines indicate that the multiplicities are larger than 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.

Sections 4.4.3 and 4.4.4 Study the preimages of fixed point loci of anti-Poissons for Kleinian singularities of type D_n, with n odd. We have $X = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy^2 - z(z - x^{\frac{n-1}{2}}))$.

4.4.3 Type $D_n,\ n$ odd, case I

In this section, we consider the anti-Poisson involution

$$\theta(x) = x,\ \theta(y) = -y,\ \theta(z) = z.$$



Picture 4.9: $D_n,\ n$ even, Type II

The fixed point locus $X^\theta = \operatorname{Spec}\mathbb{C}[X]/(y) = \operatorname{Spec}\mathbb{C}[x,z]/(z(z-x^{\frac{n-1}{2}}))$ has two irreducible components $L_j,\ j = 1, 2$, and each of them is an $\mathbb{A}^1$. Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}_1 \cup \tilde{L}_2$. This section is very similar to Section 4.4.1 with some minor changes due to the parity. We state the results, but omit the proofs.

Proposition 4.4.8. *The involution $\tilde{\theta}\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w)$ defined by*

$$\tilde{\theta}([B, B^*, l_0, k_0]) = [-B, B^*, l_0, -k_0] \tag{4.4.21}$$

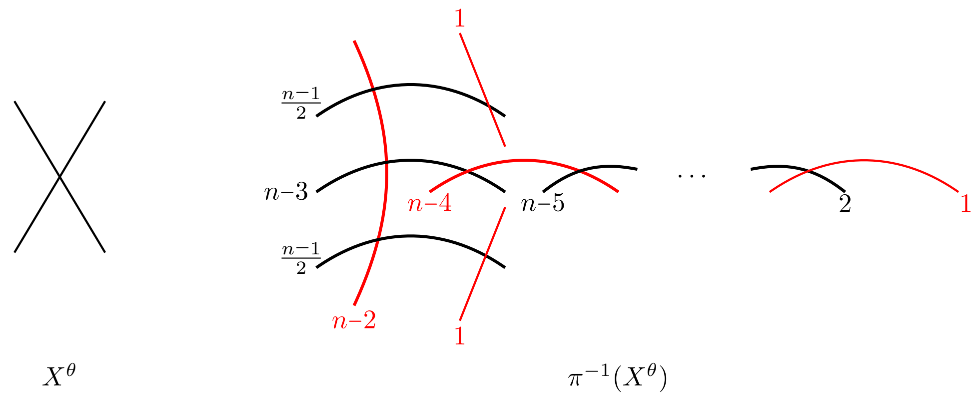
is anti-symplectic. Moreover, it is a lift of $\theta\colon X \to X$.

Proposition 4.4.9. *(1) The involution $\tilde{\theta}$ preserves all the components $C_i,\ 1 \leqslant i \leqslant n$. We have $\pi^{-1}(0)^{\tilde{\theta}} = \bigcup_{\substack{i \text{ odd} \\ 2\leqslant i\leqslant n-2}} C_i \cup \{p_1, p_2\}$ where $p_1 \in C_{n-1} \setminus C_{n-2}$ and $p_2 \in C_n \setminus C_{n-2}$.*

(2) With a suitable choice of labeling, we have $\tilde{L}_1 \cap \pi^{-1}(0) = \tilde{L}_1 \cap C_{n-1} = \{p_1\}$ and $\tilde{L}_2 \cap \pi^{-1}(0) = \tilde{L}_2 \cap C_n = \{p_2\}$.

(3) The scheme $\pi^{-1}(X^\theta)$ is not reduced. As a divisor, we have $\pi^{-1}(X^\theta) = \tilde{L}_1 + \tilde{L}_2 + \sum_{i=1}^n a_i C_i$, where $a_i = i,\ 1 \leqslant i \leqslant n-2$ and $a_{n-1} = a_n = \frac{n-1}{2}$.

The fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ are depicted in Picture 4.10. The number next to each component indicates its multiplicity. Thickened lines indicate that the multiplicities are larger than 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.



Picture 4.10: $D_n,\ n$ odd, Type I

4.4.4 Type $D_n,\ n$ odd, case II

In this section, we consider the anti-Poisson involution

$$\theta(x) = x,\ \theta(y) = y,\ \theta(z) = x^{\frac{n-1}{2}} - z.$$

The fixed point locus $X^\theta = \operatorname{Spec}\mathbb{C}[X]/(z) = \operatorname{Spec}\mathbb{C}[x,y]/(x(4y^2 + x^{\frac{n-2}{2}}))$ has two irreducible components $L_j,\ j = 1, 2$. One of them is $\mathbb{A}^1$ and the other is a cusp. Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}_1 \cup \tilde{L}_2$. This section is very similar to Section 4.4.2 with some minor changes due to the parity. We state the results, but omit the proofs. Recall the automorphism τ defined on the extended Dynkin diagram of type $\widetilde{D}_n$ in eq. (4.4.18).

Proposition 4.4.10. *The involution $\tilde{\theta}\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w)$ defined by*

$$\tilde{\theta}([B_a, B_a^*, l_0, k_0]) = [-B_{\tau(a)}, B_{\tau(a)}^*, l_0, -k_0] \tag{4.4.22}$$

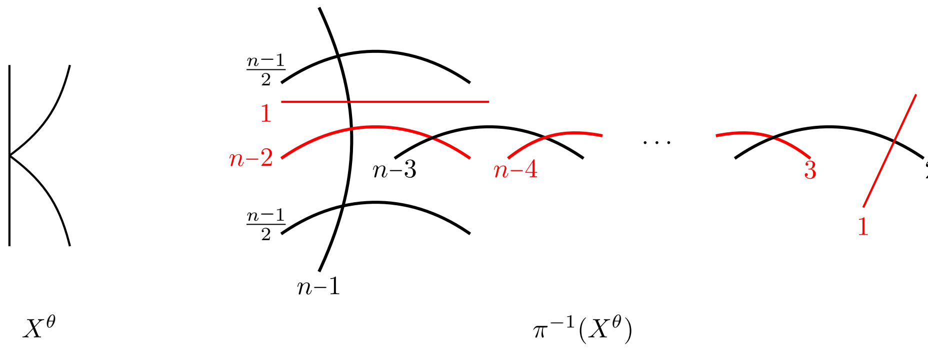
is anti-symplectic. Moreover, it is a lift of $\theta\colon X \to X$.

Proposition 4.4.11. *(1) The involution $\tilde{\theta}$ preserves the components $C_i,\ 1 \leqslant i \leqslant n-2$, and swaps the two components C_{n-1}, C_n. We have $\pi^{-1}(0)^{\tilde{\theta}} = \bigcup_{\substack{i \text{ even} \\ 2 \leqslant i \leqslant n-3}} C_i \cup \{p_1, p_2\}$ where $p_1 \in C_1 \setminus C_2$ and $p_2 \in C_{n-2} \setminus (C_{n-3} \cup C_{n-1} \cup C_n)$.*

(2) With a suitable choice of labeling, $\tilde{L}_1 \cap \pi^{-1}(0) = \tilde{L}_1 \cap C_1 = \{p_1\},\ \tilde{L}_2 \cap \pi^{-1}(0) = \tilde{L}_2 \cap C_{n-2} = \{p_2\}$.

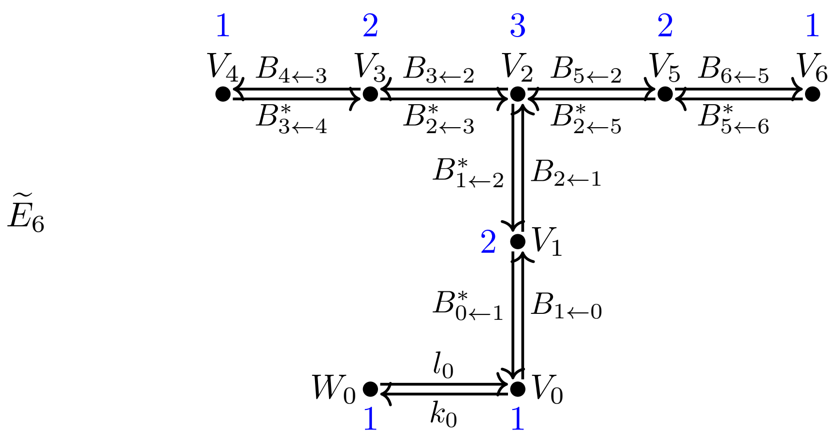
(3) The scheme $\pi^{-1}(X^\theta)$ is not reduced. As a divisor, we have $\pi^{-1}(X^\theta) = \tilde{L}_1 + \tilde{L}_2 + \sum_{i=1}^n a_i C_i$, where $a_i = i+1,\ 1 \leqslant i \leqslant n-2$ and $a_{n-1} = a_n = \frac{n-1}{2}$.

The fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ are depicted in Picture 4.11. The number next to each component indicates its multiplicity. Thickened lines indicate that the multiplicities are larger than 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.



Picture 4.11: $D_n,\ n$ odd, Type II

4.5 Preimages of fixed point loci for type E_6 Kleinian singularity



Picture 4.12: Extended Dynkin quiver $\widetilde{E}_6$

We realize the Kleinian singularity of type E_6 : $\operatorname{Spec}\mathbb{C}[X] = \mathbb{C}[x, y, z]/(z^2 + zx^2 + y^3)$ as a Nakajima quiver variety explicitly. Recall that $\delta = (\delta_i)_{0\leqslant i\leqslant 6} = (1, 2, 3, 2, 1, 2, 1)$. We have $\dim W_0 = 1,\ \dim V_i = \delta_i$. The number (colored in blue) next to each vector space indicates its dimension. The representation space is

$$M(\delta, w) = T^*\big(\bigoplus_{a\in\Omega} \operatorname{Hom}(V_{t(a)}, V_{h(a)}) \oplus \operatorname{Hom}(W_0, V_0)\big).$$

The group $\mathrm{GL}(\delta) = \prod_{i=0}^{n} \mathrm{GL}(V_i)$ acts on $M(\delta, w)$ naturally with the moment map

$$\mu = (\mu_i)_{0 \leqslant i \leqslant n} \colon M(\delta, w) \to \mathfrak{gl}(\delta),$$

where

$$\begin{aligned}
\mu_0 &= -B^*_{0\leftarrow 1} B_{1\leftarrow 0} + l_0 k_0, \\
\mu_1 &= B_{1\leftarrow 0} B^*_{0\leftarrow 1} - B^*_{1\leftarrow 2} B_{2\leftarrow 1}, \\
\mu_2 &= B_{2\leftarrow 1} B^*_{1\leftarrow 2} - B^*_{2\leftarrow 3} B_{3\leftarrow 2} - B^*_{2\leftarrow 5} B_{5\leftarrow 2}, \\
\mu_3 &= B_{3\leftarrow 2} B^*_{2\leftarrow 3} - B^*_{3\leftarrow 4} B_{4\leftarrow 3}, \\
\mu_4 &= B_{4\leftarrow 3} B^*_{3\leftarrow 4}, \\
\mu_5 &= B_{5\leftarrow 2} B^*_{2\leftarrow 5} - B^*_{5\leftarrow 6} B_{6\leftarrow 5}, \\
\mu_6 &= B_{6\leftarrow 5} B^*_{5\leftarrow 6}.
\end{aligned} \tag{4.5.1}$$

Recall from eq. (4.1.11) that setting $\mu = 0$ implies that $B^*_{0\leftarrow 1} B_{1\leftarrow 0} = l_0 k_0 = 0$. Recall from Proposition 4.2.6, we have

$$C_i = \big\{ [B, B^*, l_0, 0] \in \mathcal{M}_\chi(\delta, w) \big| \operatorname{rank}(B_a)_{\substack{a\in\bar{\Omega} \\ t(a)=i}} < \dim V_i \big\},\ 1 \leqslant i \leqslant n.$$

According to Corollary 4.1.8 and Proposition 2.3.1, the algebra of invariant functions $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ is generated by trace functions of loops that start and end at the vertex 0 of degrees $6, 8, 12$. The explicit generators are given in Proposition 4.5.1.

Proposition 4.5.1. *We have* $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)} \simeq \mathbb{C}[x, y, z]/(z^2 + zx^2 + y^3)$*, where*

$$\begin{aligned}
x &:= B^*_{0\leftarrow 1} B^*_{1\leftarrow 2} B^*_{2\leftarrow 3} B_{3\leftarrow 2} B_{2\leftarrow 1} B_{1\leftarrow 0}, \\
y &:= B^*_{0\leftarrow 1} B^*_{1\leftarrow 2} B^*_{2\leftarrow 3} B^*_{3\leftarrow 4} B_{4\leftarrow 3} B_{3\leftarrow 2} B_{2\leftarrow 1} B_{1\leftarrow 0}, \\
z &:= B^*_{0\leftarrow 1} B^*_{1\leftarrow 2} B^*_{2\leftarrow 5} B^*_{5\leftarrow 6} B_{6\leftarrow 5} B_{5\leftarrow 2} B^*_{2\leftarrow 3} B^*_{3\leftarrow 4} B_{4\leftarrow 3} B_{3\leftarrow 2} B_{2\leftarrow 1} B_{1\leftarrow 0}.
\end{aligned} \tag{4.5.2}$$

Before proving Proposition 4.5.1, we need a lemma.

Lemma 4.5.2. *(1)* $B^*_{1\leftarrow 2} B_{2\leftarrow 1} B_{1\leftarrow 0} = B^*_{0\leftarrow 1} B^*_{1\leftarrow 2} B_{2\leftarrow 1} = 0$.

(2) $(B^*_{2\leftarrow 3} B_{3\leftarrow 2})^3 = (B^*_{2\leftarrow 5} B_{5\leftarrow 2})^3 = 0$

(3) $B_{4\leftarrow 3}B_{3\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}B_{6\leftarrow 5}B_{5\leftarrow 2}B^*_{2\leftarrow 3}B^*_{3\leftarrow 4} = B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}B_{6\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0} = y$.

(4) $B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}B^*_{2\leftarrow 5}B_{5\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}B^*_{2\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0} = z$.

Proof. (1) We have $\underline{B^*_{1\leftarrow 2}B_{2\leftarrow 1}}B_{1\leftarrow 0} \overset{\mu_1}{=} B_{1\leftarrow 0}\underline{B^*_{0\leftarrow 1}B_{1\leftarrow 0}} \overset{\mu_0}{=} 0$. Similarly, $B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B_{2\leftarrow 1} = 0$.

(2) We have $(B^*_{2\leftarrow 3}B_{3\leftarrow 2})^3 \overset{\mu_3}{=} B^*_{2\leftarrow 3}B^*_{3\leftarrow 4}\underline{B_{4\leftarrow 3}B^*_{3\leftarrow 4}}B_{4\leftarrow 3}B_{3\leftarrow 2} \overset{\mu_4}{=} 0$. Similarly, $(B^*_{2\leftarrow 5}B_{5\leftarrow 2})^3 = 0$.

(3) We have

$$\begin{aligned}
&B_{6\leftarrow 5}B_{5\leftarrow 2}B^*_{2\leftarrow 3}B^*_{3\leftarrow 4}B_{4\leftarrow 3}B_{3\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6} \xlongequal{\mu_3,\ \mu_5} \operatorname{tr}((B^*_{2\leftarrow 3}B_{3\leftarrow 2})^2(B^*_{2\leftarrow 5}B_{5\leftarrow 2})^2)\\
&\quad \xlongequal{(2),\ \mu_2} \operatorname{tr}((B^*_{2\leftarrow 3}B_{3\leftarrow 2})^2(B_{2\leftarrow 1}B^*_{1\leftarrow 2})^2) \xlongequal{\mu_1} B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}(B^*_{2\leftarrow 3}B_{3\leftarrow 2})^2B^*_{2\leftarrow 1}B_{1\leftarrow 0} = y.
\end{aligned}$$

Similarly $B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}B_{6\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0} = y$.

(4) We can compute that

$$\begin{aligned}
&B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}B^*_{2\leftarrow 5}B_{5\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}B^*_{2\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}\\
&\overset{\mu_2}{=} B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}(B_{2\leftarrow 1}B^*_{1\leftarrow 2} - B^*_{2\leftarrow 5}B_{5\leftarrow 2})B^*_{2\leftarrow 5}B_{5\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}(B_{2\leftarrow 1}B^*_{1\leftarrow 2} - B^*_{2\leftarrow 3}B_{3\leftarrow 2})B_{2\leftarrow 1}B_{1\leftarrow 0}\\
&\overset{(1)}{=} B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}(B^*_{2\leftarrow 5}B_{5\leftarrow 2})^2(B^*_{2\leftarrow 3}B_{3\leftarrow 2})^2B_{2\leftarrow 1}B_{1\leftarrow 0} \xlongequal{\mu_3,\ \mu_5} z.
\end{aligned}$$

□

Proof of Proposition 4.5.1. We first show that x, y, z generate $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$. It suffices to show that the degree d-th component $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_d$ with $d = 6, 8, 12$, is contained in the subalgebra generated by x, y, z. We only need to consider trace functions of loops such that the vertex 0 only appears once; otherwise, the function can be written as the product of two trace functions of lower degrees.

The loops of degree 6 that start and end at the vertex 0 are $0\to 1\to 2\to 3\to 2\to 1\to 0,\ 0\to 1\to 2\to 5\to 2\to 1\to 0,\ 0\to 1\to 2\to 1\to 2\to 1\to 0$. The corresponding trace functions are

$$\begin{aligned}
x &\overset{\text{def}}{=} B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0},\\
x' &:= B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0},\\
x + x' &\overset{\mu_2}{=} B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B_{2\leftarrow 1}\underline{B^*_{1\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}} \xlongequal{(1)} 0,
\end{aligned} \tag{4.5.3}$$

which not only tells us that there is a relation $x + x' = 0$, but also that $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_6 = \mathbb{C}x$.

Next, we consider the loops that start and end at the vertex 0 of degree 8. If the loop only involves vertices $0,1,2$, then one can show that the corresponding trace function is 0 by using (1) of Lemma 4.5.2. If the loop contains (exactly) one of the vertices $4,6$, then it will be of the form $0\to 1\to 2\to 3\to 4\to 3\to 2\to 1\to 0,\ 0\to 1\to 2\to 5\to 6\to 5\to 2\to 1\to 0$. The corresponding trace functions are

$$\begin{aligned} &B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 3}B^*_{3\leftarrow 4}B_{4\leftarrow 3}B_{3\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}\overset{\text{def}}{=}y,\\ &B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}B_{6\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}\overset{(3)}{=}y.\end{aligned}\tag{4.5.4}$$

Meanwhile, if the loop contains neither 4 nor 6, then it will be of the form $0\to 1\to 2\to 3\to 2\to 3\to 2\to 1\to 0,\ 0\to 1\to 2\to 3\to 2\to 5\to 2\to 1\to 0,\ 0\to 1\to 2\to 5\to 2\to 3\to 2\to 1\to 0,\ 0\to 1\to 2\to 5\to 2\to 5\to 2\to 1\to 0$. Similar to the proof of (3) in Lemma 4.5.2, one can show that the corresponding trace functions are either y or $-y$. It follows that $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}=\mathbb{C}y$.

Lastly, we consider the following two loops of degree 12 that starts and ends at 0.

$$\begin{aligned} z&\overset{\text{def}}{=}B^*_{0\leftarrow 1}B_{1\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}B_{6\leftarrow 5}B_{5\leftarrow 2}B^*_{2\leftarrow 3}B^*_{3\leftarrow 4}B_{4\leftarrow 3}B_{3\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0},\\ z'&:=B^*_{0\leftarrow 1}B_{1\leftarrow 2}B^*_{2\leftarrow 3}B^*_{3\leftarrow 4}B_{4\leftarrow 3}B_{3\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}B_{6\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}.\end{aligned}\tag{4.5.5}$$

We have $z'\xlongequal{\mu_3,\,\mu_5}B^*_{0\leftarrow 1}B_{1\leftarrow 2}(B^*_{2\leftarrow 3}B_{3\leftarrow 2})^2(B^*_{2\leftarrow 5}B_{5\leftarrow 2})^2B_{2\leftarrow 1}B_{1\leftarrow 0}$. Using (4) in Lemma 4.5.2, we get

$$\begin{aligned} z+z'&=B^*_{0\leftarrow 1}B_{1\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}(B^*_{2\leftarrow 5}B_{5\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}+B^*_{2\leftarrow 3}B_{3\leftarrow 2}B^*_{2\leftarrow 5}B_{5\leftarrow 2})B^*_{2\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}\\ &\overset{(2)}{=}B^*_{0\leftarrow 1}B_{1\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}(B^*_{2\leftarrow 3}B_{3\leftarrow 2}+B^*_{2\leftarrow 5}B_{5\leftarrow 2})^2B^*_{2\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}\\ &\overset{\mu_2}{=}B^*_{0\leftarrow 1}B_{1\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}(B_{2\leftarrow 1}B^*_{1\leftarrow 2})^2B^*_{2\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}\\ &\overset{\mu_1}{=}B^*_{0\leftarrow 1}B_{1\leftarrow 2}B^*_{2\leftarrow 3}B_{3\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}=xx'\xlongequal{eq.\ (4.5.3)}-x^2.\end{aligned}\tag{4.5.6}$$

Similarly to the case of degree 8, we can show that $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_{12}=\mathbb{C}z+\mathbb{C}x^2$. Therefore, $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ is generated by the invariant functions x,y,z defined in eq. (4.5.2). Finally, we have that

$$\begin{aligned} zz'&=B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}B_{6\leftarrow 5}B_{5\leftarrow 2}B^*_{2\leftarrow 3}B^*_{3\leftarrow 4}\underline{B_{4\leftarrow 3}B_{3\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}}\\ &\quad\cdot\underline{B^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 3}B^*_{3\leftarrow 4}}B_{4\leftarrow 3}B_{3\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}B_{6\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}\\ &\overset{\text{def}}{=}yB^*_{0\leftarrow 1}B^*_{1\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}\underline{B_{6\leftarrow 5}B_{5\leftarrow 2}B^*_{2\leftarrow 3}B^*_{3\leftarrow 4}B_{4\leftarrow 3}B_{3\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}}B_{6\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}\\ &\overset{(3)}{=}y^2B^*_{0\leftarrow 1}B_{1\leftarrow 2}B^*_{2\leftarrow 5}B^*_{5\leftarrow 6}B_{6\leftarrow 5}B_{5\leftarrow 2}B_{2\leftarrow 1}B_{1\leftarrow 0}\overset{(3)}{=}y^3,\end{aligned}\tag{4.5.7}$$

Combining eqs. (4.5.6) and (4.5.7), we obtain that $z^2 + zx^2 + y^3 = 0$. It follows that there is a surjective homomorphism

$$\mathbb{C}[x, y, z]/(z^2 + zx^2 + y^3) \twoheadrightarrow \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}, \tag{4.5.8}$$

which turns out to be an isomorphism for the same reason as for type A_n in eq. (4.3.3). □

4.5.1 Type E_6, case I

In this section, we consider the anti-Poisson involution

$$\theta(x) = -x,\ \theta(y) = y,\ \theta(z) = z.$$

The fixed point locus $X^\theta = \operatorname{Spec} \mathbb{C}[X]/(x) = \operatorname{Spec} \mathbb{C}[y, z]/(z^2 + y^3)$ is a cusp (a single irreducible component L). Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}$.

Proposition 4.5.3. *The involution $\tilde{\theta}\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w)$ defined by*

$$\tilde{\theta}([B, B^*, l_0, k_0]) = [-B, B^*, l_0, -k_0] \tag{4.5.9}$$

is anti-symplectic. Moreover, it is a lift of $\theta\colon X \to X$.

Proof. We consider the following linear anti-symplectic involution on the representation space.

$$\Theta\colon M(\delta, w) \to M(\delta, w),\ \ (B, B^*, l_0, k_0) \mapsto (-B, B^*, l_0, -k_0).$$

It satisfies the conditions listed in Proposition 4.2.1.

1. $g^\Theta = g$.

2. $\Theta(x) = (-1)^3 x = -x,\ \Theta(y) = (-1)^4 y = y,\ \Theta(z) = (-1)^6 z = z.$

The involution $\tilde{\theta}$ defined in eq. (4.5.9) is the descent of Θ to $\mathcal{M}_\chi(\delta, w)$. The rest follows from Proposition 4.2.1. □

Proposition 4.5.4. *(1) The involution $\tilde{\theta}$ preserves all the components C_i, $1 \leqslant i \leqslant 6$. We have $\pi^{-1}(0)^{\tilde{\theta}} = C_2 \cup C_4 \cup C_6 \cup \{p\}$ where $p \in C_1 \setminus C_2$.*

(2) We have $\tilde{L} \cap \pi^{-1}(0) = \tilde{L} \cap C_1 = \{p\}$.

(3) *The scheme $\pi^{-1}(X^\theta)$ is not reduced. As a divisor, we have $\pi^{-1}(X^\theta) = \tilde{L} + \sum_{i=1}^{6} \delta_i C_i = \tilde{L} + \pi^{-1}(0)$. Recall that $(\delta_i)_{1 \leqslant i \leqslant 6} = (2, 3, 2, 1, 2, 1)$.*

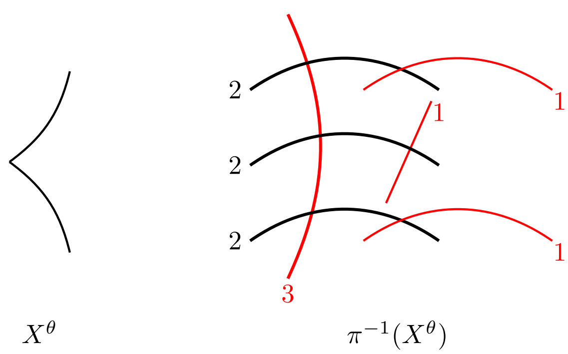
Proof. (1) It is easy to see that $\tilde{\theta}(C_i) = C_i$ thanks to Proposition 4.2.6. The component C_2 intersects with three other components C_1, C_3, C_5, so there are at least three $\tilde{\theta}$-fixed points on C_2. It follows from (2) of Lemma 4.2.3 that $\tilde{\theta}$ acts trivially on C_2. By (3) of Lemma 4.2.3, $\tilde{\theta}$ has exactly two fixed points on each of C_1, C_3, C_5, and $\tilde{\theta}$ acts trivially on the components C_4, C_6. Therefore, there is a unique isolated point in $\pi^{-1}(0)^{\tilde{\theta}}$, which lies in $C_1 \setminus C_2$.

(2) This follows from Proposition 4.2.5.

(3) Recall the definition of b_i from eq. (4.2.9). We have $(b_1, b_2, \cdots, b_6) = (1, 0, \cdots, 0)$ from (2). Then the multiplicities in $\pi^{-1}(X^\theta)$ can be determined by Proposition 4.2.8 as $X^\theta = \mathrm{div}(x)$ is a principal divisor.

□

The fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ are depicted in Picture 4.13. The number next to each component indicates its multiplicity. Thickened lines indicate that the multiplicities are larger than 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.



Picture 4.13: E_6, Type I

4.5.2 Type E_6, case II

In this section, we consider the anti-Poisson involution

$$\theta(x) = x,\ \theta(y) = y,\ \theta(z) = -z - x^2.$$

The fixed point locus $X^\theta = \operatorname{Spec}\mathbb{C}[X]/(2z+x^2) = \operatorname{Spec}\mathbb{C}[x,y]/(x^4-4y^3)$ is a cusp (a single irreducible component L). Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}$. Consider the following automorphism defined on the extended Dynkin diagram of type $\widetilde{E}_6$.

$$\tau(i) = i,\ i = 0,1,2;\ \tau(3) = 4,\ \tau(4) = 3,\ \tau(5) = 6,\ \tau(6) = 5. \tag{4.5.10}$$

It induces an automorphism of the extended Dynkin quiver $\widetilde{E}_6$ in Picture 4.12 by sending an arrow $a : i \to j$ to the arrow $\tau(a) : \tau(i) \to \tau(j)$. Note that τ coincides with the nontrivial automorphism of the McKay graph of type $\tilde{E}_6$ induced by an element $g \in N_{\mathrm{GL}_2(\mathbb{C})}(\Gamma)^-$ (c.f. Remark 3.3.12).

Proposition 4.5.5. *The involution $\tilde{\theta}\colon \mathcal{M}_\chi(\delta,w) \to \mathcal{M}_\chi(\delta,w)$ defined by*

$$\tilde{\theta}([B_a, B_a^*, l_0, k_0]) = [-B_{\tau(a)}, B^*_{\tau(a)}, l_0, -k_0] \tag{4.5.11}$$

is anti-symplectic. Moreover, it is a lift of $\theta\colon X \to X$.

Proof. We consider the following linear anti-symplectic involution on the representation space.

$$\Theta\colon M(\delta,w) \to M(\delta,w),\ \ (B_a, B_a^*, l_0, k_0) \mapsto (-B_{\tau(a)}, B^*_{\tau(a)}, l_0, -k_0).$$

It satisfies the conditions listed in Proposition 4.2.1.

1. $(g_i)^\Theta_{i\in I} = (g_{\tau(i)})_{i\in I}$.
2. $\Theta(x) = (-1)^3 x' \overset{\text{eq. (4.5.3)}}{=\!=\!=} x,\ \Theta(y) \overset{\text{Lemma 4.5.2, (3)}}{=\!=\!=} (-1)^4 y = y,\ \Theta(z) = (-1)^6 z' \overset{\text{eq. (4.5.6)}}{=\!=\!=} -z - x^2$.

The involution $\tilde{\theta}$ defined in eq. (4.5.11) is the descent of Θ to $\mathcal{M}_\chi(\delta,w)$. The rest follows from Proposition 4.2.1. $\square$

Proposition 4.5.6. *(1) The involution $\tilde{\theta}$ preserves the components C_1, C_2. It swaps C_3 with C_5, and C_4 with C_6. We have $\pi^{-1}(0)^{\tilde{\theta}} = C_1 \cup \{p\}$ where $p \in C_2 \setminus (C_1 \cup C_3 \cup C_5)$.*

(2) We have $\tilde{L} \cap \pi^{-1}(0) = \tilde{L} \cap C_2 = \{p\}$.

(3) The scheme $\pi^{-1}(X^\theta)$ is not reduced. As a divisor, we have $\pi^{-1}(X^\theta) = \tilde{L} + \sum_{i=1}^n a_i C_i$, where $(a_i)_{1\leqslant i\leqslant 6} = (3,6,4,2,4,2)$.

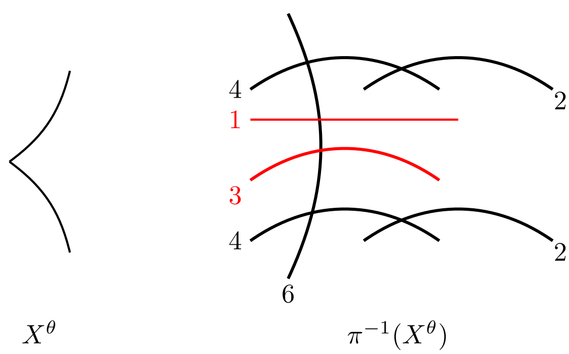
Proof. (1) It is easy to see that $\tilde{\theta}(C_i) = C_{\tau(i)}$ thanks to Proposition 4.2.6. The intersection points $C_2 \cap C_3,\ C_2 \cap C_5$ are permuted by $\tilde{\theta}$. It follows from (2) of Lemma 4.2.3 that there are exactly two $\tilde{\theta}$-fixed points on C_2. The intersection point $C_1 \cap C_2$ is one of the $\tilde{\theta}$-fixed points on C_2, and the other one is an isolated point in $\pi^{-1}(0)^{\tilde{\theta}}$.

(2) This follows from Proposition 4.2.5.

(3) Recall the definition of b_i in eq. (4.2.9). We have $(b_1, b_2, b_3 \cdots, b_6) = (0, 1, 0 \cdots, 0)$ by (2). Moreover, $X^\theta = \operatorname{div}(2z + x^2)$ is a principal divisor, then a_i's can be determined by Proposition 4.2.8.

□

The fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ are depicted in Picture 4.14. The number next to each component indicates its multiplicity. Thickened lines indicate that the multiplicities are larger than 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.



Picture 4.14: E_6, Type II

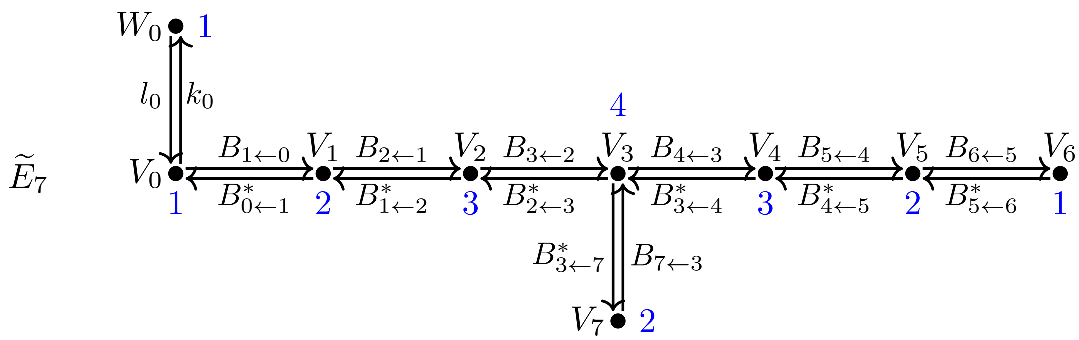
4.6 Preimages of fixed point loci for type E_7 Kleinian singularity

We realize the Kleinian singularity of type E_7 : $\operatorname{Spec}\mathbb{C}[X] = \mathbb{C}[x, y, z]/(x^3y + y^3 + z^2)$ as a Nakajima quiver variety explicitly. Recall that $\delta = (\delta_i)_{0\leqslant i\leqslant 7} = (1, 2, 3, 4, 3, 2, 1, 2)$. We have $\dim W_0 = 1,\ \dim V_i = \delta_i$. The number (colored in blue) next to each vector space indicates its dimension. The representation space is

$$M(\delta, w) = T^*\big(\bigoplus_{a\in\Omega} \operatorname{Hom}(V_{t(a)}, V_{h(a)}) \oplus \operatorname{Hom}(W_0, V_0)\big).$$

The group $\mathrm{GL}(\delta) = \prod_{i=0}^n \mathrm{GL}(V_i)$ acts on $M(\delta, w)$ naturally with the moment map

$$\mu = (\mu_i)_{0\leqslant i\leqslant n} \colon M(\delta, w) \to \mathfrak{gl}(\delta).$$



Picture 4.15: Extended Dynkin quiver $\widetilde{E}_7$

According to Corollary 4.1.8 and Proposition 2.3.1, the algebra of invariant functions $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ is generated by trace functions x, y, z with $\deg x = 8,\ \deg y = 12,\ \deg z = 18.$ We would like to point out that the lift of the unique anti-Poisson involution in type E_7 can be constructed without knowing the explicit forms of x, y, z, thanks to Lemma 4.6.1 and the observation on the gradings of x, y, z. The lift will be constructed in Proposition 4.6.2.

Though the explicit forms of x, y, z as trace functions is not needed for Objective (2) of this thesis, it is still nice to have some pictorial description of the functions on Kleinian singularity as trace functions of loops on the extended Dynkin quiver. This will be discussed in Proposition 4.6.4.

Lemma 4.6.1. *Let $Q = (I, \Omega)$ be a quiver without loops (for example, extended Dynkin quiver of types D or E). Let q be a loop on the doubled quiver $\bar{Q} = (I, \bar{\Omega})$. Then q has $\frac{\deg q}{2}$ number of arrows from Ω (resp. Ω^*).*

Proof. Omitted. □

4.6.1 Lift of anti-Poisson involution

Recall from Proposition 3.3.14 that the Kleinian singularity of type E_7 has a unique conjugacy class of anti-Poisson involution, which is given by

$$\theta(x) = x,\ \theta(y) = y,\ \theta(z) = -z.$$

The fixed point locus $X^\theta = \operatorname{Spec}\mathbb{C}[x, y]/(y(x^3 + y^2))$ has two irreducible components $L_j,\ j = 1, 2$ with an $\mathbb{A}^1$ and a cusp. Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}_1 \cup \tilde{L}_2$.

Proposition 4.6.2. *The involution $\tilde{\theta}\colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w)$ defined by*

$$\tilde{\theta}([B, B^*, l_0, k_0]) = [-B, B^*, l_0, -k_0] \tag{4.6.1}$$

is anti-symplectic. Moreover, it is a lift of $\theta\colon X \to X$.

Proof. We consider the following linear anti-symplectic involution on the representation space.

$$\Theta\colon M(\delta, w) \to M(\delta, w),\ \ (B, B^*, l_0, k_0) \mapsto (-B, B^*, l_0, -k_0).$$

It satisfies the conditions listed in Proposition 4.2.1.

1. $g^\Theta = g$.
2. Recall that $\deg x = 8,\ \deg y = 12,\ \deg z = 18$. By Lemma 4.6.1, we have $\Theta(x) = (-1)^4 x = x,\ \Theta(y) = (-1)^6 y = y,\ \Theta(z) = (-1)^9 z = -z$.

The involution $\tilde{\theta}$ defined in eq. (4.6.1), is the descent of Θ to $\mathcal{M}_\chi(\delta, w)$. The rest follows from Proposition 4.2.1. □

Proposition 4.6.3. *(1) The involution $\tilde{\theta}$ preserves all the components $C_i,\ 1 \leqslant i \leqslant 7$. We have $\pi^{-1}(0)^{\tilde{\theta}} = C_1 \cup C_3 \cup C_5 \cup \{p_1, p_2\}$ where $p_1 \in C_6 \setminus C_5,\ p_2 \in C_7 \setminus C_3$.*

(2) With a suitable choice of labeling, $\tilde{L}_1 \cap \pi^{-1}(0) = \tilde{L}_1 \cap C_6 = \{p_1\},\ \tilde{L}_2 \cap \pi^{-1}(0) = \tilde{L}_2 \cap C_7 = \{p_2\}$.

(3) The scheme $\pi^{-1}(X^\theta)$ is not reduced. As a divisor, we have $\pi^{-1}(X^\theta) = \tilde{L}_1 + \tilde{L}_2 + \sum_{i=1}^{7} a_i C_i$, where $(a_i)_{1 \leqslant i \leqslant 7} = (3, 6, 9, 7, 5, 3, 5)$.

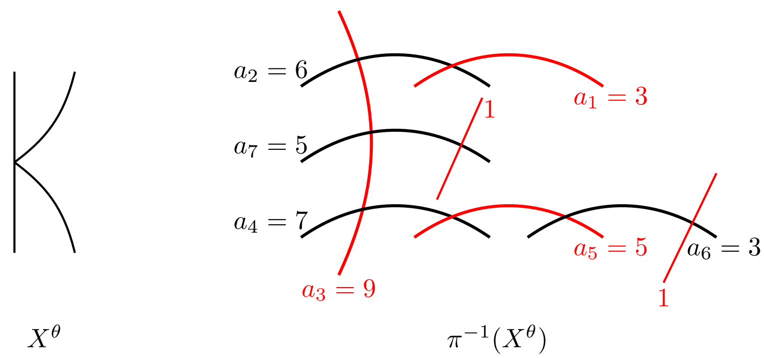
Proof. (1) It is easy to see that $\tilde{\theta}(C_i) = C_i$ thanks to Proposition 4.2.6. The component C_3 intersects with three other components $C_2,\ C_4,\ C_7$, so there are at least three $\tilde{\theta}$-fixed points on C_3. It follows from (2) of Lemma 4.2.3 that $\tilde{\theta}$ acts trivially on C_3. By (3) of Lemma 4.2.3, $\tilde{\theta}$ has exactly two fixed points on each of the components C_2, C_4, C_6, C_7, and $\tilde{\theta}$ acts trivially on the components C_1, C_5. Therefore, there are two isolated points in $\pi^{-1}(0)^{\tilde{\theta}}$. One of them lies in $C_6 \setminus C_5$, and the other lies in $C_7 \setminus C_3$.

(2) This follows from Proposition 4.2.5.

(3) Recall the definition of b_i from eq. (4.2.9). We have $(b_1, \cdots, b_5, b_6, b_7) = (0, \cdots, 0, 1, 1)$ from (2). Moreover, $X^\theta = \operatorname{div}(z)$ is a principal divisor, then a_i's can be determined by Proposition 4.2.8.

□

The fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ are depicted in Picture 4.16. The number next to each component indicates its multiplicity. Thickened lines indicate that the multiplicities are larger than 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.



Picture 4.16: E_7, fixed point locus and its preimage

4.6.2 Explicit generators x, y, z for type E_7

In this section, we discuss how to find the explicit forms of the generators x, y, z of $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ as trace functions.

Proposition 4.6.4. *We have $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)} \simeq \mathbb{C}[x,y,z]/(x^3y + y^3 + z^2)$, where*

$$
\begin{aligned}
x &:= B^*_{0\leftarrow 1} B^*_{1\leftarrow 2} B^*_{2\leftarrow 3} B^*_{3\leftarrow 4} B_{4\leftarrow 3} B_{3\leftarrow 2} B_{2\leftarrow 1} B_{1\leftarrow 0}, \\
y &:= B^*_{0\leftarrow 1} B^*_{1\leftarrow 2} B^*_{2\leftarrow 3} (B^*_{3\leftarrow 4} B_{4\leftarrow 3})^3 B_{3\leftarrow 2} B_{2\leftarrow 1} B_{1\leftarrow 0}, \\
z &:= B^*_{0\leftarrow 1} B^*_{1\leftarrow 2} B^*_{2\leftarrow 3} (B^*_{3\leftarrow 4} B_{4\leftarrow 3})^2 B^*_{3\leftarrow 7} B_{7\leftarrow 3} (B^*_{3\leftarrow 4} B_{4\leftarrow 3})^3 B_{3\leftarrow 2} B_{2\leftarrow 1} B_{1\leftarrow 0}.
\end{aligned}
\tag{4.6.2}
$$

We will give a proof that is different from types D_n and E_6 (c.f. Propositions 4.4.1 and 4.5.1). The reason is that there are too few 1-dimensional vector spaces among the vector spaces $V_i,\ 0 \leqslant i \leqslant 7$ in type E_7, making it very hard to show that x, y, z generate $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ and to find the relation among x, y, z by manipulating with the ADHM equation in eq. (4.1.12). Lemma 4.6.5 will be needed in the proof Proposition 4.6.4. Consider the following two points $(B^i, B^{*,i}, 1, 0) \in M(\delta, w),\ i = 1, 2$, given in Table 4.1. Let x_i, y_i, z_i denote values of the functions x, y, z at the points $(B^i, B^{*,i}, 1, 0),\ i = 1, 2$.

Lemma 4.6.5. *1.* $(B^i, B^{*,i}, 1, 0) \in \mu^{-1}(0),\ i = 1, 2.$

2. $x_1 = -8,\ y_1 = 16,\ z_1 = 64.$

3. $x_2 = 4,\ y_2 = -8,\ z_2 = -32.$

$(B^1, B^{*,1}, 1, 0)$	$(B^2, B^{*,2}, 1, 0)$
$B^1_{1\leftarrow 0} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},\ B^{*,1}_{0\leftarrow 1} = \begin{pmatrix} -8 & 0 \end{pmatrix}$	$B^2_{1\leftarrow 0} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},\ B^{*,2}_{0\leftarrow 1} = \begin{pmatrix} 4 & 0 \end{pmatrix}$
$B^1_{2\leftarrow 1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix},\ B^{*,1}_{1\leftarrow 2} = \begin{pmatrix} 0 & 1 & 0 \\ -8 & 0 & 0 \end{pmatrix}$	$B^2_{2\leftarrow 1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix},\ B^{*,2}_{1\leftarrow 2} = \begin{pmatrix} 0 & 1 & 0 \\ 4 & 0 & 0 \end{pmatrix}$
$B^1_{3\leftarrow 2} = \begin{pmatrix} 2 & 0 & 0 \\ -2 & 1 & 0 \\ 4 & 0 & 0 \\ -4 & 0 & 1 \end{pmatrix},\ B^{*,1}_{2\leftarrow 3} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ -4 & 0 & 0 & 0 \end{pmatrix}$	$B^2_{3\leftarrow 2} = \begin{pmatrix} -1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 0 & 0 \\ -4 & 0 & 1 \end{pmatrix},\ B^{*,2}_{2\leftarrow 3} = \begin{pmatrix} -2 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ -4 & 0 & 0 & 0 \end{pmatrix}$
$B^1_{4\leftarrow 3} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,1}_{3\leftarrow 4} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$	$B^2_{4\leftarrow 3} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,2}_{3\leftarrow 4} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$
$B^1_{5\leftarrow 4} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},\ B^{*,1}_{4\leftarrow 5} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$	$B^2_{5\leftarrow 4} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},\ B^{*,2}_{4\leftarrow 5} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$
$B^1_{6\leftarrow 5} = \begin{pmatrix} 0 & 1 \end{pmatrix},\ B^{*,1}_{5\leftarrow 6} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$	$B^2_{6\leftarrow 5} = \begin{pmatrix} 0 & 1 \end{pmatrix},\ B^{*,2}_{5\leftarrow 6} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
$B^1_{7\leftarrow 3} = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 4 & 0 & 0 & 1 \end{pmatrix},\ B^{*,1}_{3\leftarrow 7} = \begin{pmatrix} 1 & 0 \\ -2 & 0 \\ 4 & -1 \\ -4 & 0 \end{pmatrix}$	$B^2_{7\leftarrow 3} = \begin{pmatrix} -1 & 1 & 0 & 0 \\ -2 & 0 & 0 & 1 \end{pmatrix},\ B^{*,2}_{3\leftarrow 7} = \begin{pmatrix} -2 & 0 \\ -2 & 0 \\ -2 & -1 \\ -4 & 0 \end{pmatrix}$

Table 4.1: Two generic points in E_7

Proof. The proof is just multiplication of matrices, using the definitions of the moment maps in eq. (4.1.12) and the functions x, y, z in Proposition 4.6.4. $\square$

Proof of Proposition 4.6.4. Recall from Corollary 4.1.8 and Proposition 2.3.1 that the algebra of invariant functions $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ is generated by trace functions of degrees $8, 12, 18$. Note that none of the degrees can be written as a positive integral linear combination of the others. It follows that

$$\dim \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_8 = \dim \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_{12} = \dim \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_{18} = 1.$$

We know that x, y, z are all nonzero functions in $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ from Lemma 4.6.5, and that $\deg x = 8, \deg y = 12, \deg z = 18$ from their definitions in eq. (4.6.2). Therefore,

$$\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_8 = \mathbb{C}x,\ \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_{12} = \mathbb{C}y,\ \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}_{18} = \mathbb{C}z.$$

It follows that $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ is generated by x, y, z. Still by Corollary 4.1.8 and Proposition 2.3.1, there must be a unique (up to scaling) nontrivial relation $F(x, y, z) = 0$ of degree 36 among x, y, z. The only monomials of degree 36 are x^3y, y^3, z^2, thus we can write

$$F(x,y,z) = ax^3y + by^3 + cz^2 = 0,\ a,b,c \in \mathbb{C},\ \text{not all zero} \tag{4.6.3}$$

Plugging the points $(x_i, y_i, z_i),\ i = 1,2$ from Lemma 4.6.5 into eq. (4.6.3), we get

$$F(x_1,y_1,z_1) = 2^{12}(-2a+b+c) = 0,$$
$$F(x_2,y_2,z_2) = 2^{9}(-a-b+2c) = 0.$$

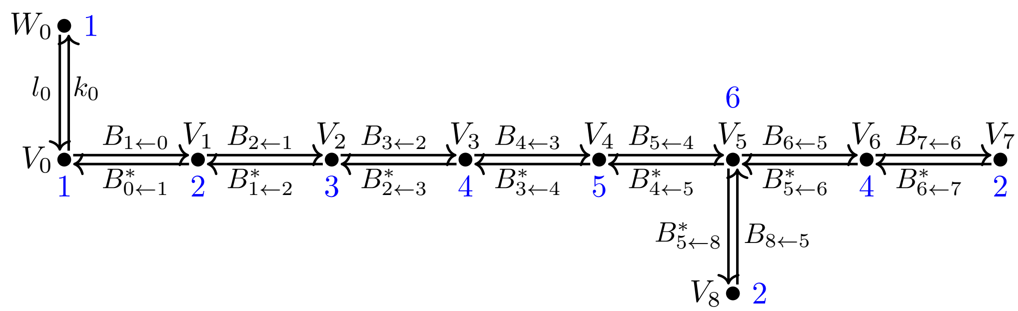
It follows that $a = b = c$. We may take $a = b = c = 1$. Then we get a surjective homomorphism

$$\mathbb{C}[x,y,z]/(x^3y+y^3+z^2) \twoheadrightarrow \mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}, \tag{4.6.4}$$

which turns out to be an isomorphism because both sides are integral and of dimension 2. $\square$

4.7 Preimages of fixed point loci for type E_8 Kleinian singularity

$\widetilde{E}_8$



Picture 4.17: Extended Dynkin quiver $\widetilde{E}_8$

We realize the Kleinian singularity of type $E_8 : \operatorname{Spec}\mathbb{C}[X] = \mathbb{C}[x,y,z]/(x^5+y^3+z^2)$ as a Nakajima quiver variety explicitly. Recall $\delta = (\delta_i)_{0\leqslant i\leqslant 8} = (1,2,3,4,5,6,4,2,3)$. We have $\dim W_0 = 1,\ \dim V_i = \delta_i$. The number (colored in blue) next to each vector space indicates its dimension. The representation space is

$$M(\delta, w) = T^*\big(\bigoplus_{a\in\Omega} \operatorname{Hom}(V_{t(a)}, V_{h(a)}) \oplus \operatorname{Hom}(W_0, V_0)\big).$$

The group $\mathrm{GL}(\delta) = \prod_{i=0}^{n} \mathrm{GL}(V_i)$ acts on $M(\delta, w)$ naturally with the moment map

$$\mu = (\mu_i)_{0 \leqslant i \leqslant n} \colon M(\delta, w) \to \mathfrak{gl}(\delta).$$

According to Corollary 4.1.8 and Proposition 2.3.1, the algebra of invariant functions $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ is generated by trace functions x, y, z with $\deg x = 12,\ \deg y = 20,\ \deg z = 30$. Similar to the case of type E_7, we can construct the lift of the unique anti-Poisson involution in type E_8 without knowing the explicit forms of x, y, z (c.f. Proposition 4.7.1). The explicit forms of the trace functions x, y, z will be discussed in Proposition 4.7.3.

4.7.1 Lift of anti-Poisson involution

Recall from Proposition 3.3.17 that the Kleinian singularity of type E_8 has a unique conjugacy class of anti-Poisson involution, which is given by

$$\theta(x) = x,\ \theta(y) = y,\ \theta(z) = -z.$$

The fixed point locus $X^\theta = \operatorname{Spec} \mathbb{C}[x, y]/(x^5 + y^3)$ is a cusp (a single irreducible component L). Following the notations in Section 4.2, we have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{L}$. The proofs of Propositions 4.7.1 and 4.7.2 repeat that of type E_7. We leave them to the reader.

Proposition 4.7.1. *The involution $\tilde{\theta} \colon \mathcal{M}_\chi(\delta, w) \to \mathcal{M}_\chi(\delta, w)$ defined by*

$$\tilde{\theta}([B, B^*, l_0, k_0]) = [-B, B^*, l_0, -k_0] \tag{4.7.1}$$

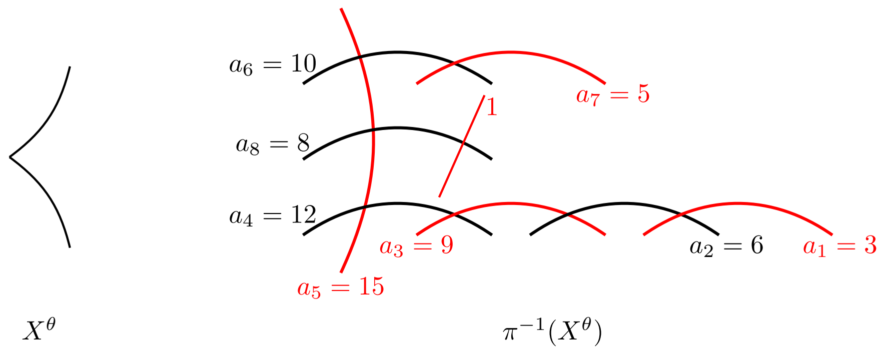
is anti-symplectic. Moreover, it is a lift of $\theta \colon X \to X$.

Proposition 4.7.2. *(1) The involution $\tilde{\theta}$ preserves all the components $C_i,\ 1 \leqslant i \leqslant 8$. We have $\pi^{-1}(0)^{\tilde{\theta}} = C_1 \cup C_3 \cup C_5 \cup C_7 \cup \{p\}$ where $p \in C_8 \setminus C_5$.*

(2) We have $\tilde{L} \cap \pi^{-1}(0) = \tilde{L} \cap C_8 = \{p\}$.

(3) The scheme $\pi^{-1}(X^\theta)$ is not reduced. As a divisor, we have $\pi^{-1}(X^\theta) = \tilde{L} + \sum_{i=1}^{8} a_i C_i$, where $(a_i)_{1 \leqslant i \leqslant 8} = (3, 6, 9, 12, 15, 10, 5, 8)$.

The fixed point locus X^θ and the preimage $\pi^{-1}(X^\theta)$ are depicted in Picture 4.18. The number next to each component indicates its multiplicity. Thickened lines indicate that the multiplicities are larger than 1. The $\tilde{\theta}$-fixed components of $\pi^{-1}(X^\theta)$ are colored in red.



Picture 4.18: E_8, fixed point locus and its preimage

4.7.2 Explicit generators x, y, z for type E_8

In this section, we discuss how to find the explicit forms of the generators x, y, z of $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)}$ as trace functions.

Proposition 4.7.3. *We have $\mathbb{C}[\mu^{-1}(0)]^{\mathrm{GL}(\delta)} \simeq \mathbb{C}[x, y, z]/(x^5 + y^3 + z^2)$, where*

$$\begin{aligned}
x :=& B^*_{0\leftarrow 1} B^*_{1\leftarrow 2} B^*_{2\leftarrow 3} B^*_{3\leftarrow 4} B^*_{4\leftarrow 5} B^*_{5\leftarrow 8} B_{8\leftarrow 5} B_{5\leftarrow 4} B_{4\leftarrow 3} B_{3\leftarrow 2} B_{2\leftarrow 1} B_{1\leftarrow 0},\\
y :=& B^*_{0\leftarrow 1} B^*_{1\leftarrow 2} B^*_{2\leftarrow 3} B^*_{3\leftarrow 4} B^*_{4\leftarrow 5} (B^*_{5\leftarrow 6} B_{6\leftarrow 5})^2 B^*_{5\leftarrow 8} B_{8\leftarrow 5} (B^*_{5\leftarrow 6} B_{6\leftarrow 5})^2 B_{5\leftarrow 4} B_{4\leftarrow 3} B_{3\leftarrow 2} B_{2\leftarrow 1} B_{1\leftarrow 0},\\
z :=& B^*_{0\leftarrow 1} B^*_{1\leftarrow 2} B^*_{2\leftarrow 3} B^*_{3\leftarrow 4} B^*_{4\leftarrow 5} (B^*_{5\leftarrow 6} B_{6\leftarrow 5})^2 (B_{5\leftarrow 4} B^*_{4\leftarrow 5})^3\\
&\cdot (B^*_{5\leftarrow 6} B_{6\leftarrow 5})^2 B^*_{5\leftarrow 8} B_{8\leftarrow 5} (B^*_{5\leftarrow 6} B_{6\leftarrow 5})^2 B_{5\leftarrow 4} B_{4\leftarrow 3} B_{3\leftarrow 2} B_{2\leftarrow 1} B_{1\leftarrow 0}.
\end{aligned} \tag{4.7.2}$$

The proof of Proposition 4.7.3 is identical to that of Proposition 4.6.4, with the help of Lemma 4.7.4. Consider the following two points $(B^i, B^{*,i}, 1, 0) \in M(\delta, w),\ i = 1, 2$, given in Table 4.2. Let x_i, y_i, z_i denote the values of the functions x, y, z at the points $(B^i, B^{*,i}, 1, 0),\ i = 1, 2$.

Lemma 4.7.4. *1.* $(B^i, B^{*,i}, 1, 0) \in \mu^{-1}(0),\ i = 1, 2.$

2. $x_1 = -2^5,\ y_1 = 2^8,\ z_1 = -2^{12}.$

3. $x_2 = -2^3,\ y_2 = -2^5,\ z_2 = -2^8.$

$(B^1, B^{*,1}, 1, 0)$	$(B^2, B^{*,2}, 1, 0)$
$B^1_{1\leftarrow 0} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},\ B^{*,1}_{0\leftarrow 1} = \begin{pmatrix} 32 & 0 \end{pmatrix}$	$B^2_{1\leftarrow 0} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},\ B^{*,2}_{0\leftarrow 1} = \begin{pmatrix} 8 & 0 \end{pmatrix}$
$B^1_{2\leftarrow 1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix},\ B^{*,1}_{1\leftarrow 2} = \begin{pmatrix} 0 & 1 & 0 \\ 32 & 0 & 0 \end{pmatrix}$	$B^2_{2\leftarrow 1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix},\ B^{*,2}_{1\leftarrow 2} = \begin{pmatrix} 0 & 1 & 0 \\ 8 & 0 & 0 \end{pmatrix}$
$B^1_{3\leftarrow 2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},\ B^{*,1}_{2\leftarrow 3} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 32 & 0 & 0 & 0 \end{pmatrix}$	$B^2_{3\leftarrow 2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},\ B^{*,2}_{2\leftarrow 3} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 8 & 0 & 0 & 0 \end{pmatrix}$
$B^1_{4\leftarrow 3} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,1}_{3\leftarrow 4} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 32 & 0 & 0 & 0 & 0 \end{pmatrix}$	$B^2_{4\leftarrow 3} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,2}_{3\leftarrow 4} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 8 & 0 & 0 & 0 & 0 \end{pmatrix}$
$B^1_{5\leftarrow 4} = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 & 0 \\ 4 & 0 & 1 & 0 & 0 \\ 8 & 0 & 0 & 1 & 0 \\ -16 & 0 & 0 & 0 & 0 \\ 16 & 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,1}_{4\leftarrow 5} = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ -2 & 0 & 1 & 0 & 0 & 0 \\ -4 & 0 & 0 & 1 & 0 & 0 \\ 8 & 0 & 0 & 0 & 1 & 0 \\ 16 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$	$B^2_{5\leftarrow 4} = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 & 0 \\ -2 & 0 & 1 & 0 & 0 \\ -4 & 0 & 0 & 1 & 0 \\ -4 & 0 & 0 & 0 & 0 \\ -8 & 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,2}_{4\leftarrow 5} = \begin{pmatrix} -2 & 1 & 0 & 0 & 0 & 0 \\ -2 & 0 & 1 & 0 & 0 & 0 \\ -4 & 0 & 0 & 1 & 0 & 0 \\ -4 & 0 & 0 & 0 & 1 & 0 \\ -8 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$
$B^1_{6\leftarrow 5} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,1}_{5\leftarrow 6} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	$B^2_{6\leftarrow 5} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,2}_{5\leftarrow 6} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
$B^1_{7\leftarrow 6} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,1}_{6\leftarrow 7} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$	$B^2_{7\leftarrow 6} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,2}_{6\leftarrow 7} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$
$B^1_{8\leftarrow 5} = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ -8 & 0 & 0 & 1 & 0 & 0 \\ -16 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,1}_{5\leftarrow 8} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 0 & 0 \\ 4 & 1 & 0 \\ 8 & 0 & 0 \\ -16 & 0 & -1 \\ 16 & 0 & 0 \end{pmatrix}$	$B^2_{8\leftarrow 5} = \begin{pmatrix} -1 & 1 & 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 1 & 0 & 0 \\ -4 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},\ B^{*,2}_{5\leftarrow 8} = \begin{pmatrix} -2 & 0 & 0 \\ -2 & 0 & 0 \\ -2 & 1 & 0 \\ -4 & 0 & 0 \\ -4 & 0 & -1 \\ -8 & 0 & 0 \end{pmatrix}$

Table 4.2: Two generic points in E_8

Chapter 5

Anti-Poisson Involutions over Filtered Poisson Deformations of Kleinian Singularities

5.1 Reminder on conical symplectic singularities and their Poisson deformations

We give an over review of conical symplectic singularities and universal graded Poisson deformations in this section, following [Los22, Sections 2–3]. We also discuss anti-Poisson involutions of universal (resp. filtered) Poisson deformations at the end of this secition.

5.1.1 Conical symplectic singularities

Definition 5.1.1. Let X be a normal Poisson algebraic variety over $\mathbb{C}$ such that the smooth locus X_{reg} is a symplectic variety. Let ω_{reg} denote the symplectic form on X_{reg}. Following [Bea00, Definition 1.1], we say that X has *symplectic singularities* if there is a projective resolution of singularities

$$\pi : Y \to X$$

such that $\pi^*(\omega_{\mathrm{reg}})$ extends to a regular (but not necessarily symplectic) 2-form on Y. We say π is a *symplectic resolution of singularities* if $\pi^*\omega_{\mathrm{reg}}$ extends to a symplectic form on Y.

Definition 5.1.2. Let $A = \bigoplus_{i\geqslant 0} A_i$ be a finitely generated graded Poisson algebra. We say that A is *positively graded* if there exists a positive integer d such that

- $A_0 = \mathbb{C}$,

- $\{A_i, A_j\} \subset A_{i+j-d}$, i.e., the Poisson bracket is of degree $-d$.

Definition 5.1.3. Let A be a positively graded Poisson algebra. If $X = \operatorname{Spec} A$ is a symplectic singularity, we say that X is a *conical* symplectic singularity.

Let us give two classical examples of conical symplectic singularities, see [Bea00, (2.5), (2.6)].

Example 5.1.4. Let V be a symplectic vector space and $\Gamma \subset \mathrm{Sp}(V)$ a finite subgroup. Then V/Γ is a conical symplectic singularity with $d = 2$. Kleinian singularities are special cases of symplectic quotient singularities in the case of $V = \mathbb{C}^2$.

Example 5.1.5. Let $\mathfrak{g}$ be the Lie algebra of an algebraic group. Then $\mathfrak{g}^*$ is a Poisson algebraic variety whose symplectic leaves are coadjoint orbits. Assume that $\mathfrak{g}$ is semisimple. Let $\mathbb{O}$ be a nilpotent orbit in $\mathfrak{g}^*$. The algebra $\mathbb{C}[\mathbb{O}]$ is finitely generated; in fact, it is the normalization of $\mathbb{C}[\overline{\mathbb{O}}]$, where $\overline{\mathbb{O}}$ is the closure of $\mathbb{O}$ in $\mathfrak{g}$. The variety $X := \operatorname{Spec}(\mathbb{C}[\mathbb{O}])$ is a conical symplectic singularity with $d = 1$.

5.1.2 Universal graded Poisson deformations

Definition 5.1.6. Let X be a conical symplectic singularity. Suppose the Poisson bracket has degree $-d$ for some $d \in \mathbb{Z}_{>0}$. A *graded Poisson deformation* of X is the data $(\mathcal{X}, B, j)$, where:

(1) $B = \bigoplus_{i \geqslant 0} B_i$ is a finitely generated graded $\mathbb{C}$-algebra such that the degree 0 piece $B_0 = \mathbb{C}$.

(2) $\mathcal{X}$ is an affine Poisson B-variety equipped with a $\mathbb{C}^*$-action, the Poisson structure on $\mathbb{C}[\mathcal{X}]$ is B-linear, and the structure morphism $\varphi : \mathcal{X} \to \operatorname{Spec} B$ is $\mathbb{C}^*$-equivariant and flat.

(3) $j : X \xrightarrow{\sim} \varphi^{-1}(0)$ is a $\mathbb{C}^*$-equivariant isomorphism of Poisson varieties, where $0 \in \operatorname{Spec}(B)$ corresponds to the maximal ideal $\bigoplus_{i>0} B_i$.

We say a graded Poisson deformation $(\mathcal{X}, B, j)$ is *universal* if for any graded Poisson deformation $(\mathcal{X}', B', j')$, there is a unique morphism of graded Poisson deformations from $(\mathcal{X}', B', j')$ to $(\mathcal{X}, B, j)$.

Namikawa has proved that the universal graded Poisson deformation always exists for conical symplectic singularities. (c.f. [Nam11, Theorem 5.1]). Let X be a conical symplectic variety, and let $\pi : Y \to X$ be a symplectic resolution of singularities.

Theorem 5.1.7 ([Nam11, Theorem 5.5]). *There is a commutative diagram*

$$\begin{tikzcd} \mathcal{Y} \arrow[r] \arrow[d, "\varphi_Y"'] & \mathcal{X} \arrow[d, "\varphi_X"] \\ H^2(Y, \mathbb{C}) \arrow[r, "q"] & \mathbb{A}^n \end{tikzcd} \tag{5.1.1}$$

where $n = \dim H^2(Y, \mathbb{C})$, and φ_X, φ_Y are universal graded Poisson deformations of X and Y, respectively, with $\varphi_X^{-1}(0) = X,\ \varphi_Y^{-1}(0) = Y$.

The space $\mathfrak{h}_X := H^2(Y, \mathbb{C})$ is called is called the *Namikawa-Cartan space* of X. Moreover, there exists a finite reflection group $W_X \subset \mathrm{GL}(\mathfrak{h}_X)$, called the *Namikawa Weyl group*, such that the map $q\colon \mathfrak{h}_X = H^2(Y, \mathbb{C}) \to \mathbb{A}^d$ in eq. (5.1.1) is the quotient map of the action of W_X ([Nam10][Theorem 1.1]).

Definition 5.1.8. Let $A = \bigoplus_{i \geqslant 0} A_i$ be a positively graded Poisson algebra with Poisson bracket having degree $-d$. A *filtered Poisson deformation* is a commutative algebra $\mathcal{A}$ equipped with a Poisson bracket decreasing the filtration degree by d such that $\operatorname{gr} \mathcal{A} \xrightarrow{\sim} A$ becomes a Poisson algebra isomorphism.

Furthermore, we say that the Poisson bracket $\{\cdot, \cdot\}$ on $\mathcal{A}$ having *filtration degree* $-d$ if

- $\{\mathcal{A}_{\leqslant i}, \mathcal{A}_{\leqslant i}\} \subset \mathcal{A}_{\leqslant i+j-d}$;
- There exist $i, j \geqslant 0$ such that $\{\mathcal{A}_{\leqslant i}, \mathcal{A}_{\leqslant i}\} \not\subset \mathcal{A}_{\leqslant i+j-d-1}$.

Let X be a conical symplectic singularity, and $(\mathcal{X}, \mathfrak{h}_X/W_X, j)$ denote its universal graded Poisson deformation.

Proposition 5.1.9. *[Nam11, Theorem 5.1, Theorem 5.5] The filtered Poisson deformations of X are canonically indexed by the points of the quotient $\mathfrak{h}_X/W_X$. In other words, they can all be obtained through the pullback of the structure morphism $\varphi\colon \mathcal{X} \to \mathfrak{h}_X/W_X$ to a particular point $\lambda \in \mathfrak{h}_X/W_X$. Moreover, $X_\lambda := \varphi^{-1}(\lambda)$ is generically symplectic, i.e., is a symplectic variety on an open dense subset, for any λ.*

5.1.3 Anti-Poisson involutions of Poisson deformations

In Definition 3.1.1, we defined anti-Poisson involutions of the alegbra of regular functions on a Kleinain singularity. Similarly, one can define anti-Poisson involutions for any Poisson algebra.

Definition 5.1.10. Let A be a Poisson algebra. An *anti-Poisson involution* of A is an algebra involution $\theta\colon A \to A$ such that

$$\theta(\{f_1, f_2\}) = -\{\theta(f_1), \theta(f_2)\},\ \ \forall\, f_1, f_2 \in A.$$

Moreover, if A is graded (resp. filtered), we further require that θ preserves the grading (resp. filtration).

Remark 5.1.11. Let $A = \bigoplus_{i \geqslant 0} A_i$ be a graded Poisson algebra, and $\mathcal{A} = \bigcup_{i \geqslant 0} \mathcal{A}_i$ a filtered Poisson deformation of A. If ϑ is an anti-Poisson involution of $\mathcal{A}$, then $\theta := \operatorname{gr} \vartheta$ is automatically an anti-Poisson involution of A.

Let X be a conical symplectic singularity and take a filtered Poisson deformation $X_\lambda,\ \lambda \in \mathfrak{h}_X/W_X$. According to Remark 5.1.11, the associated graded of any anti-Poisson involution of X_λ will produce an anti-Poisson involution of X. We would like to ask the converse question: fix an anti-Poisson involution $\theta\colon X \to X$, does there exist an anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$ such that $\operatorname{gr}\theta_\lambda = \theta$? This question will be answered in Proposition 5.1.17.

It is helpful to study the lift of anti-Poisson involutions of X to the universal graded Poisson deformation $(\mathcal{X}, \mathfrak{h}_X/W_X, j)$, which we refer to as a *UPD-lift* of θ and denote by Θ. Note that the symbol Θ also appears in Proposition 4.2.1 and elsewhere in Chapter 4 to denote an anti-symplectic involution on the cotangent bundle of the representation space of a quiver. However, in this chapter, we use Θ to denote the lift of θ to the universal Poisson deformation $\mathcal{X}$. The term *UPD-lift* is used to distinguish this notion from the lift of θ to the minimal resolutions of Kleinian singularities, which was introduced in Section 4.2 and plays an essential role in Chapter 4.

Recall that $\varphi\colon \mathcal{X} \to \mathfrak{h}_X/W_X$ denote the structure morphism. It follows from Proposition 5.1.9 that φ is surjective, hence dominant. Therefore, the algebra homomorphism $\varphi^*\colon \mathbb{C}[\mathfrak{h}_X/W_X] \to \mathbb{C}[\mathcal{X}]$ is injective. Thus, we can view $\mathbb{C}[\mathfrak{h}_X/W_X]$ as a subalgebra of $\mathbb{C}[\mathcal{X}]$. Moreover, by (2) of Definition 5.1.6, we have that the Poisson bracket of $\mathbb{C}[\mathcal{X}]$ is $\mathbb{C}[\mathfrak{h}_X/W_X]$-linear, which means that that $\mathbb{C}[\mathfrak{h}_X/W_X]$ is contained in the Poisson center of $\mathbb{C}[\mathcal{X}]$. Actually, we have the Proposition 5.1.12 below.

Proposition 5.1.12. *The algebra $\mathbb{C}[\mathfrak{h}_X/W_X]$ coincides with the Poisson center of $\mathbb{C}[\mathcal{X}]$.*

To prove Proposition 5.1.12, we need Lemma 5.1.13 below.

Lemma 5.1.13. *Let M be an affine Poisson variety that is generically symplectic. Then the Poisson center of $\mathbb{C}[M]$ consists only of constant functions.*

Proof. Let $f \in \mathbb{C}[M]$ be a regular function that lies in the Poisson center. It suffices to show that f is constant on an open dense subset of M. Hence, we may assume that M is symplectic.

Let ω denote the symplectic form of M. Then ω induces the Poisson bracket of $\mathbb{C}[M]$:

$$\{f, g\} = \omega(df, dg),\ \forall f, g \in \mathbb{C}[M],$$

Therefore, f lies in the Poisson center if and only if $\{f, -\} = \omega(df, -) = 0$. It follows that $df = 0$ by the nondegeneracy of $\omega(-,-)$. Hence, f is a constant function. $\square$

Proof of Proposition 5.1.12. Let $f \in \mathbb{C}[\mathcal{X}]$ be a regular function that lies in the Poisson center. We would like to show that f is contained in the image of the injective homormorphism $\varphi^*\colon \mathbb{C}[\mathfrak{h}_X/W_X] \hookrightarrow \mathbb{C}[\mathcal{X}]$.

According to Lemma 5.1.13, the restriction of f to each fiber $X_\lambda := \varphi^{-1}(\lambda)$, $\lambda \in \mathfrak{h}_X/W_X$ is a constant function as X_λ is generically symplectic by Proposition 5.1.9. By [OV90, Theorem 2.1.8], the function $f\colon \mathcal{X} \to \mathbb{C}$ factors through $\mathfrak{h}/W$; that is, there exists a rational morphism $\bar{f}\colon \mathfrak{h}/W \to \mathbb{C}$ such that $f = \bar{f} \circ \varphi$.

In fact, $\bar{f}$ is a regular function on $\mathfrak{h}/W$. If $\bar{f}$ has a pole at a point $\lambda' \in \mathfrak{h}/W$, then f will have poles on the fiber $X_{\lambda'}$. However, f is regular function on $\mathcal{X}$, and therefore also regular on the fiber $X_{\lambda'}$. Hence $f \in \mathbb{C}[\mathfrak{h}_X/W_X]$. Finally, note $f = \bar{f} \circ \varphi = \varphi^*(\bar{f})$ means exactly that f lies in the image of the pullback of regular functions $\varphi^*\colon \mathbb{C}[\mathfrak{h}_X/W_X] \hookrightarrow \mathbb{C}[\mathcal{X}]$.

$\square$

Let $\mathfrak{m}_\lambda \subset \mathbb{C}[\mathfrak{h}_X/W_X]$ denote the maximal ideal corresponding to the point $\lambda \in \mathfrak{h}_X/W_X$. Then we have $X_\lambda := \varphi^{-1}(\lambda) = \operatorname{Spec} \mathbb{C}[\mathcal{X}]/\mathfrak{m}_\lambda \mathbb{C}[\mathcal{X}]$. Let $\Theta\colon \mathbb{C}[\mathcal{X}] \to \mathbb{C}[\mathcal{X}]$ be a graded Poisson (resp. anti-Poisson) algebra automorphism, then Θ must preserve the Poisson center $\mathbb{C}[\mathfrak{h}_X/W_X]$ (c.f. Proposition 5.1.12). Therefore, Θ induces a graded algebra automorphism of $\mathbb{C}[\mathfrak{h}_X/W_X]$, and Θ preserves the ideal $\mathfrak{m}_0$.

We have $\mathbb{C}[\varphi^{-1}(0)] = \mathbb{C}[\mathcal{X}]/\mathfrak{m}_0\mathbb{C}[\mathcal{X}]$ and we obtain an induced morphism

$$\Theta_0 : \varphi^{-1}(0) \to \varphi^{-1}(0), \tag{5.1.2}$$

whose pullback to the algebra of regular functions $\mathbb{C}[\varphi^{-1}(0)]$ is a graded Poisson (resp. anti-Poisson) automorphism. Recall that $j\colon X \xrightarrow{\sim} \varphi^{-1}(0)$ is a $\mathbb{C}^*$-equivariant isomorphism of Poisson varieties (c.f. (3) of Definition 5.1.6). Therefore, the composition

$$j^{-1} \circ \Theta_0 \circ j\colon X \to X \tag{5.1.3}$$

is also $\mathbb{C}^*$-equivariant, and its pullback to the algebra of regular functions $\mathbb{C}[X]$ is a graded Poisson (resp. anti-Poisson) algebra automorphism.

Conversely, given a Poisson (resp. anti-Poisson) automorphism of X, we can also produce a corresponding Poisson (resp. anti-Poisson) automorphism of the universal graded Poisson deformation $\mathcal{X}$, according to the following Proposition 5.1.14.

Proposition 5.1.14. *For any Poisson (resp. anti-Poisson) automorphism $\theta\colon X \to X$, there exists a unique*

Poisson (resp. anti-Poisson) automorphism $\Theta\colon \mathcal{X} \to \mathcal{X}$ *such that*

$$\Theta_0 \circ j = j \circ \theta, \tag{5.1.4}$$

where Θ_0 *denotes the restriction of* Θ *to the zero fiber* $\varphi^{-1}(0)$ *and* j *is the identification* $j\colon X \xrightarrow{\sim} \varphi^{-1}(0)$*. We call* Θ *a lift to the universal graded Poisson deformation* $\mathcal{X}$ *(abbreviated as UPD-lift).*

Proof. We denote Poisson automorphisms of X by τ and anti-Poisson automorphisms by θ, as their proofs differ and it is convenient to give them different notations in the proof.

We first consider the Poisson automorphism case. Let $\tau\colon X \to X$ be a Poisson involution of X. Then the composition $j \circ \tau\colon X \to \varphi^{-1}(0)$ is a $\mathbb{C}^*$-equivariant isomorphism of Poisson varieties. Consider the graded Poisson deformation $(\mathcal{X}, \mathfrak{h}_X/W_X, j \circ \tau)$ of X. By the universal property of the universal graded Poisson deformation $(\mathcal{X}, \mathfrak{h}_X/W_X, j)$ (c.f. [Los22, Proposition 2.12]), there is a unique graded Poisson automorphism $T\colon \mathcal{X} \to \mathcal{X}$ intertwining the isomorphisms of zero fibers

$$\varphi^{-1}(0) \xrightarrow[\sim]{j^{-1}} X \xrightarrow[\sim]{j\circ\tau} \varphi^{-1}(0).$$

In other words, the restriction of T to the zero fiber, i.e. $T_0\colon \varphi^{-1}(0) \to \varphi^{-1}(0)$ coincides with the composition $j \circ \tau \circ j^{-1}$. Hence, we obtain

$$T_0 \circ j = j \circ \tau.$$

For the anti-Poisson automorphism case, we consider a new bracket on $\mathcal{X}$ (resp. X) defined from the original bracket,

$$\{\cdot,\cdot\}' := -\{\cdot,\cdot\}.$$

Then the anti-Poisson automorphism θ of $(X, \{\cdot,\cdot\})$ will become a Poisson automorphism of $(X, \{\cdot,\cdot\}')$, and $(\mathcal{X}, \{\cdot,\cdot\}', \mathfrak{h}/W, j \circ \theta)$ is a graded Poisson deformation of $(X, \{\cdot,\cdot\}')$. We reduce to the previous case. $\square$

Let $\mathrm{API}(X)$ denote the set of anti-Poisson involutions of X, and similarly define $\mathrm{API}(\mathcal{X})$.

Proposition 5.1.15. *There exists a bijection*

$$\begin{aligned} \mathrm{API}(\mathcal{X}) &\to \mathrm{API}(X) \\ \Theta &\mapsto j^{-1} \circ \Theta_0 \circ j. \end{aligned} \tag{5.1.5}$$

Proof. It is clear that eq. (5.1.5) is well-defined through our discussion on the composition eq. (5.1.3). The fact that eq. (5.1.5) is a bijection follows from a variant of the universal property of the universal graded Poisson deformation $(\mathcal{X}, \mathfrak{h}_X/W_X, j)$, given in Proposition 5.1.14. $\square$

Let $\mathrm{API}(X)/\sim$ denotes the set of conjugacy classes of anti-Poisson involutions of X by Poisson automorphisms, and similarly define $\mathrm{API}(\mathcal{X})/\sim$. As a corollary of Propositions 5.1.14 and 5.1.15, we have the following result.

Corollary 5.1.16. *The bijection eq.* (5.1.5) *induces a bijection between conjugacy classes.*

$$\mathrm{API}(\mathcal{X})/\sim \xleftrightarrow{1-1} \mathrm{API}(X)/\sim \tag{5.1.6}$$

Let X be a conical symplectic singularity and $\theta\colon X \to X$ an anti-Poisson involution. Let $\Theta\colon \mathcal{X} \to \mathcal{X}$ denote the UPD-lift of θ as in Proposition 5.1.14. Let $X_\lambda = \varphi^{-1}(\lambda)$ be a filtered Poisson deformation with the parameter $\lambda \in \mathfrak{h}_X/W_X$.

Proposition 5.1.17. *Fix X and θ. There exists an anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$ such that $\operatorname{gr}\theta_\lambda = \theta$ if and only if $\lambda \in (\mathfrak{h}/W)^\Theta$. Moreover, if θ_λ exists, then it is induced by the restriction of Θ to the λ-fiber:*

$$\theta_\lambda\colon \mathbb{C}[\mathcal{X}]/\mathfrak{m}_\lambda\mathbb{C}[\mathcal{X}] \to \mathbb{C}[\mathcal{X}]/\mathfrak{m}_\lambda\mathbb{C}[\mathcal{X}],$$

where $\mathfrak{m}_\lambda \subset \mathbb{C}[\mathfrak{h}_X/W_X]$ is the maximal ideal corresponding to the point $\lambda \in \mathfrak{h}_X/W_X$.

Proof. This more or less follows from the variant of the universal property in Proposition 5.1.14, and we leave it to the reader. $\square$

5.2 Universal graded Poisson deformations of Kleinian singularities

According to Example 5.1.4, Kleinian singularities $X = \mathbb{C}^2/\Gamma$ are conical symplectic singularities in dimension 2. Therefore, all the discussions in section 5.1.1 for universal (resp. filtered) deformations of conical symplectic singularties apply to those of Kleinian singularities as well.

From now on, let $X = \mathbb{C}^2/\Gamma$ denote a Klienian singularity. The Namikawa-Cartan space $\mathfrak{h}_X = \mathfrak{h}$ and Namikawa-Weyl group $W_X = W$ coincide with the usual Cartan space and Weyl group of the corresponding types ADE. We will review the result in this section (c.f. Theorem 5.2.5).

5.2.1 An alternative way to construct Poisson brackets on Kleinian singularities

Recall that the Kleinian singularity attached to a finite subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$ is the quotient singularity

$$X := \mathbb{C}^2/\Gamma = \operatorname{Spec} \mathbb{C}[u,v]^\Gamma$$

Klein showed that X can be viewed as a hypersurface in $\mathbb{C}^3$

$$X \simeq \{(x,y,z) \in \mathbb{C}^3 \mid F(x,y,z) = 0\}$$

with an isolated singularity at 0 [Kle93]. Here F denotes the relation among the three fundamental invariants x, y, z in $\mathbb{C}[u,v]^\Gamma$. Note that $\mathbb{C}[X] = \mathbb{C}[u,v]^\Gamma$ is graded with the grading given by $\deg u = \deg v = 1$ and $F(x,y,z)$ is homogeneous in $\mathbb{C}[x,y,z]$.

$$\begin{aligned}
&(A_n,\ n \geqslant 1) - xy + z^{n+1} = 0 \text{ with} \deg x = \deg y = n+1,\ \deg z = 2.\\
&(D_n,\ n \geqslant 4)\ z^2 + xy^2 - x^{n-1} = 0 \text{ with} \deg x = 4,\ \deg y = 2n-4,\ \deg z = 2n-2.\\
&(E_6) - z^2 - x^2 z + y^3 = 0 \text{ with} \deg x = 6,\ \deg y = 8,\ \deg z = 12.\\
&(E_7) - z^2 - y^3 + 16x^3 y = 0 \text{ with} \deg x = 8,\ \deg y = 12,\ \deg z = 18.\\
&(E_8) - z^2 + y^3 - x^5 = 0 \text{ with} \deg x = 12,\ \deg y = 20,\ \deg z = 30.
\end{aligned} \tag{5.2.1}$$

Note the relation $F(x,y,z) = 0$ chosen here is slightly different from the normalization given in Proposition 2.3.1, which will align better with the relation of the universal graded Poisson deformation given in Proposition 5.2.8.

There is a standard Poisson bracket of degree (-2) on $\mathbb{C}[X] = \mathbb{C}[u,v]^\Gamma$ given in eq. (2.3.1):

$$\{f_1, f_2\} = \frac{\partial f_1}{\partial u}\frac{\partial f_2}{\partial v} - \frac{\partial f_1}{\partial v}\frac{\partial f_2}{\partial u}.$$

We have classified anti-Poisson involutions of $\mathbb{C}[X]$ up to conjugation with respect to the above Poisson bracket case by case in Section 3.3. Next, in Proposition 5.2.1 below, we explain an alternative way to construct a Poisson bracket on $\mathbb{C}[X]$ that is equivalent, up to rescaling, to the bracket in eq. (2.3.1). This construction was first introduced by Nambu [Nam73, (2)] in the study of generalized Hamiltonian systems in physics. A treatment in the algebraic setting, particularly for Kleinian singularities, can be found in [ES18, Section 8.1].

Proposition 5.2.1. *Consider the three-variable polynomial algebra $\mathbb{C}[x, y, z]$ with x, y, z having possibly different gradings. Let $F(x, y, z)$ be a homogeneous polynomial in $\mathbb{C}[x, y, z]$. Then $\mathbb{C}[x, y, z]/(F)$ is a graded Poisson algebra with the following Poisson bracket*

$$\{f, g\}_F = \det \Big| \frac{\partial(f, g, F)}{\partial(x, y, z)} \Big|, \tag{5.2.2}$$

of degree $\deg x + \deg y + \deg z - \deg F$.

Proof. It is easy to see that $\{\cdot, \cdot\}_F$ is skew-symmetric and is extended from

$$\{x, y\}_F = F_z, \ \{x, z\}_F = -F_y, \ \{y, z\}_F = F_x, \tag{5.2.3}$$

following the Leibniz identity. One can check that $\{\cdot, \cdot\}_F$ satisfies the Jacobian identity on the generators x, y, z using eq. (5.2.3). Therefore, $\{\cdot, \cdot\}_F$ defines a Poisson bracket on $\mathbb{C}[x, y, z]$. It is also easy to check that the bracket $\{\cdot, \cdot\}_F$ is of degree $\deg x + \deg y + \deg z - \deg F$.

Recall that F is homogeneous, hence $\mathbb{C}[x, y, z]/(F)$ is graded. Next, one verify that F is in the Poisson center of $\mathbb{C}[x, y, z]$, based on the definition given in eq. (5.2.2). Therefore, $\{\cdot, \cdot\}_F$ descends to a Poisson bracket on the quotient algebra $\mathbb{C}[x, y, z]/(F)$. $\square$

We equip the algebra of functions on Kleinain singularities $\mathbb{C}[X] = \mathbb{C}[x, y, z]/(F)$ with the Poisson bracket given in eq. (5.2.2) (equivalently, eq. (5.2.3)), where F is listed in eq. (5.2.1). Based on a case by case calculation, we can see that $\{\cdot, \cdot\}_F$ is of degree

$$\deg x + \deg y + \deg z - \deg F = -2$$

in all types of ADE. According to [EG02, Lemma 2.23, (i)], the Poisson bracket on X of degree -2 is unique up to rescaling, as $X = \mathbb{C}^2/\Gamma$ is a symplectic quotient singularity and the tautological representation $\mathbb{C}^2$ is an irreducible Γ-representation of quaternionic type, i.e. $(\wedge^2\mathbb{C}^2)^\Gamma = \wedge^2\mathbb{C}^2 \neq 0$. It follows that

$$\{\cdot, \cdot\}_F = c\{\cdot, \cdot\}$$

for some nonzero scalar $c \in \mathbb{C}$. From now on, we will mainly use $\{\cdot, \cdot\}_F$ instead of $\{\cdot, \cdot\}$, as it is easier to compute. We may simply write $\{\cdot, \cdot\}$ instead of $\{\cdot, \cdot\}_F$ if no confusion will be caused.

5.2.2 Graded universal deformations of Kleinian singularities

Next, we study the universal graded Poisson deformations of a Kleinian singularity X. Recall that $X := \mathbb{C}^2/\Gamma$ admits a natural $\mathbb{C}^*$-action induced from that of $\mathbb{C}^2$. We recall the definition of graded universal deformation of X, and the fact that the universal graded Poisson deformation of X coincides with its graded universal deformation. (See Proposition 5.2.7.)

We first review the definition of semiuniversal deformation for a general variety X.

Definition 5.2.2. [Slo80, Definition 2.3] Let X be an algebraic variety and $x_0 \in X$. A deformation of (X, x_0) is a flat morphism $\varphi : (\mathcal{X}, x) \to (U, u)$ with $(\varphi^{-1}(u), x) \simeq (X, x_0)$. U is called the *base* of the deformation.

A deformation $\varphi\colon \mathcal{X} \to U$ of (X, x_0) is *semiuniversal* if any other deformation $\varphi' : \mathcal{X}' \to U'$ of (X, x_0) is induced from φ by a base change $f : (U', u') \to (U, u)$. Moreover, if f_1, f_2 are two such morphisms, then $d_{u'} f_1 = d_{u'} f_2$.

Remark 5.2.3. (1) The "semi-" part in the "semiuniversal" means that the flat deformation is only unique at the level of Zariski tangent spaces.

(2) If the variety X admits a $\mathbb{C}^*$- action, equivalently, a grading, then the semiuniversal deformation $\mathcal{X}$ will also admit a grading, which we call $\mathbb{C}^*$-semiuniversal deformation. ([Pin74, Proposition 2.3])

When the $\mathbb{C}^*$-semiuniversal deformation is unique, we also refer to it as the *graded universal deformation* for brevity. This is the case for Kleinian singularities, and we will use this term throughout the rest of this Chapter. The classical Theorem 5.2.5, due to Brieskorn and Slodowy, gives a characterization of the graded universal deformation of Kleinian singularities.

Let $\mathfrak{g}$ be a simple complex Lie algebra and consider the quotient map that is also called characteristic map $\varphi\colon \mathfrak{g} \to \mathfrak{g}/\!/G \simeq \mathfrak{h}/W$ for the action of the adjoint group G of $\mathfrak{g}$, where $\mathfrak{h}$ is the Cartan space and W is the Weyl group. Let $\mathcal{N} := \varphi^{-1}(0)$ consists of the nilpotent elements in $\mathfrak{g}$ and is called the *nilpotent* cone of $\mathfrak{g}$. Let $e \in \mathfrak{g}$ be a subregular nilpotent element, and consider a $\mathfrak{sl}_2$-triple $\{e, h, f\}$. The *Slodowy slice* to e is the affine space

$$S = e + \ker(\operatorname{ad} f). \tag{5.2.4}$$

Construction 5.2.4. One can define the *Kazhdan action* of $\mathbb{C}^*$ on S in the following way. Define the Kazhdan action of $\mathbb{C}^*$ on $\mathfrak{g}$ which stabilizes the Slodowy slice $S = e + \ker \operatorname{ad} f$. First, recall that $\{e, h, f\}$

is a $\mathfrak{sl}_2$-triple. Consider the Lie algebra homomorphism $\mathfrak{sl}_2(\mathbb{C}) \to \mathfrak{g}$ defined by

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \mapsto e, \qquad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \mapsto h, \qquad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \mapsto f.$$

This Lie algebra homomorphism exponentiates to a rational homomorphism $\tilde{\gamma} : \mathrm{SL}_2(\mathbb{C}) \to G$. We put

$$\gamma : \mathbb{C}^\times \to G, \qquad \gamma(t) = \tilde{\gamma} \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}, \quad \forall\, t \in \mathbb{C}^*.$$

Note that$\mathrm{Ad}\,\big(\gamma(t)\big)(e) = t^2 e$. The desired action of $\mathbb{C}^*$ on $\mathfrak{g}$, to be denoted by ρ, is defined by

$$\rho(t)(x) = t^{-2}\,\mathrm{Ad}\,\big(\gamma(t)\big)(x), \qquad \forall\, x \in \mathfrak{g}.$$

Note that $\rho(t)(e + x) = e + \rho(t)(x)$. Thus, since $\rho(t)$ stabilizes $\ker \operatorname{ad} f$, it also stabilizes the Slodowy slice $S = e + \ker \operatorname{ad} f$.

Theorem 5.2.5. *[Slo80, Theorem 8.7] Let $\mathfrak{g}$ be a simple Lie algebra of type ADE, let $\mathcal{N}$ be its nilpotent cone, and let S be a slice to the orbit of a subregular nilpotent element e in $\mathfrak{g}$. Then the germ $(S \cap \mathcal{N}, e)$ is a Kleinian singularity with the same Dynkin diagram as $\mathfrak{g}$, and the restriction $\varphi : (S, e) \to (\mathfrak{h}/W, 0)$ of the characteristic map $\mathfrak{g} \to \mathfrak{g}/\!/G \simeq \mathfrak{h}/W$, equipped with Kazhdan grading, is isomorphic to the graded universal deformation of the Kleinian singularity $(S \cap \mathcal{N}, e)$. Moreover, the fibers $\varphi^{-1}(\lambda)$ are normal and two-dimensional for any $\lambda \in \mathfrak{h}/W$.*

Remark 5.2.6. Note that, under the Kazhdan grading constructed in Construction 5.2.4, one has $\deg \mathfrak{h} = 2$. Moreover, the graded universal deformation $\mathcal{X}$ is a hypersurface in $\mathbb{C}^{n+3}$ cut out by a homogeneous polynomial (c.f. Proposition 5.2.8), where $n = \dim \mathfrak{h}/W = \operatorname{rank}(\mathfrak{g})$. Actually, one has that $\mathcal{X}$ is isomorphic to the affine space $\mathbb{C}^{n+2}$ (c.f. [Slo80, Section 7.4]).

There is a standard Poisson bracket $\{\cdot, \cdot\}$ on the Slodowy slice S induced from the Lie bracket of $\mathfrak{g}$ via Hamiltonian reduction (cf. the article of Gan and Ginzburg [GG02, Section 3]). Note that with the Kazhdan grading in Construction 5.2.4, the Poisson bracket $\{\cdot, \cdot\}$ on S is of degree -2. Moreover, $\mathbb{C}[\mathfrak{h}]^W$ coincides with the Poisson center of $\mathbb{C}[S]$. The result below, which can be found in [LNS12, Proposition 6.2], states that the graded universal deformation $\varphi\colon S \to \mathfrak{h}/W$ is also the universal graded Poisson deformation of the Kleinian singularity $S \cap \mathcal{N} \simeq X$.

Proposition 5.2.7. *[LNS12, Proposition 6.2] Let $\mathfrak{g}$ be of types ADE. Then the restriction of the characterstic map $\mathfrak{g} \to \mathfrak{g}//G \simeq \mathfrak{h}/W$ to the Slodowy slice S of subregular nilpotent element $\varphi : S \to \mathfrak{h}/W$ is not only the graded universal deformation, but also the the universal graded Poisson deformation of the Kleinian singularity $S \cap \mathcal{N} \simeq X$.*

5.2.3 Poisson brackets on universal graded Poisson deformations

As we explained above Proposition 5.2.7, there is a standard way to equip the Slodowy slice S with a degree -2 Poisson bracket from Hamiltonian reductions. However, this construction is not very good for computation.

In this subsection, we take the graded universal deformation $\mathcal{X}$ of X listed in Proposition 5.2.8, following [KM92, Table 3]. One can identify $\mathcal{X} \simeq S$, as they are both the graded universal deformations of X. The Poisson brackets on S will transform to a Poisson bracket on $\mathcal{X}$. We will see how the Poisson brackets on $\mathcal{X}$ look like in Proposition 5.2.12.

Let $X = \mathbb{C}^2/\Gamma \simeq \operatorname{Spec}\mathbb{C}[x,y,z]/(F(x,y,z))$ be a Kleinian singularity, with defining equation $F(x,y,z)$ as in eq. (5.2.1). Let $G_1, \cdots, G_n$ be monomials forming a basis of the vector space $\mathbb{C}[x,y,z]/(F_x, F_y, F_z)$ and let $\mu_1, \cdots, \mu_n$ be the coordinate functions on the base $\mathfrak{h}/W$. Recall that $\deg \mathfrak{h} = 2$ from the Kazhdan grading in Construction 5.2.4, which induces a grading on $\mathfrak{h}/W$. Then the graded universal deformation of X is given by

$$\mathcal{X} = \operatorname{Spec}\mathbb{C}[x,y,z,\mu_1,\cdots,\mu_n]/\big(G(x,y,z,\mu_1,\cdots,\mu_n)\big), \tag{5.2.5}$$

where G is a homogeneous polynomial defined by

$$G(x,y,z,\mu_1,\cdots,\mu_n) = F(x,y,z) + \mu_1 G_1 + \cdots \mu_n G_n. \tag{5.2.6}$$

In Proposition 5.2.8 below, we provide explicit forms of the function G for each ADE type. Note we use $s_i, \delta_i, \gamma_n, \epsilon_i$ instead of μ_i to denote the coordinates of $\mathfrak{h}/W$ in each ADE type. These choices align better with the invariant functions in $\mathbb{C}[\mathfrak{h}]^W$ coming from simple Lie algebras (c.f. Remark 5.2.9) and also ensure that the superscripts reflect appropriate gradings.

Proposition 5.2.8. *[KM92, Table 3] The defining equations G of a graded universal deformation $\mathcal{X}$ of a Kleinian singularity is given below. The variables other than x, y, z are coordinate functions of the base $\mathfrak{h}/W$, and the term with subscript i indicates their gradings is $2i$.*

- A_n $(n \geqslant 1) : -xy + z^{n+1} + \sum_{i=2}^{n+1} s_i z^{n+1-i}$.

- D_n $(n \geqslant 4) : z^2 + xy^2 - x^{n-1} - \sum_{i=1}^{n-1} \delta_{2i} x^{\,n-i-1} + 2\gamma_n y$.

- $E_6 : -z^2 - zx^2 + y^3 + \epsilon_2 yx^2 + \epsilon_5 yx + \epsilon_6 x^2 + \epsilon_8 y + \epsilon_9 x + \epsilon_{12}$.

- $E_7 : -z^2 - y^3 + 16yx^3 + \epsilon_2 y^2 x + \epsilon_6 y^2 + \epsilon_8 yx + \epsilon_{10} x^2 + \epsilon_{12} y + \epsilon_{14} x + \epsilon_{18}$.

- $E_8 : -z^2 + y^3 - x^5 + \epsilon_2 yx^3 + \epsilon_8 yx^2 + \epsilon_{12} x^3 + \epsilon_{14} yx + \epsilon_{18} x^2 + \epsilon_{20} y + \epsilon_{24} x + \epsilon_{30}$.

Remark 5.2.9. If we fix a basis of the Cartan space $\mathfrak{h}$, then one can also write down the generators of $\mathbb{C}[\mathfrak{h}]^W = \mathbb{C}[\mu_1, \cdots, \mu_n]$ in types A and D easily. For type A_n, let

$$\mathfrak{h} = \{(t_0, t_1, \cdots, t_n) \in \mathbb{C}^{n+1} | \sum_{i=0}^{n} t_i = 0\} \simeq \mathbb{C}^n,$$

then s_i = i-th elementary symmetric polynomials in $t_0, \cdots, t_n$.

For type D_n, let

$$\mathfrak{h} = \{(t_1, \cdots, t_n, -t_1, \cdots, -t_n)\} \simeq \mathbb{C}^n,$$

then δ_i = i-the symmetric polynomial in $t_1^2, \cdots, t_n^2$, and $\gamma_n = t_1 \cdots t_n$.

However, it is harder to find the generators of $\mathbb{C}[\mathfrak{h}]^W$ in types E. For details, see [KM92, Appendices 0-2].

Remark 5.2.10. When discussing the general theory, for example, the construction of Poisson bracket of $\mathcal{X}$ in Proposition 5.2.12, we will continue to use the notation μ_i as the coordinate functions on $\mathfrak{h}/W$ for simplicity. However, in Section 5.3, where we study the UPD-lift of anti-Poisson involutions for each ADE type case by case, we will instead use the parameters $s_i, \delta_i, \gamma_n, \epsilon_i$ as the coordinates of $\mathfrak{h}/W$, as introduced in Proposition 5.2.8.

Now assume the graded universal deformation $\mathcal{X}$ is Poisson, with Poisson brackets inherited from the Slodowy slice S. We would like to figure out how the Poisson brackets look like in terms of generators of $\mathbb{C}[\mathcal{X}]$. Note that $\mathbb{C}[\mathfrak{h}/W]$ is the Poisson center of $\mathbb{C}[\mathcal{X}]$ (c.f. Proposition 5.1.12). Therefore,

$$\{x, \mu_i\} = \{y, \mu_i\} = \{z, \mu_i\} = 0. \tag{5.2.7}$$

It remains to figure out the brackets

$$\{x, y\},\ \{x, z\},\ \{y, z\}.$$

These brackets will be given in Proposition 5.2.12, for which we need the help of Lemma 5.2.11

Lemma 5.2.11. *Take $\lambda = (\lambda_1, \cdots, \lambda_n) \in \mathfrak{h}/W$, and define $G_\lambda(x,y,z) := G(x,y,z,\lambda) = F(x,y,z) + \sum_{i=1}^n \lambda_i G_i(x,y,z)$, then the filtered Poisson deformation*

$$X_\lambda := \operatorname{Spec} \mathbb{C}[\varphi^{-1}(\lambda)] = \operatorname{Spec} \mathbb{C}[x,y,z]/(G_\lambda)$$

has Poisson brackets

$$\{x,y\} = G_{\lambda,z} := \frac{\partial G_\lambda}{\partial z},\ \{x,z\} = -G_{\lambda,y} := -\frac{\partial G_\lambda}{\partial y},\ \{y,z\} = G_{\lambda,x} := \frac{\partial G_\lambda}{\partial x}. \tag{5.2.8}$$

Proof. It is easy to check that eq. (5.2.8) is a well-defined Poisson brackets on $\mathbb{C}[X_\lambda]$, following similar argument as one checks that eq. (5.2.2) (equivalently eq. (5.2.3)) defines a Poisson bracket on $\mathbb{C}[X]$. Moreover, it is easy to see that one has $\operatorname{gr} \mathbb{C}[X_\lambda] \xrightarrow{\sim} \mathbb{C}[X]$ is an isomorphism of Poisson algebras because the highest degree term in $G_{\lambda,x}$ (resp. $G_{\lambda,y},\ G_{\lambda,z}$) is F_x (resp. $F_y,\ F_z$).

Lastly, note that the Poisson brackets on $\mathbb{C}[X_\lambda]$ of filtration degree -2 is unique up to rescaling ([EG02, Lemma 2.26]). Therefore, the Poisson brackets given in eq. (5.2.8) is the only possible choice making $\operatorname{gr} \mathbb{C}[X_\lambda] \xrightarrow{\sim} \mathbb{C}[X]$ a graded Poisson algebra isomorphism.

□

Proposition 5.2.12. *The Poisson bracket on $\mathcal{X}$ is given by*

$$\{x,y\} = G_z,\ \{x,z\} = -G_y,\ \{y,z\} = G_x, \tag{5.2.9}$$

$$\{x,\mu_i\} = \{y,\mu_i\} = \{z,\mu_i\} = 0,\ \forall i. \tag{5.2.10}$$

Proof. eq. (5.2.10) is the same as eq. (5.2.7), which has already shown to be true. To prove eq. (5.2.9), we may assume

$$\{x,y\} = H_1,\ \{x,z\} = H_2,\ \{y,z\} = H_3,$$

where H_1, H_2, H_3 are polynomials in $x, y, z, \mu_i,\ 1 \leqslant i \leqslant n$. Take a particular point $\lambda = (\lambda_1, \cdots, \lambda_n) \in \mathfrak{h}/W$, the Poisson brackets of $\mathbb{C}[\mathcal{X}]$ descend to that of the filtered Poisson deformation $\mathbb{C}[X_\lambda]$, which are given by

$$\{x,y\} = H_1(x,y,z,\lambda),\ \{x,z\} = H_2(x,y,z,\lambda),\ \{y,z\} = H_3(x,y,z,\lambda). \tag{5.2.11}$$

Comparing eq. (5.2.11) with eq. (5.2.8), one obtain that

$$H_1(x, y, z, \lambda) - G_{\lambda,z} = aG_\lambda$$

for some scalar $a \in \mathbb{C}$ as $G_\lambda = 0$ is the only relation in $\mathbb{C}[X_\lambda] = \mathbb{C}[x, y, z]/(G_\lambda)$ and X_λ is normal, hence integral (c.f. Theorem 5.2.5). Note that $\deg H_1(x, y, z, \lambda) = \deg x + \deg y - 2 = \deg G_{\lambda,z} < \deg G_\lambda$. Therefore, we must have

$$H_1(x, y, z, \lambda) = G_{\lambda,x} \tag{5.2.12}$$

as polynomials in x, y, z.

Note that eq. (5.2.12) is valid for any choice of $\lambda \in \mathfrak{h}/W$. It follows that

$$H_1(x, y, z, \mu) = G_x(x, y, z, \mu)$$

as polynomials in x, y, z, μ. Similarly, one can show that

$$H_2 = -G_y,\ H_3 = G_x.$$

$\square$

5.3 Lifting anti-Poisson involutions to the universal graded Poisson deformations

Recall $X = \mathbb{C}^2/\Gamma \simeq \operatorname{Spec}\mathbb{C}[x, y, z]/(F)$ denotes a Kleinian singularity, whose defining equation F is given in eq. (5.2.1). Moreover, $\mathcal{X} = \operatorname{Spec}\mathbb{C}[x, y, z, \mu_1, \cdots, \mu_n]/(G)$ denotes its universal graded Poisson deformation, whose defining equation G is given in Proposition 5.2.8, and Poisson brackets given in eq. (5.2.3)

$$\{x, y\} = G_z,\ \{x, z\} = -G_y,\ \{y, z\} = G_x.$$

In this section, we classify $\mathfrak{h}/W$-linear anti-Poisson involutions of $\mathcal{X}$ (c.f. Proposition 5.1.14), and study when Θ restricts to an anti-Poisson involution of the filtered Poisson deformation $X_\lambda := \varphi^{-1}(\lambda)$ (c.f. Proposition 5.1.17), where $\varphi\colon \mathcal{X} \to \mathfrak{h}/W$ denotes the deformation morphism.

5.3.1 Type A

Let X be the Kleinian singularity of type A_n, and $\mathcal{X}$ denote its universal graded Poisson deformation. Recall from Proposition 5.2.8 that we have

$$\mathbb{C}[\mathcal{X}] = \mathbb{C}[x, y, z, s_2, \cdots, s_{n+2}]/(-xy + z^{n+1} + \sum_{i=2}^{n+1} s_i z^{n+1-i}), \tag{5.3.1}$$

with gradings $\deg x = \deg y = n+1,\ \deg z = 2,\ \deg s_i = 2i,\ 2 \leqslant i \leqslant n+1$. Let us denote

$$f(z, s) = z^{n+1} + \sum_{i=2}^{n+1} s_i z^{n+1-i}. \tag{5.3.2}$$

According to Proposition 5.2.12, the Poisson brackets on $\mathbb{C}[\mathcal{X}]$ are given by

$$\begin{aligned} &\{x, y\} = f_z(z, s),\ \{x, z\} = x,\ \{y, z\} = -y, \\ &\{x, s_i\} = \{y, s_i\} = \{z, s_i\} = 0,\ \forall i. \end{aligned} \tag{5.3.3}$$

The deformation parameter space $\mathfrak{h}/W = \operatorname{Spec} \mathbb{C}[s_2, \cdots, s_{n+1}]$ is an affine n-space. The deformation morphism is the projection.

$$\begin{aligned} \varphi \colon \mathcal{X} &\to \mathfrak{h}/W \\ (x, y, z, s_2 \cdots, s_{n+1}) &\mapsto (s_2 \cdots, s_{n+1}). \end{aligned} \tag{5.3.4}$$

Note that we recover the Kleinian singularity of type $A_n \colon \mathbb{C}[X] \simeq \mathbb{C}[x, y, z]/(-xy + z^{n+1})$ by considering the fiber $j \colon X \simeq \varphi^{-1}(0)$.

According to Proposition 3.3.1, there are three types of anti-Poisson involutions of X in type A_n when n is odd.

Proposition 5.3.1. *The universal graded Poisson deformation $\mathcal{X}$ of Kleinian singularity of Type A_n, with n odd, has three conjugacy classes of anti-Poisson involutions.*

I. $\Theta(x) = y,\ \Theta(y) = x,\ \Theta(z) = z,\ \Theta(s_i) = s_i,\ 2 \leqslant i \leqslant n+1.$

II. $\Theta(x) = x,\ \Theta(y) = y,\ \Theta(z) = -z,\ \Theta(s_i) = (-1)^i s_i,\ 2 \leqslant i \leqslant n+1.$

III. $\Theta(x) = -x,\ \Theta(y) = -y,\ \Theta(z) = -z,\ \Theta(s_i) = (-1)^i s_i,\ 2 \leqslant i \leqslant n+1.$

The Roman Numeral Labels match with those of Proposition 3.3.1 under the bijection eq. (5.1.5).

Proof. First, it is easy to verify that the involutions Θ of Types I-III are indeed anti-Poisson using the Poisson brackets given in eq. (5.3.3). Next, their image under the bijection eq. (5.1.5) are exactly the three non-conjugate types of anti-Poisson involutions of the Kleinian singularities X of type A_n given in Proposition 3.3.1. Then it is a direct application of Proposition 5.1.15 that the listed three types of anti-Poisson involutions Θ of Types I-III exhaust all conjugacy classes of anti-Poisson involutions of $\mathcal{X}$. $\square$

Fix $\lambda = (s_2, \cdots, s_{n+1}) \in \mathfrak{h}/W$, and consider the filtered Poisson deformation

$$X_\lambda = \varphi^{-1}(s) = \operatorname{Spec} \mathbb{C}[x, y, z]/(-xy + z^{n+1} + \sum_{i=2}^{n} s_i z^{n+1-i}). \tag{5.3.5}$$

Next, given an anti-Poisson involution $\theta\colon X \to X$ listed in Proposition 3.3.1, we study when does there exist an anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$ such that $\operatorname{gr}\theta_\lambda = \theta$.

Proposition 5.3.2. *Let X be a Kleinian singularity of type A_n, with n odd and X_λ a filtered Poisson deformation of X. Let θ be an anti-Poisson involution of X listed in Proposition 3.3.1. We have*

I. Consider the anti-Poisson involution $\theta\colon X \to X$ defined by $x \mapsto y,\ y \mapsto x,\ z \mapsto z$. Then

$\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto y,\ y \mapsto x,\ z \mapsto z$ defines an anti-Poisson involution of X_λ such that $\operatorname{gr}\theta_\lambda = \theta$ for any $\lambda \in \mathfrak{h}/W$.

II. Consider the anti-Poisson involution $\theta\colon X \to X$ defined by $x \mapsto x,\ y \mapsto y,\ z \mapsto -z$. Then $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ defines an anti-Poisson involution of X_λ such that $\operatorname{gr}\theta_\lambda = \theta$ if and only if $\lambda \in (\mathfrak{h}/W)^\Theta = \{(s_i)_{2\leqslant i\leqslant n+1} \mid s_i = 0,\ i \text{ odd}\}$.

III. Consider the anti-Poisson involution $\theta\colon X \to X$ defined by $x \mapsto -x,\ y \mapsto -y,\ z \mapsto -z$. Then $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto -x,\ y \mapsto -y,\ z \mapsto -z$ defines an anti-Poisson involution of X_λ such that $\operatorname{gr}\theta_\lambda = \theta$ if and only if $\lambda \in (\mathfrak{h}/W)^\Theta = \{(s_i)_{2\leqslant i\leqslant n+1} \mid s_i = 0,\ i \text{ odd}\}$.

The Roman Numeral Labels match with those of Proposition 3.3.1.

Proof. For any anti-Poisson involution $\theta\colon X \to X$, there exists a unique anti-Poisson involution $\Theta\colon \mathbb{C}[\mathcal{X}] \to \mathbb{C}[\mathcal{X}]$ that preserves the graded subalgebra $\mathbb{C}[\mathfrak{h}]^W \subset \mathbb{C}[\mathcal{X}]$ and whose restriction to $X \simeq \varphi^{-1}(0)$ recovers θ, given in Proposition 5.3.1. According to Proposition 5.1.17, there exists an anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$ such that $\operatorname{gr}\theta_\lambda = \theta$ if and only if $\lambda \in (\mathfrak{h}/W)^\Theta$ and θ_λ is uniquely given by the restriction of Θ to the fiber $X_\lambda = \varphi^{-1}(\lambda)$. $\square$

Parallel to Propositions 5.3.1 and 5.3.2, one has similar results for universal (resp. filtered) Poisson deformations of Kleinian singularities of type A_n, with n even. We listed them in Propositions 5.3.3 and 5.3.4, but omit the proof.

Proposition 5.3.3. *The universal graded Poisson deformation $\mathcal{X}$ of Kleinian singularity of Type A_n, with n even, has two conjugacy classes of anti-Poisson involutions.*

I. $\Theta(x) = y,\ \Theta(y) = x,\ \Theta(z) = z,\ \Theta(s_i) = s_i,\ 2 \leqslant i \leqslant n+1.$

II. $\Theta(x) = -x,\ \Theta(y) = y,\ \Theta(z) = -z,\ \Theta(s_i) = (-1)^i s_i,\ 2 \leqslant i \leqslant n+1.$

The Roman Numeral Labels match with those of Proposition 3.3.2 under the bijection eq. (5.1.5).

Proposition 5.3.4. *Let X be a Kleinian singularity of type A_n, with n even and X_λ a filtered Poisson deformation of X. Let θ be an anti-Poisson involution of X listed in Proposition 3.3.2. We have*

I. *Consider the anti-Poisson involution $\theta\colon X \to X$ defined by $x \mapsto y,\ y \mapsto x,\ z \mapsto z$. Then $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto y,\ y \mapsto x,\ z \mapsto z$ defines an anti-Poisson involution of X_λ such that $\operatorname{gr}\theta_\lambda = \theta$ for any $\lambda \in \mathfrak{h}/W$.*

II. *Consider the anti-Poisson involution $\theta\colon X \to X$ defined by $x \mapsto x,\ y \mapsto y,\ z \mapsto -z$. Then $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto -x,\ y \mapsto y,\ z \mapsto -z$ defines an anti-Poisson involution of X_λ such that $\operatorname{gr}\theta_\lambda = \theta$ if and only if $\lambda \in (\mathfrak{h}/W)^\Theta = \{(s_i)_{2\leqslant i\leqslant n+1} \mid s_i = 0,\ i \text{ odd}\}$.*

5.3.2 Type D

Let X be the Kleinian singularity of type D_n $n \geqslant 4$, and $\mathcal{X}$ denote its universal graded Poisson deformation. Recall from Proposition 5.2.8 that we have

$$\mathbb{C}[\mathcal{X}] = \mathbb{C}[x, y, z, \delta_2, \delta_4, \cdots, \delta_{2n-2}, \gamma_n]/(z^2 + xy^2 - x^{n-1} - \sum_{i=1}^{n-1} \delta_{2i} x^{n-i-1} + 2\gamma_n y), \tag{5.3.6}$$

with gradings $\deg x = 4,\ \deg y = 2n-4,\ \deg z = 2n-2,\ \deg \delta_{2i} = 4i,\ 2 \leqslant i \leqslant n-1,\ \deg \gamma_n = 2n$. Let us denote

$$g(x, \delta) = x^{n-1} + \sum_{i=1}^{n-1} \delta_{2i} x^{n-i-1}. \tag{5.3.7}$$

According to Proposition 5.2.12, the Poisson brackets on $\mathbb{C}[\mathcal{X}]$ are given by

$$\begin{aligned} \{x,y\}=2z,\ \{x,z\}=-2xy+2\gamma_n,\ \{y,z\}=y^2-g_x(x,\delta),\\ \{x,\delta_{2i}\}=\{y,\delta_{2i}\}=\{z,\delta_{2i}\}=0,\ \forall i,\\ \{x,\gamma_n\}=\{y,\gamma_n\}=\{z,\gamma_n\}. \end{aligned} \tag{5.3.8}$$

The deformation parameter space $\mathfrak{h}/W=\operatorname{Spec}\mathbb{C}[\delta_2,\delta_4,\cdots,\delta_{2n-2},\gamma_n]$ is an affine n-space. The deformation morphism is the projection.

$$\begin{aligned} \varphi\colon \mathcal{X}\to\mathfrak{h}/W\\ (x,y,z,\delta_2,\delta_4,\cdots,\delta_{2n-2},\gamma_n)\mapsto(\delta_2,\delta_4,\cdots,\delta_{2n-2},\gamma_n). \end{aligned} \tag{5.3.9}$$

Note that we recover the Kleinian singularity of type $D_n\colon \mathbb{C}[X]\simeq\mathbb{C}[x,y,z]/(z^2+xy^2-x^{n-1})$ by considering the fiber $j\colon X\simeq\varphi^{-1}(0)$. Note that the defining relation of X chosen in this subsection is different from those of Sections 3.3.2 and 3.3.3, and we will discuss $D_n,\ n$ even and odd cases together in this section.

Proposition 5.3.5. *The universal graded Poisson deformation $\mathcal{X}$ of Kleinian singularity of Type D_n has two conjugacy classes of anti-Poisson involutions.*

I. $\Theta(x)=x,\ \Theta(y)=y,\ \Theta(z)=-z,\ \Theta(\gamma_n)=\gamma_n,\Theta(\delta_{2i})=\delta_{2i},\ 1\leqslant i\leqslant n-1.$

II. $\Theta(x)=x,\ \Theta(y)=-y,\ \Theta(z)=z,\ \Theta(\gamma_n)=-\gamma_n,\Theta(\delta_{2i})=\delta_{2i},\ 1\leqslant i\leqslant n-1.$

The Roman Numeral Labels match with those of Propositions 3.3.5 and 3.3.8 respectively, under the bijection eq. (5.1.5)*.*

Fix $\lambda=(\delta,\gamma_n)=(\delta_2,\cdots,\delta_{2n-2},\gamma_n)\in\mathfrak{h}/W$, and consider the filtered Poisson deformation

$$X_\lambda=\varphi^{-1}(\lambda)=\operatorname{Spec}\mathbb{C}[x,y,z]/(z^2+xy^2-x^{n-1}-\sum_{i=1}^{n-1}\delta_{2i}x^{n-i-1}+2\gamma_n y). \tag{5.3.10}$$

Next, given an anti-Poisson involution $\theta\colon X\to X$ listed in Propositions 3.3.5 and 3.3.8, we study when does there exist an anti-Poisson involution $\theta_\lambda\colon X_\lambda\to X_\lambda$ such that $\operatorname{gr}\theta_\lambda=\theta$.

Proposition 5.3.6. *Let X be a Kleinian singularity of type D_n and X_λ a filtered Poisson deformation of X. Let θ be an anti-Poisson involution of X. We have*

I. Consider the anti-Poisson involution $\theta\colon X\to X$ defined by $x\mapsto x,\ y\mapsto y,\ z\mapsto -z$. Then

$\theta_\lambda \colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ defines an anti-Poisson involution of X_λ such that $\operatorname{gr}\theta_\lambda = \theta$ for any $\lambda \in \mathfrak{h}/W$.

II. *Consider the anti-Poisson involution $\theta \colon X \to X$ defined by $x \mapsto x,\ y \mapsto y,\ z \mapsto -z$. Then $\theta_\lambda \colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto x,\ y \mapsto -y,\ z \mapsto z$ defines an anti-Poisson involution of X_λ such that $\operatorname{gr}\theta_\lambda = \theta$ if and only if $\lambda \in (\mathfrak{h}/W)^\Theta$, where $(\mathfrak{h}/W)^\Theta = (\gamma_n = 0) \subset \mathfrak{h}/W$.*

The Roman Numeral Labels match with those of Propositions 3.3.5 and 3.3.8.

The proofs of Propositions 5.3.5 and 5.3.6 similar to those of Propositions 5.3.1 and 5.3.2. We leave the proofs to the reader.

5.3.3 Type E_6

Let X be the Kleinian singularity of type E_6, and $\mathcal{X}$ denote its universal graded Poisson deformation. Recall from Proposition 5.2.8 that we have

$$\mathbb{C}[\mathcal{X}] = \frac{\mathbb{C}[x,y,z,\epsilon_2,\epsilon_5,\epsilon_6,\epsilon_8,\epsilon_9,\epsilon_{12}]}{(-z^2 - zx^2 + y^3 + \epsilon_2 yx^2 + \epsilon_5 yx + \epsilon_6 x^2 + \epsilon_8 y + \epsilon_9 x + \epsilon_{12})}, \tag{5.3.11}$$

with gradings $\deg x = 6,\ \deg y = 8,\ \deg z = 12,\ \deg \epsilon_i = 2i,\ i = 2,5,6,8,9,12$. According to Proposition 5.2.12, the Poisson brackets on $\mathbb{C}[\mathcal{X}]$ are given by

$$\begin{aligned}
\{x,y\} &= -2z - x^2,\\
\{x,z\} &= -3y^2 - \epsilon_2 x^2 - \epsilon_5 x - \epsilon_8,\\
\{y,z\} &= -2xz + 2\epsilon_2 xy + \epsilon_5 y + 2\epsilon_6 x + \epsilon_9,\\
\{x,\epsilon_i\} &= \{y,\epsilon_i\} = \{z,\epsilon_i\} = 0,\ \forall i.
\end{aligned} \tag{5.3.12}$$

The deformation parameter space $\mathfrak{h}/W = \operatorname{Spec}\mathbb{C}[\epsilon_2,\epsilon_5,\epsilon_6,\epsilon_8,\epsilon_9,\epsilon_{12}] \simeq \mathbb{A}^6$. The deformation morphism is the projection.

$$\begin{aligned}
\varphi \colon \mathcal{X} &\to \mathfrak{h}/W\\
(x,y,z,\epsilon_2,\epsilon_5,\epsilon_6,\epsilon_8,\epsilon_9,\epsilon_{12}) &\mapsto (\epsilon_2,\epsilon_5,\epsilon_6,\epsilon_8,\epsilon_9,\epsilon_{12}).
\end{aligned} \tag{5.3.13}$$

Note that we recover the Kleinian singularity of type $E_6 \colon \mathbb{C}[X] \simeq \mathbb{C}[x,y,z]/(-z^2 - zx^2 + y^3)$ by considering the fiber $j \colon X \simeq \varphi^{-1}(0)$. Note that the defining relation of X chosen in this subsection is different from that of Section 3.3.4, particularly with a sign change in front of the term y^3.

Proposition 5.3.7. *The universal graded Poisson deformation $\mathcal{X}$ of Kleinian singularity of Type E_6 has two conjugacy classes of anti-Poisson involutions.*

I. $\Theta(x) = -x,\ \Theta(y) = y,\ \Theta(z) = z,\ \Theta(\epsilon_i) = (-1)^i\epsilon_i,\ i = 2, 5, 6, 8, 9, 12.$

II. $\Theta(x) = x,\ \Theta(y) = y,\ \Theta(z) = -z - x^2,\ \Theta(\epsilon_i) = \epsilon_i,\ i = 2, 5, 6, 8, 9, 12.$

The Roman Numeral Labels match with those of Proposition 3.3.11 under the bijection eq. (5.1.5).

Fix $\lambda = (\epsilon_2, \epsilon_5, \epsilon_6, \epsilon_8, \epsilon_9, \epsilon_{12}) \in \mathfrak{h}/W$, and consider the filtered Poisson deformation

$$X_\lambda = \varphi^{-1}(\lambda) = \operatorname{Spec}\mathbb{C}[x, y, z]/(-z^2 - zx^2 + y^3 + \epsilon_2 yx^2 + \epsilon_5 yx + \epsilon_6 x^2 + \epsilon_8 y + \epsilon_9 x + \epsilon_{12}). \quad (5.3.14)$$

Next, given an anti-Poisson involution $\theta\colon X \to X$ listed in Proposition 3.3.11, we study when does there exist an anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$ such that $\operatorname{gr}\theta_\lambda = \theta$.

Proposition 5.3.8. *Let X be a Kleinian singularity of type E_6 and X_λ a filtered Poisson deformation of X. Let θ be an anti-Poisson involution of X listed in Proposition 3.3.11. We have*

I. Consider the anti-Poisson involution $\theta\colon X \to X$ defined by $x \mapsto -x,\ y \mapsto y,\ z \mapsto z$. Then $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto -x,\ y \mapsto y,\ z \mapsto z$ defines an anti-Poisson involution of X_λ such that $\operatorname{gr}\theta_s = \theta$ if and only if $\lambda \in (\mathfrak{h}/W)^\Theta$, where $(\mathfrak{h}/W)^\Theta = (\epsilon_5 = \epsilon_9 = 0) \subset \mathfrak{h}/W$.

II. Consider the anti-Poisson involution $\theta\colon X \to X$ defined by $x \mapsto x,\ y \mapsto y,\ z \mapsto -z - x^2$. Then $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto x,\ y \mapsto y,\ z \mapsto -z - x^2$ defines an anti-Poisson involution of X_λ for any $\lambda \in \mathfrak{h}/W$.

The Roman Numeral Labels match with those of Proposition 3.3.11.

The proofs of Propositions 5.3.7 and 5.3.8 similar to those of Propositions 5.3.1 and 5.3.2. We leave the proofs to the reader.

5.3.4 Type E_7

Let X be the Kleinian singularity of type E_7, and $\mathcal{X}$ denote its universal graded Poisson deformation. Recall from Proposition 5.2.8 that we have

$$\mathbb{C}[\mathcal{X}] = \frac{\mathbb{C}[x, y, z, \epsilon_2, \epsilon_6, \epsilon_8, \epsilon_{10}, \epsilon_{12}, \epsilon_{14}, \epsilon_{18}]}{(-z^2 - y^3 + 16yx^3 + \epsilon_2 y^2 x + \epsilon_6 y^2 + \epsilon_8 yx + \epsilon_{10} x^2 + \epsilon_{12} y + \epsilon_{14} x + \epsilon_{18})}, \quad (5.3.15)$$

with gradings $\deg x = 8,\ \deg y = 12,\ \deg z = 18,\ \deg \epsilon_i = 2i,\ i = 2, 6, 8, 10, 12, 14, 18$. According to Proposition 5.2.12, the Poisson brackets on $\mathbb{C}[\mathcal{X}]$ are given by

$$\begin{aligned}
\{x, y\} &= -2z, \\
\{x, z\} &= 3y^2 - 16x^3 - 2\epsilon_6 y - \epsilon_8 x - \epsilon_{12}, \\
\{y, z\} &= 48yx^2 + \epsilon_8 y + 2\epsilon_{10} x + \epsilon_{14}, \\
\{x, \epsilon_i\} &= \{y, \epsilon_i\} = \{z, \epsilon_i\} = 0,\ \forall i.
\end{aligned} \tag{5.3.16}$$

The deformation parameter space $\mathfrak{h}/W = \operatorname{Spec}\mathbb{C}[\epsilon_2, \epsilon_6, \epsilon_8, \epsilon_{10}, \epsilon_{12}, \epsilon_{14}, \epsilon_{18}] \simeq \mathbb{A}^7$. The deformation morphism is the projection.

$$\begin{gathered}
\varphi\colon \mathcal{X} \to \mathfrak{h}/W \\
(x, y, z, \epsilon_2, \epsilon_6, \epsilon_8, \epsilon_{10}, \epsilon_{12}, \epsilon_{14}, \epsilon_{18}) \mapsto (\epsilon_2, \epsilon_6, \epsilon_8, \epsilon_{10}, \epsilon_{12}, \epsilon_{14}, \epsilon_{18}).
\end{gathered} \tag{5.3.17}$$

Note that we recover the Kleinian singularity of type E_7: $\mathbb{C}[X] \simeq \mathbb{C}[x, y, z]/(-z^2 - y^3 + 16yx^3)$ by considering the fiber $j\colon X \simeq \varphi^{-1}(0)$. Note that the defining relation of X chosen in this subsection is slightly different from that of Section 3.3.5, particularly the coefficient 16 in front of the term yx^3.

Proposition 5.3.9. *The universal graded Poisson deformation $\mathcal{X}$ of Kleinian singularity of Type E_7 has a unique conjugacy class of anti-Poisson involutions.*

$$\Theta(x) = x,\ \Theta(y) = y,\ \Theta(z) = -z,\ \Theta(\epsilon_i) = \epsilon_i,\ i = 2, 6, 8, 10, 12, 14, 18.$$

Fix $\lambda = (\epsilon_2, \epsilon_6, \epsilon_8, \epsilon_{10}, \epsilon_{12}, \epsilon_{14}, \epsilon_{18}) \in \mathfrak{h}/W$, and consider the filtered Poisson deformation

$$X_\lambda = \varphi^{-1}(\lambda) = \operatorname{Spec}\mathbb{C}[x, y, z]/(-z^2 - y^3 + 16yx^3 + \epsilon_2 y^2 x + \epsilon_6 y^2 + \epsilon_8 yx + \epsilon_{10} x^2 + \epsilon_{12} y + \epsilon_{14} x + \epsilon_{18}). \tag{5.3.18}$$

Next, given the unique anti-Poisson involution $\theta\colon X \to X$ listed in Proposition 3.3.14, we study when does there exist an anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$ such that $\operatorname{gr}\theta_\lambda = \theta$.

Proposition 5.3.10. *Let X be a Kleinian singularity of type E_7 and X_λ a filtered Poisson deformation of X. Let $\theta\colon X \to X,\ x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ be the unique anti-Poisson involution given in Proposition 3.3.14. Then $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ defines an anti-Poisson involution of X_λ for any $\lambda \in \mathfrak{h}/W$.*

The proofs of Propositions 5.3.9 and 5.3.10 similar to those of Propositions 5.3.1 and 5.3.2. We leave the proofs to the reader.

5.3.5 Type E_8

Let X be the Kleinian singularity of type E_8, and $\mathcal{X}$ denote its universal graded Poisson deformation. Recall from Proposition 5.2.8 that we have

$$\mathbb{C}[\mathcal{X}] = \frac{\mathbb{C}[x, y, z, \epsilon_2, \epsilon_8, \epsilon_{12}, \epsilon_{14}, \epsilon_{18}, \epsilon_{20}, \epsilon_{24}, \epsilon_{30}]}{(-z^2 + y^3 - x^5 + \epsilon_2 yx^3 + \epsilon_8 yx^2 + \epsilon_{12} x^3 + \epsilon_{14} yx + \epsilon_{18} x^2 + \epsilon_{20} y + \epsilon_{24} x + \epsilon_{30})}, \tag{5.3.19}$$

with gradings $\deg x = 12,\ \deg y = 20,\ \deg z = 30,\ \deg \epsilon_i = 2i,\ i = 2, 8, 12, 14, 18, 20, 24, 30$. According to Proposition 5.2.12, the Poisson brackets on $\mathbb{C}[\mathcal{X}]$ are given by

$$\begin{aligned}
\{x, y\} &= -2z, \\
\{x, z\} &= -(3y^2 + \epsilon_2 x^3 + \epsilon_8 x^2 + \epsilon_{14} x + \epsilon_{20}), \\
\{y, z\} &= -5x^4 + 3\epsilon_2 yx^2 + 2\epsilon_8 yx + 3\epsilon_{12} x^2 + \epsilon_{14} y + 2\epsilon_{18} x + \epsilon_{24}, \\
\{x, \epsilon_i\} &= \{y, \epsilon_i\} = \{z, \epsilon_i\} = 0,\ \forall i.
\end{aligned} \tag{5.3.20}$$

The deformation parameter space $\mathfrak{h}/W = \operatorname{Spec} \mathbb{C}[\epsilon_2, \epsilon_8, \epsilon_{12}, \epsilon_{14}, \epsilon_{18}, \epsilon_{20}, \epsilon_{24}, \epsilon_{30}] \simeq \mathbb{A}^8$. The deformation morphism is the projection.

$$\begin{gathered}
\varphi\colon \mathcal{X} \to \mathfrak{h}/W \\
(x, y, z, \epsilon_2, \epsilon_8, \epsilon_{12}, \epsilon_{14}, \epsilon_{18}, \epsilon_{20}, \epsilon_{24}, \epsilon_{30}) \mapsto (\epsilon_2, \epsilon_8, \epsilon_{12}, \epsilon_{14}, \epsilon_{18}, \epsilon_{20}, \epsilon_{24}, \epsilon_{30}).
\end{gathered} \tag{5.3.21}$$

Note that we recover the Kleinian singularity of type E_8: $\mathbb{C}[X] \simeq \mathbb{C}[x, y, z]/(-z^2 + y^3 - x^5)$ by considering the fiber $j\colon X \simeq \varphi^{-1}(0)$. Note that the defining relation of X chosen in this subsection is slightly different from that of Section 3.3.6, particularly with a sign change in front of the term y^3.

Proposition 5.3.11. *The universal graded Poisson deformation $\mathcal{X}$ of Kleinian singularity of Type E_8 has a unique conjugacy class of anti-Poisson involutions.*

$$\Theta(x) = x,\ \Theta(y) = y,\ \Theta(z) = -z,\ \Theta(\epsilon_i) = \epsilon_i,\ i = 2, 8, 12, 14, 18, 20, 24, 30.$$

Fix $\lambda = (\epsilon_2, \epsilon_8, \epsilon_{12}, \epsilon_{14}, \epsilon_{18}, \epsilon_{20}, \epsilon_{24}, \epsilon_{30}) \in \mathfrak{h}/W$, and consider the filtered Poisson deformation

$$X_\lambda = \varphi^{-1}(s) = \operatorname{Spec}\mathbb{C}[x,y,z]/(-z^2+y^3-x^5+\epsilon_2 yx^3+\epsilon_8 yx^2+\epsilon_{12}x^3+\epsilon_{14}yx+\epsilon_{18}x^2+\epsilon_{20}y+\epsilon_{24}x+\epsilon_{30}). \tag{5.3.22}$$

Next, given the unique anti-Poisson involution $\theta\colon X \to X$ listed in Proposition 3.3.17, we study when does there exist an anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$ such that $\operatorname{gr}\theta_\lambda = \theta$.

Proposition 5.3.12. *Let X be a Kleinian singularity of type E_8 and X_λ a filtered Poisson deformation of X. Let $\theta\colon X \to X,\ x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ be the unique anti-Poisson involution given in Proposition 3.3.17. Then $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda],\ x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ defines an anti-Poisson involution of X_λ for any $\lambda \in \mathfrak{h}/W$.*

The proofs of Propositions 5.3.11 and 5.3.12 similar to those of Propositions 5.3.1 and 5.3.2. We leave the proofs to the reader.

5.4 Restricting anti-Poisson involutions to singular points in filtered Poisson deformations

Let X be a Kleinain singularity, and X_λ a filtered Poisson deformation of X. Then X_λ have finitely many singular points, and the completion of X_λ at each singular point p, which we denote by $\widehat{X}_{\lambda,p}$, is isomorphic to the completion of a Klenian singularity at the unique singular point 0 as Poisson varieties. There is a Lie-theoretic way to determine the type of the singularity $\widehat{X}_{\lambda,p}$ in [Slo80, Section 6.5]. We will review this story in Section 5.4.1. We will also give a more direct way to determine the type of the neighboring singularities for type A in Proposition 5.4.7 and for type D in Proposition 5.4.12.

For an anti-Poisson involution $\theta\colon X \to X$, let $\Theta\colon \mathcal{X} \to \mathcal{X}$ denote its UPD-lift (c.f. Propositions 5.3.1, 5.3.3, 5.3.5, 5.3.7, 5.3.9 and 5.3.11). Assume $\lambda \in (\mathfrak{h}/W)^\Theta$, then Θ restricts to an anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$ (c.f. Propositions 5.3.2, 5.3.4, 5.3.5, 5.3.8, 5.3.10 and 5.3.12). Let p be a singular point of X_λ. Then θ_λ will either preserve p, or swap p with another singular points as θ_λ is an involution. Let us assume θ_λ preserves p. Then we can consider the restriction of θ_λ to the formal neighborhood (i.e. completion)

$$\widehat{\theta}_{\lambda,p}\colon \widehat{X}_{\lambda,p} \to \widehat{X}_{\lambda,p}. \tag{5.4.1}$$

Then $\widehat{\theta}_{\lambda,p}$ is an anti-Poisson involution of the Kleinian singularity $\widehat{X}_{\lambda,p}$, and we would like to determine the type of $\widehat{\theta}_{\lambda,p}$ in terms of the classifications in Section 3.3.

Consider the triple (X_λ, ι, p), where X_λ is a filtered deformation of a Kleinian singularity, ι is the Roman Numeral encoding the Type of an anti-Poisson involution of the universal graded Poisson deformation $\Theta\colon \mathcal{X} \to \mathcal{X}$, and p is a singular point of X_λ.

Definition 5.4.1. We call the triple (X_λ, ι, p) an *Involution-Singularity Triple*, abbreviated as IST, if they satisfy:

(1) $\lambda \in (\mathfrak{h}/W)^\Theta$, so that Θ restricts to an anti-Poisson involution $\theta_\lambda\colon X_\lambda \to X_\lambda$.

(2) p is a singular point of X_λ that is fixed by θ_λ.

Consider the pair (Q, ι'), where Q encodes the type of a Kleinian singularity of type ADE, and ι' is the Roman Numeral of an anti-Poisson involution of the underlying Kleinian singularity.

Definition 5.4.2. We say the Involution-Singularity Triple (IST) (X_λ, ι, p) is of type (Q, ι') and we write $(X_\lambda, \iota, p) \sim (Q, \iota')$ if

(1) The completion $\widehat{X}_{\lambda,p}$ is isomorphic to the completion of a Kleinain singularity of type Q. (In what follows, we abbreviate this by saying that "the completion $\widehat{X}_{\lambda,p}$ is a Kleinain singularity of type Q".)

(2) The restriction of the anti-Poisson involution θ_λ to the formal neighborhood $\widehat{X}_{\lambda,p}$ is an anti-Poisson involution of Type ι' for the singularity Q.

Example 5.4.3. (1) Recall that when $\lambda = 0$, we have $X_0 \simeq X$ is a Kleinian singularity with the unique singularity at 0. We have $(X_0, \iota, 0) \sim (Q, \iota)$, where Q is the corresponding type of the Kleinian singularity X.

(2) Consider $X_\lambda = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - (z-1)^2(z+2))$, which is a deformation of Kleinian singularity of type A_2. Let θ_λ be the Type I anti-Poisson involution of X_λ, i.e., θ_λ swaps x and y (c.f. Proposition 5.3.4).

Let $p = (0, 0, 1)$ denote the singular point of X_λ. It is easy to see that the completion $\widehat{X}_{\lambda,p}$ is a Kleinian singularity of type A_1, and the restriction of θ_λ to a formal neighborhood of p is the Type I anti-Poisson involution of A_1 Kleinain singularity (c.f. Proposition 3.3.1). Therefore we have $(X_\lambda, \mathrm{I}, p) \sim (A_1, \mathrm{I})$.

For an Involution-Singularity Triple (X_λ, ι, p), we have a complete solution to determine the type (Q, ι') for type A (c.f. Propositions 5.4.8 and 5.4.9) and type D (c.f. Proposition 5.4.13).

5.4.1 Singularities of filtered Poisson deformations: Lie-theoretic approach

We describe a Lie-theoretic approach to determine the type of the singularity $\widehat{X}_{\lambda,p}$ in this subsection, following [Slo80, Section 6.5]. In other words, how to read out the type Q in the pair (Q, ι') in terms of λ and p in the triple (X_λ, ι, p), assuming $(X_\lambda, \iota, p) \sim (Q, \iota')$.

Let $\mathfrak{g}$ be a simple Lie algebra of types ADE, and Δ denote its Dynkin diagram. Let X be the corresponding Kleinian singularity, and $\varphi : S \to \mathfrak{h}/W$ the universal graded Poisson deformation X, as in Section 5.2. We identified the singularities of the fiber $\varphi^{-1}(0) = S \cap \mathcal{N} \simeq X$ as Kleinain singularities of the type associated to $\mathfrak{g}$. Now we consider the singularities of the other fibers of φ, i.e., the filtered Poisson deformations $X_\lambda = \varphi^{-1}(\lambda)$, which are also normal and two-dimensional.

Let us write $\mathcal{X}_\mathfrak{h} := \mathcal{X} \times_{\mathfrak{h}/W} \mathfrak{h}$ and X_t for the fiber of $\mathcal{X}_\mathfrak{h} \to \mathfrak{h}$ over t. Note that $t \in \mathfrak{h},\ \lambda \in \mathfrak{h}/W$. We use the following convention.

Convention 5.4.4. Let the Dynkin diagram of a reductive Lie algebra be that of its semisimple commutator.

Proposition 5.4.5. *[Slo80, Lemma 6.5.1, Proposition 6.6.2] Let $t \in \mathfrak{h}$. Let $\{p_1, \cdots, p_m\}$ denote the set of singular points of the filtered Poisson deformation X_t. Let $\Delta(t) = \Delta_1 \cup \ldots \Delta_{m'}$ be the decomposition of the Dynkin diagram of the centralizer $Z_\mathfrak{g}(t)$ into connected components. Then $m' = m$, and there exists a bijection*

$$\Delta_j \mapsto p_j,\ 1 \leqslant j \leqslant m$$

from the connected components of $\Delta(t)$ to the singular points of X_t such that the completion of $\widehat{X}_{t,p_j}$ is a Kleinian singularity of type Δ_j. We say that X_t has the singular configuration $\Delta(t)$.

Corollary 5.4.6. *[Slo80, Corollary 8.10.2] X_λ, for some λ, has singular configuration Δ' if and only if Δ' is a subdiagram of the Dynkin diagram Δ of $\mathfrak{g}$.*

5.4.2 Type A

We study deformations and restrictions of anti-Poisson involutions in type A_n in this subsection. Recall from Remark 5.2.9 that we fix a basis of the Cartan space $\mathfrak{h} = \{(t_0, t_1, \cdots, t_n) \in \mathbb{C}^{n+1} | \sum_{i=0}^n t_i = 0\}$. The Weyl group $W \simeq S_{n+1}$ acts by permuting factors. We have $\mathbb{C}[\mathfrak{h}]^W = \mathbb{C}[s_2, \cdots, s_{n+1}]$, where $s_i = i$-th elementary symmetric polynomials in $t_0, \cdots, t_n$.

Recall from Proposition 5.2.8 that a filtered Poisson deformation of type A_n Kleinian singularity is given by $X_\lambda = \varphi^{-1}(\lambda) = \operatorname{Spec} \mathbb{C}[x, y, z]/(-xy + z^{n+1} + \sum_{i=2}^n s_i z^{n+1-i})$ with $\lambda = (s_2, \cdots, s_{n+1}) \in \mathfrak{h}/W$.

Let $t = (t_0, \cdots, t_n)$ be a preimage of λ under the quotient $\mathfrak{h} \to \mathfrak{h}/W$. Define

$$f_t(z) := \prod_{i=0}^{n}(z + t_i). \tag{5.4.2}$$

Then we have

$$f_t(z) = z^{n+1} + \sum_{i=2}^{n} s_i z^{n+1-i}.$$

using that $s_i = i$-th symmetric function on $t_0, \cdots, t_n$, and the fact that $\sum_{i=0}^{n} t_i = 0$. It follows that $X_t = \operatorname{Spec}\mathbb{C}[x, y, z]/(-xy + \prod_{i=0}^{n}(z + t_i)) = \operatorname{Spec}\mathbb{C}[x, y, z]/(-xy + f_t(z))$, where X_t is the fiber of $\mathcal{X}_{\mathfrak{h}} = \mathcal{X} \times_{\mathfrak{h}/W} \mathfrak{h} \to \mathfrak{h}$ over t, defined in Section 5.4.1.

If $r \in \mathbb{C}$ is a root of $f_t(z)$, we denote the *multiplicity* of $f_t(z)$ at r by $\operatorname{mult}_{f_t}(r)$. It is clear that the roots of $f_t(z)$ are $-t_0, \cdots, -t_n$, counted with multiplicities. We set $\operatorname{mult}_{f_t}(r) = 0$ if r is not a root of $f_t(z)$.

Proposition 5.4.7. *The singular points of the filtered Poisson deformation* $X_t = \operatorname{Spec}\mathbb{C}[x, y, z]/(-xy + f_t(z)) = \operatorname{Spec}\mathbb{C}[x, y, z]/(-xy + \prod_{i=0}^{n}(z + t_i))$ *are*

$$p = (0, 0, -t_i), \textit{ for those } t_i \textit{ with} \operatorname{mult}_{f_t}(-t_i) > 1.$$

Proof. The Jacobi criterion implies that the singular locus of X_t is defined by the ideal

$$(-y, -x, f_t'(z), f_t(z)),$$

which gives the claimed set of points because the common roots of $f_t'(z), f_t(z)$ are the roots of $f_t(z)$ with multiplicity greater than 1.

Take a singular point $p = (0, 0, -t_i)$. We have

$$\mathbb{C}[\widehat{X}_{t,p}] = \frac{\mathbb{C}[[x, y, z + t_i]]}{(xy - f_t(z))} \simeq \frac{\mathbb{C}[[x, y, z + t_i]]}{(xy - (z + t_i)^{\operatorname{mult}_{f_t}(-t_i)})} \simeq \frac{\mathbb{C}[[x, y, z]]}{(xy - z^{\operatorname{mult}_{f_t}(-t_i)})}, \tag{5.4.3}$$

where the first isomorphism follows from the fact that $z + t_j$ is invertible in the completion $\widehat{X}_{t,p}$ if $j \neq i$. The last ring is manifestly the completion of the coordinate ring of the Kleinian singularity of type $A_{\operatorname{mult}_{f_t}(-t_i)-1}$ at the unique singular point 0 (c.f. eq. (5.2.1)). $\square$

Recall the Involution-Singularity Triple (IST) (X_λ, ι, p) that we defined in Definition 5.4.1, where ι encodes the type of an anti-Poisson involution θ_λ of X_λ, and p is a singular point of X_λ that is being fixed by θ_λ. Recall the discussion at the beginning of Section 5.4, the goal of this section is to determine the type (Q, ι') of the IST (X_λ, ι, p); that is, to find a pair (Q, ι'), where Q is the type of a Kleinian singularity of type ADE, and ι' is the Roman Numeral encoding the Type of the restriction of the anti-Poisson involution θ_λ to the formal neighborhood $\widehat{X}_{\lambda,p}$ (c.f. Definition 5.4.2).

Let $t = (t_0, t_1, \cdots, t_n)$ be a preimage of λ under the quotient $\mathfrak{h} \to \mathfrak{h}/W$. We have $\lambda = (s_2, \cdots, s_{n+1}) \in \mathfrak{h}/W$, where s_j is the j-th symmetric polynomial on $t = (t_0, t_1, \cdots, t_n) \in \mathfrak{h}$. We write $\lambda = \bar{t}$ to denote that λ is the image of t under the quotient $\mathfrak{h} \to \mathfrak{h}/W$. We have $X_\lambda \simeq X_t = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - f_t(z))$. Proposition 5.4.7 tells us how to determine the singularity Q in terms of λ and p. Propositions 5.4.8 and 5.4.9 will tell us how to determine the anti-Poisson involution Type ι' in terms of the IST (X_t, ι, p).

Proposition 5.4.8. *Consider $X_\lambda \simeq X_t = \operatorname{Spec} \mathbb{C}[x, y, z]/(-xy + f_t(z))$, where $f_t(z) = \prod_{i=0}^{n}(x + t_i)$, a filtered deformation of Kleinian singularity of type A_n, with n odd. Take t_i with $\operatorname{mult}_{f_t}(-t_i) > 1$ and let $p = (0, 0, -t_i)$ be a singular point of X_t.*

(i) The triple (X_t, I, p) is an IST for any $\lambda = \bar{t} \in \mathfrak{h}/W$. Moreover, we have $(X_t, \mathrm{I}, p) \sim (A_{\operatorname{mult}_{f_t}(-t_i)-1}, \mathrm{I})$.

(ii) The triple (X_t, II, p) is an IST if and only if $p = (0, 0, 0)$ and $s_j(t) = 0$ for j odd and $j = n + 1$. (It follows that $\operatorname{mult}_{f_t}(0)$ is even.) Moreover, we have $(X_t, \mathrm{II}, 0) \sim (A_{\operatorname{mult}_{f_t}(0)-1}, \mathrm{II})$.

(iii) The triple (X_t, III, p) is an IST if and only if $p = (0, 0, 0)$ and $s_j(t) = 0$ for j odd and $j = n + 1$. (It follows that $\operatorname{mult}_{f_t}(0)$ is even.) Moreover, we have $(X_t, \mathrm{III}, 0) \sim (A_{\operatorname{mult}_{f_t}(0)-1}, \mathrm{III})$.

The Roman Numerals in the triples match with those of Proposition 5.3.2, and the Roman Numerals in the pairs match with those of Propositions 3.3.1 and 3.3.2.

Proof. Recall the three types of anti-Poisson involutions I-III of $\mathbb{C}[X_t] = \mathbb{C}[x, y, z]/(xy - f_t(z))$ given in Proposition 5.3.2.

(i) Type I. $\theta_\lambda \colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda]$ $x \mapsto y,\ y \mapsto x,\ z \mapsto z$ defines an anti-Poisson involution for any $\lambda \in \mathfrak{h}/W$ by Proposition 5.3.2. Moreover, it is clear that $p = (0, 0, -t_i)$ is being fixed by θ_λ. Thus, (X_t, I, p) is an IST by Definition 5.4.1.

According to Proposition 5.4.7, we have $\mathbb{C}[\widehat{X}_{t,p}] \simeq \mathbb{C}[[x, y, z]]/(xy - z^{\operatorname{mult}_{f_t}(-t_i)})$ is the singularity of type $A_{\operatorname{mult}_{f_t}(-t_i)-1}$ and one can compute that the restriction $\widehat{\theta}_{\lambda,p} \colon \mathbb{C}[\widehat{X}_{t,p}] \to \mathbb{C}[\widehat{X}_{t,p}]$ is given by

$$x \mapsto y,\ y \mapsto x,\ z \mapsto z. \tag{5.4.4}$$

which is the Type I anti-Poisson involution of $A_{\mathrm{mult}_{f_t}(-t_i)-1}$ singularity (c.f. Propositions 3.3.1 and 3.3.2).

(ii) Type II. $\theta_\lambda \colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda]$ $x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ defines an anti-Poisson involution if and only if $s_j(t) = 0$ for j odd, where s_j is the j-th symmetric polynomial on $t = (t_0, t_1, \cdots, t_n)$. Note that $p = (0, 0, -t_i)$ is being fixed by θ_λ if and only if $t_i = 0$, which is equivalent to saying that $s_{n+1} = t_0 t_1 \cdots t_{n+1} = 0$. Note that $s_j(t) = 0$ for j odd implies that $s_n(t) = 0$ as n is odd. Furthermore, $p = (0, 0, 0)$ is singular if and and only if $\mathrm{mult}_{f_t}(0) > 1$, which is equivalent to $s_n(t) = s_{n+1}(t) = 0$. Note that n is odd, so $s_j(t) = 0$ for j odd implies that $s_n(t) = 0$. Therefore, (X_t, II, p) is an IST if and only if $p = (0, 0, 0)$ and $s_j(t) = 0$ for j odd and $j = n + 1$. Moreover, it follows from that $\mathrm{mult}_{f_t}(0)$ must be even.

According to Proposition 5.4.7, we have $\mathbb{C}[\widehat{X}_{t,0}] \simeq \mathbb{C}[[x, y, z]]/(xy - z^{\mathrm{mult}_{f_t}(0)})$ is the singularity of type $A_{\mathrm{mult}_{f_t}(0)-1}$ and one can compute that the restriction $\widehat{\theta}_{\lambda,0} \colon \mathbb{C}[\widehat{X}_{t,p}] \to \mathbb{C}[\widehat{X}_{t,p}]$ is given by

$$x \mapsto y,\ y \mapsto x,\ z \mapsto -z. \tag{5.4.5}$$

which is the Type II anti-Poisson involution of $A_{\mathrm{mult}_{f_t}(0)-1}$ singularity (c.f. Proposition 3.3.1).

(iii) Type III. $\theta_\lambda \colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda]$ $x \mapsto -x,\ y \mapsto -y,\ z \mapsto -z$. The proof is similar to that of Type II anti-Poisson involution in (ii) above, and we leave it to the reader.

$\square$

Proposition 5.4.9. *Consider $X_\lambda \simeq X_t = \operatorname{Spec} \mathbb{C}[x, y, z]/(-xy + f_t(z))$, where $f_t(z) = \prod_{i=0}^{n}(x + t_i)$, a filtered deformation of Kleinian singularity of type A_n, with n even. Take t_i with $\mathrm{mult}_{f_t}(-t_i) > 1$ and let $p = (0, 0, -t_i) \in X_t$ be a singular point of X_t.*

(1) The triple (X_t, I, p) is an IST for any $\lambda = \bar{t} \in \mathfrak{h}$. Moreover, we have $(X_t, \mathrm{I}, p) \sim (A_{\mathrm{mult}_{f_t}(-t_i)-1}, \mathrm{I})$.

(2) The triple (X_t, II, p) is an IST if and only if $p = 0$ and $s_j(t) = 0$ for j odd and $j = n$. (It follows that $\mathrm{mult}_{f_t}(0)$ is odd.) Moreover, we have $(X_t, \mathrm{II}, 0) \sim (A_{\mathrm{mult}_{f_t}(0)-1}, \mathrm{II})$.

The Roman Numerals in the triples match with those of Proposition 5.3.4, and the Roman Numerals in the pairs match with those of Propositions 3.3.1 and 3.3.2.

Proof. The proof is similar to that of Proposition 5.4.8, and we leave it to the reader. $\square$

5.4.3 Type D

We study deformations and restrictions of anti-Poisson involutions in type D_n in this subsection. Recall from Remark 5.2.9 that we fix a basis of the Cartan space $\mathfrak{h} = \{(t_1, \cdots, t_n, -t_1, \cdots, -t_n)\} \simeq \mathbb{C}^n$. The Weyl group $W = (\mathbb{Z}/2\mathbb{Z})^{n-1} \rtimes S_n$. The symmetric group S_n acts by permutations on $\{\lambda_1, \cdots, \lambda_n\}$ We have $(\mathbb{Z}/2\mathbb{Z})^{n-1} = \oplus_{i=1}^{n-1}\mathbb{Z}\tau_i$ with $\tau_i^2 = \mathrm{id}$. The generator τ_i sends $t_i \mapsto -t_i,\ t_n \mapsto -t_n$, and fixes t_j for $j \neq i, n$. We have $\mathbb{C}[\mathfrak{h}]^W = \mathbb{C}[\delta_2, \delta_4, \cdots, \delta_{2n-2}, \gamma_n]$, where $\delta_i = i$-the symmetric polynomial in $t_1^2, \cdots, t_n^2$, and $\gamma_n = t_1 \cdots t_n$. Note that $\delta_{2n} = \gamma_n^2$.

Recall from Proposition 5.2.8 that a filtered Poisson deformation of type D_n Kleinian singularity is given by $X_\lambda = \varphi^{-1}(\lambda) = \operatorname{Spec}\mathbb{C}[x,y,z]/(z^2 + xy^2 - x^{n-1} - \sum_{i=1}^{n-1}\delta_{2i}x^{n-i-1} + 2\gamma_n y)$ with $\lambda = (\delta_2, \delta_4 \cdots, \delta_{2n-2}, \gamma_n) \in \mathfrak{h}/W$. Let $t = (t_1, \cdots, t_n)$ be a preimage of λ under the quotient $\mathfrak{h} \to \mathfrak{h}/W$. Define

$$g_t(x) := \prod_{i=1}^{n}(x + t_i^2), \tag{5.4.6}$$

and

$$h_t(x) := x^{n-1} + \sum_{i=1}^{n-1}\delta_{2i}x^{n-i-1}. \tag{5.4.7}$$

We then have

$$X_t = \operatorname{Spec}\mathbb{C}[x,y,z]/(z^2 + xy^2 - h_t(x) + 2\gamma_n y),$$

where X_t is the fiber of $\mathcal{X}_{\mathfrak{h}} = \mathcal{X} \times_{\mathfrak{h}/W} \mathfrak{h} \to \mathfrak{h}$ over t, defined in Section 5.4.1. We also have

$$g_t(x) = xh_t(x) + \gamma_n^2, \tag{5.4.8}$$

using that $\delta_i = i$-th symmetric function on $t_0^2, \cdots, t_n^2$, and $\gamma_n = t_1 \cdots t_n,\ \delta_{2n} = \gamma_n^2$.

In Proposition 5.4.12, we determine the type of singularities of X_t. Before going there, we need the following Lemmas 5.4.10 and 5.4.11.

Lemma 5.4.10. $X = \operatorname{Spec}\mathbb{C}[x,y,z]/(z^2 + y^2 - x^{n+1})$ *is a Kleinian singularity of type* A_n. *Moreover,* $\theta\colon \mathbb{C}[X] \to \mathbb{C}[X]$

$$x \mapsto x,\ y \mapsto y, z \mapsto -z \tag{5.4.9}$$

is the Type I *anti-Poisson involution.*

Proof. Consider the change of coordinates $x' = y - iz,\ y' = y + iz,\ z' = x$, then we obtain

$$\mathbb{C}[X] = \mathbb{C}[x, y, z]/(z^2 + y^2 - x^{n+1}) = \mathbb{C}[x', y', z']/(x'y' - (z')^{n+1}),$$

which is the Kleinian singularity of type A_n. Moreover, the anti-Poisson involution θ given in eq. (5.4.9) has been transformed into

$$x' \mapsto y',\ y' \mapsto x', z' \mapsto z',$$

which is exactly the Type I anti-Poisson involution in type A_n, as listed in Propositions 3.3.1 and 3.3.2.

$\square$

Recall from eq. (5.2.1) that the defining equation of type $D_n,\ n \geqslant 4$ singularity is $z^2 + xy^2 - x^{n-1}$. We can also consider type D_3 singularity with defining equation $z^2 + xy^2 - x^2$ and the corresponding anti-Poisson involutions given in Proposition 5.3.5. In fact, the D_3 singularity is isomorphic to the A_3 singularity, and the Roman Numeral Labels in their classification of anti-Poisson involutions also match, due to Lemma 5.4.11 below. Hence, we adopt the convention $D_3 = A_3$ when considering Kleinian singularities and anti-Poisson involutions.

Lemma 5.4.11. *The type D_3 Kleinian singularity $X = \operatorname{Spec} \mathbb{C}[x, y, z]/(z^2 + xy^2 - x^2)$ is isomorphic to the singularity of type A_3. Moreover,*

(1) The Type I *anti-Poisson involution $\theta\colon \mathbb{C}[X] \to \mathbb{C}[X],\ x \mapsto x,\ y \mapsto y,\ z \to -z$ of X (considered as D_3 singularity) is also of Type* I *when consider X as A_3 singularity.*

(2) The Type II *anti-Poisson involution $\theta\colon \mathbb{C}[X] \mapsto \mathbb{C}[X],\ x \mapsto x,\ y \mapsto -y,\ z \mapsto z$ of X (considered as D_3 singularity) is also of Type* II *when consider X as A_3 singularity.*

Proof. Consider the change of coordinates $x' = (x - \frac{y^2}{2}) - z,\ y' = (x - \frac{y^2}{2}) + z,\ z' = \frac{y}{\sqrt{2}}$, then we have

$$z^2 + xy^2 - x^2 = z^2 - (x - \frac{y^2}{2})^2 + \frac{y^4}{4} = -(x - \frac{y^2}{2} - z)(x + \frac{y^2}{2} + z) + \frac{y^4}{4} = -x'y' + (z')^4. \tag{5.4.10}$$

Thus we have $\mathbb{C}[X] = \mathbb{C}[x, y, z]/(z^2 + xy^2 - x^3) = \mathbb{C}[x', y', z']/(-x'y' + (z')^4)$ is the Kleinian singularity of type A_3, according to the defining equation of type A_n Kleinian singularity given in eq. (5.2.1). Moreover,

I The Type I anti-Poisson involution $\theta\colon \mathbb{C}[X] \mapsto \mathbb{C}[X],\ x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ has been transformed into $x' \mapsto y',\ y' \mapsto x',\ z \mapsto z'$ under the new coordinates, which is the Type I anti-Poisson involution of A_3 singularity (c.f. Proposition 3.3.1).

II The Type II anti-Poisson involution $\theta\colon \mathbb{C}[X] \mapsto \mathbb{C}[X],\ x \mapsto x,\ y \mapsto -y,\ z \mapsto z$ has been transformed into $x' \mapsto x',\ y' \mapsto y',\ z \mapsto -z'$ under the new coordinates, which is the Type I anti-Poisson involution of A_3 singularity (c.f. Proposition 3.3.1).

□

Proposition 5.4.12. *Let $X_t = \operatorname{Spec} \mathbb{C}[x,y,z]/(z^2+xy^2-h_t(x)+2\gamma_n y)$ be a filtered Poisson deformation of type D_n Kleinain singularity, where $h_t(x) = x^{n-1}+\sum_{i=1}^{n-1} \delta_{2i} x^{n-i-1}$, and consider $g_t(x) = \prod_{i=1}^{n}(x+t_i^2)$. Then X_t has three types of singularities.*

(1) For each i with $t_i \neq 0$ and $\operatorname{mult}_{g_t}(-t_i^2) > 1$, the point $p = (-t_i^2, -\frac{\gamma_n}{t_i^2}, 0)$ is singular. The completion $\widehat{X}_{t,p}$ is a Kleinain singularity of type $A_{\operatorname{mult}_{g_t}(-t_i^2)-1}$.

(2) If $\gamma_n = \delta_{2n-2} = 0,\ \delta_{2n-4} \neq 0$, the points $q_\pm = \pm(0, \sqrt{\delta_{2n-4}}, 0)$ are singular. The completions $\widehat{X}_{t,q_\pm}$ are Kleinain singularities of type A_1.

(3) If $\gamma_n = \delta_{2n-2} = \delta_{2n-4} = 0$, the point $(0,0,0)$ is singular. The completion $\widehat{X}_{t,0}$ is a Kleinain singlarity of type $D_{\operatorname{mult}_{g_t}(0)}$, with the convention $D_3 = A_3$.

Proof. The Jacobi criterion implies that the singular locus of X_t are solutions to the following system of equations eqs. (5.4.11) to (5.4.14).

$$y^2 - h_t'(x) = 0, \tag{5.4.11}$$

$$2xy + 2\gamma_n = 0, \tag{5.4.12}$$

$$2z = 0, \tag{5.4.13}$$

$$xy^2 - h_t(x) + 2\gamma_n y = 0. \tag{5.4.14}$$

It follows from eq. (5.4.13) that the z-coordinate of a singular point has to be zero. Let $p = (x_p, y_p, 0) \in X_t$ be a singular point. To determine x_p and y_p, we consider the cases $\gamma_n \neq 0$ and $\gamma_n = 0$ separately.

If $\gamma_n \neq 0$, we have $x_p \neq 0,\ y_p \neq 0$ and $y_p = -\frac{\gamma_n}{x_p}$ by eq. (5.4.12). Plugging this into eqs. (5.4.11) and (5.4.14) and upon simplificaition, we obtain that

$$\begin{aligned} \frac{\gamma_n^2}{x_p^2} - h_t'(x_p) = 0, \\ -\frac{\gamma_n^2}{x_p} - h_t(x_p) = 0. \end{aligned} \tag{5.4.15}$$

Recall from eq. (5.4.8) that $g_t(x) = xh_t(x) + \gamma_n^2$. It follows from eq. (5.4.15) that

$$\begin{aligned} g_t(x_p) &= x_p h_t(x_p) + \gamma_n^2 = 0, \\ g_t'(x_p) &= x_p h_t'(x) + h_t(x_p) = 0. \end{aligned} \tag{5.4.16}$$

It follows that $0 \neq x_p$ is a root of $g_t(x)$ with multiplicity greater than 1. Hence

$$x_p = -t_i^2 \text{ for some } i \text{ with } t_i \neq 0 \text{ and } \operatorname{mult}_{g_t}(-t_i^2) > 1.$$

Therefore, $p = (-t_i^2, -\frac{\gamma_n}{t_i^2}, 0)$, which are exactly the claimed points in (1) in the case of $\gamma_n \neq 0$.

If $\gamma_n = 0$, we have $X_t = \operatorname{Spec}\mathbb{C}[x,y,z]/(z^2 + xy^2 - h_t(x))$. The equations eqs. (5.4.11), (5.4.12) and (5.4.14) are simplified into

$$y^2 - h_t'(x) = 0, \tag{5.4.17}$$

$$2xy = 0, \tag{5.4.18}$$

$$xy^2 - h_t(x) = 0. \tag{5.4.19}$$

If $p = (x_p, y_p, 0)$ is a singular point, it follows from eq. (5.4.18) that either $x_p = 0$ or $y_p = 0$.

(i) If $y_p = 0$, it follows from eqs. (5.4.17) and (5.4.19) that $h_t'(x_p) = h_t(x_p) = 0$, i.e. x_p is a root of $h_t(x)$ with multiplicity greater than 1. Note that we have $g_t(x) = xh_t(x)$ by eq. (5.4.15) as $\gamma_n = 0$. There are two possibilities.

- $x_p \neq 0$. It follows that x_p is a nonzero of $g_t(x)$, i.e. $x_p = -t_i^2$ for some i with $t_i \neq 0$ and $\operatorname{mult}_{g_t}(-t_i^2) > 1$. Thus $p = (-t_i^2, 0, 0)$, which are exactly the claimed points in (1) in the case of $\gamma_n = 0$.
- $x_p = 0$. Then we must have $h_t(0) = \delta_{2n-2} = 0$ and $h_t'(0) = \delta_{2n-4} = 0$. This gives rise to the singular point $(0,0,0)$ in (3).

(ii) If $y_p \neq 0$, then we must have $x_p = 0$. Then it follows from eq. (5.4.19) that $h_t(0) = \delta_{2n-2} = 0$, and it follows from eq. (5.4.17) that $y_p^2 = h_t'(0) = \delta_{2n-4}$. This gives rise to the singular point $q_\pm = \pm(0, \sqrt{\delta_{2n-4}}, 0)$ in (2) if $\delta_{2n-4} \neq 0$, and gives rise to the singular point $(0,0,0)$ in in (3) if $\delta_{2n-4} = 0$.

Conversely, one can verify that the claimed points in (1)-(3) satisfy all equations in the system eqs. (5.4.11) to (5.4.14) with a similar process, so they are indeed singular. Next, we determine the type of the completions at each case of the singularities.

(1) Consider $p = (-t_i^2, -\frac{\gamma_n}{t_i^2}, 0)$ with $t_i \neq 0$. In a formal neighborhood of p, the function x is invertible. Therefore, by eq. (5.4.8), we have $h_t(x) = \frac{g_t(x)-\gamma_n^2}{x}$ as functions on the completion $\widehat{X}_{t,p}$. Hence

$$\begin{aligned}\mathbb{C}[\widehat{X}_{t,p}] =&\frac{\mathbb{C}[[x+t_i^2, y+\frac{\gamma_n}{t_i^2}, z]]}{(z^2+xy^2-\frac{g_t(x)-\gamma_n^2}{x}+2\gamma_n y)} = \frac{\mathbb{C}[[x+t_i^2, y+\frac{\gamma_n}{t_i^2}, z]]}{(\frac{z^2}{x}-\frac{g_t(x)}{x^2}+(y+\frac{\gamma_n}{x})^2)} \\ \simeq&\frac{\mathbb{C}[[x+t_i^2, y+\frac{\gamma_n}{t_i^2}, z]]}{(z^2-(x+t_i^2)^{\mathrm{mult}_{g_t}(-t_i^2)}+(y+\frac{\gamma_n}{x})^2)} \simeq \frac{\mathbb{C}[[x,y,z]]}{(z^2+y^2-x^{\mathrm{mult}_{g_t}(-t_i^2)})},\end{aligned} \tag{5.4.20}$$

where the first isomorphism in the second line follows from the definition $g_t(x) = \prod_{i=1}^{n}(x+t_i^2)$ and the fact that $x+t_j^2$ is invertible in the completion if $j \neq i$. The last ring is the completion of the coordinate ring of Kleinain singularity of type $A_{\mathrm{mult}_{g_t}(-t_i^2)-1}$ at the singular point $(0,0,0)$, according to Lemma 5.4.10.

(2) Assume $\gamma_n = \delta_{2n-2} = 0, \delta_{2n-4} \neq 0$. Then we have $x \mid h_t(x)$, but $x^2 \mid h_t(x)$ as $h_t(x) = x^{n-1} + \sum_{i=1}^{n-1}\delta_{2i}x^{n-i-1}$. However, $x^2 \mid h_t(x) - \delta_{2n-4}x$. Set $k_t(x) := \frac{h_t(x)-\delta_{2n-4}x}{x^2}$. Consider the point $q_+ = (0, \sqrt{\delta_{2n-4}}, 0)$. Note that the function $y + \sqrt{\delta_{2n-4}}$ is invertible on the completion $\widehat{X}_{t,q_+}$. We have

$$\begin{aligned}\mathbb{C}[\widehat{X}_{t,q_+}] =&\frac{\mathbb{C}[[x, y-\sqrt{\delta_{2n-4}}, z]]}{(z^2+xy^2-h_t(x))} = \frac{\mathbb{C}[[x, y-\sqrt{\delta_{2n-4}}, z]]}{(z^2+x(y^2-\delta_{2n-4})-x^2k_t(x))} \\ \simeq&\frac{\mathbb{C}[[x, y-\sqrt{\delta_{2n-4}}, z]]}{(z^2+x(y-\sqrt{\delta_{2n-4}})-x^2k_t(x))} \simeq \frac{\mathbb{C}[[x,y,z]]}{(z^2+xy-x^2k_t(x))} \\ =&\frac{\mathbb{C}[[x,y,z]]}{(z^2+x(y-xk_t(x))} \simeq \frac{\mathbb{C}[[x,y,z]]}{(z^2+xy)},\end{aligned} \tag{5.4.21}$$

where the isomorphism in the second line comes from dividing the equation in the denominator the invertible element $y + \sqrt{\delta_{2n-4}}$, and the isomorphism in the third line comes from the change of coordinates $y \mapsto y - xk_t(x)$. (Note that the x in the term $xk_t(x)$ is necessary to guarantee that it vanishes at the point $(0,0,0)$, so that this change of coordinates do not shift the center of the power series.) The last ring is the completion of the coordinate ring of Kleinain singularity of type A_1 at the singular point $(0,0,0)$. Similarly, for the singular point $q_- = (0, -\sqrt{\delta_{2n-4}}, 0)$, one can show that $\widehat{X}_{t,q_-}$ is also of type A_1.

(3) Assume $\gamma_n = \delta_{2n-2} = \delta_{2n-4} = 0$. Consider the singular point $(0,0,0)$. We have

$$\mathbb{C}[\widehat{X}_{t,0}] = \frac{\mathbb{C}[[x,y,z]]}{(z^2+xy^2-h_t(x))} \simeq \frac{\mathbb{C}[[x,y,z]]}{(z^2+xy^2-x^{\mathrm{mult}_{h_t}(0)})}, \tag{5.4.22}$$

where the isomorphism follows from that $\frac{h_t(x)}{x^2}$ are invertible in a formal neighborhood of 0. Note that we have $x^2 \mid h_t(x)$ because $\delta_{2n-2} = \delta_{2n-4} = 0$. Thus $\mathrm{mult}_{h_t}(0) \geqslant 2$ (c.f. eq. (5.4.7)). Moreover, we have $g_t(x) = xh_t(x)$ because $\gamma_n = 0$ (c.f. eq. (5.4.8)), thus $\mathrm{mult}_{g_t}(0) = \mathrm{mult}_{h_t}(0) + 1$. If $\mathrm{mult}_{h_t}(0) \geqslant 3$, it is clear that $\widehat{X}_{t,0}$ is of type $D_{\mathrm{mult}_{h_t}(0)+1} = D_{\mathrm{mult}_{g_t}(0)}$, by the defining equation of type D_n Kleinain singularities in eq. (5.2.1). If $\mathrm{mult}_{h_t}(0) = 2$, the singularity $\widehat{X}_{t,0}$ is of type $D_3 = A_3$ (c.f. discussion in Lemma 5.4.11.

□

Recall the Involution-Singularity Triple (IST) (X_λ, ι, p) that we defined in Definition 5.4.1, where ι encodes the type of an anti-Poisson involution θ_λ of X_λ, and p is a singular point of X_λ that is being fixed by θ_λ. Recall the discussion at the beginning of Section 5.4, the goal of this section is to determine the type (Q, ι') of the IST (X_λ, ι, p); that is, to find a pair (Q, ι'), where Q is the type of a Kleinian singularity of type ADE, and ι' is the Roman Numeral encoding the Type of the restriction of the anti-Poisson involution θ_λ to the formal neighborhood $\widehat{X}_{\lambda,p}$ (c.f. Definition 5.4.2).

Let $t = (t_1, \cdots, t_n)$ be a preimage of λ under the quotient $\mathfrak{h} \to \mathfrak{h}/W$. We have $\lambda = (\delta_2, \delta_4, \cdots, \delta_{2n-2}, \gamma) \in \mathfrak{h}/W$, where δ_{2j} is the j-th symmetric polynomial on $(t_1^2, \cdots, t_n^2)$. We write $\lambda = \bar{t}$ to denote that λ is the image of t under the quotient $\mathfrak{h} \to \mathfrak{h}/W$. We have $X_\lambda \simeq X_t = \mathrm{Spec}\,\mathbb{C}[x,y,z]/(z^2 + xy^2 - h_t(z) + 2\gamma_n y)$ and $g_t(x) = \prod_{i=1}^n (x + t_i^2) = xh_t(x) + \gamma_n^2$. Proposition 5.4.12 tells us how to determine the singularity Q in terms of t and p. Proposition 5.4.13 will tell us how to determine the anti-Poisson involution Type ι' in terms of the IST (X_t, ι, p).

Proposition 5.4.13. *Let $X_\lambda = X_t = \mathrm{Spec}\,\mathbb{C}[x,y,z]/(z^2 + xy^2 - h_t(z) + 2\gamma_n y)$ be a filtered deformation of Kleinian singularity of type D_n and $g_t(x) = \prod_{i=1}^n (x + t_i^2)$. Recall from Proposition 5.4.12 that X_t has three types of singularities.*

(1) Consider the singular point $p = (-t_i^2, -\frac{\gamma_n}{t_i^2}, 0)$ with $t_i \neq 0$ and $\mathrm{mult}_{g_t}(-t_i^2) > 1$.

(i) The triple (X_t, I, p) is an IST for any $\lambda = \bar{t} \in \mathfrak{h}/W$. Furthermore, we have $(X_t, \mathrm{I}, p) \sim (A_{\mathrm{mult}_{g_t}(-t_i^2)-1}, \mathrm{I})$.

(ii) The triple (X_t, II, p) is an IST if and only if $\gamma_n(t) = 0$. Furthermore, we have $(X_t, \mathrm{II}, p) \sim (A_{\mathrm{mult}_{g_t}(-t_i^2)-1}, \mathrm{I})$.

(2) Assume $\gamma_n = \delta_{2n-2} = 0,\ \delta_{2n-4} \neq 0$. Consider the singularity $q_\pm = \pm(0, \sqrt{\delta_{2n-4}}, 0)$.

(i) The triple $(X_t, \mathrm{I}, q_\pm)$ is an IST as we assumed $\gamma_n(t) = 0$. Furthermore, we have $(X_t, \mathrm{I}, q_\pm) \sim (A_1, \mathrm{I})$.

(ii) The triples $(X_t, \mathrm{II}, q_\pm)$ are not ISTs because θ_λ swaps $q_\pm$.

(3) Assume $\gamma_n = \delta_{2n-2} = \delta_{2n-4} = 0$. Consider the singularity $(0, 0, 0)$.

(i) The triple $(X_t, \mathrm{I}, 0)$ is an IST for any $\lambda = \bar{t} \in \mathfrak{h}/W$. Furthermore, we have $(X_t, \mathrm{I}, 0) \sim (D_{\mathrm{mult}_{g_t}(0)}, \mathrm{I})$.

(ii) The triple $(X_t, \mathrm{II}, 0)$ is an IST for any t as we assumed $\gamma_n(t) = 0$. Furthermore, we have $(X_t, \mathrm{II}, 0) \sim (D_{\mathrm{mult}_{g_t}(0)}, \mathrm{II})$.

The Roman Numerals in the triples match with those of Proposition 5.3.6, and the Roman Numerals in the pairs match with those of Propositions 3.3.1, 3.3.2, 3.3.5 and 3.3.8.

Proof. (1) Consider the singular point $p = (-t_i^2, -\frac{\gamma_n}{t_i^2}, 0)$ with $t_i \neq 0$ and $\mathrm{mult}_{g_t}(-t_i^2) > 1$.

(i) Type I. $\theta_\lambda \colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda]$ $x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ defines an anti-Poisson involution for any $\lambda \in \mathfrak{h}/W$ by Proposition 5.3.6. Moreover, it is clear that $p = (-t_i^2, -\frac{\gamma_n}{t_i^2}, 0)$ is being fixed by θ_λ. Thus, (X_t, I, p) forms an IST by Definition 5.4.1.

According to eq. (5.4.20) in the proof of Proposition 5.4.12, we have $\mathbb{C}[\widehat{X}_{t,p}] \simeq \mathbb{C}[[x, y, z]]/(z^2 + y^2 - x^{\mathrm{mult}_{g_t}(-t_i^2)})$, and one can compute that the restriction $\widehat{\theta}_{\lambda,p} \colon \mathbb{C}[\widehat{X}_{t,p}] \to \mathbb{C}[\widehat{X}_{t,p}]$ is given by

$$x \mapsto x,\ y \mapsto y,\ z \mapsto -z. \tag{5.4.23}$$

which is the Type I anti-Poisson involution of $A_{\mathrm{mult}_{g_t}(-t_i)-1}$ singularity by Lemma 5.4.10.

(ii) Type II. $\theta_\lambda \colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda]$ $x \mapsto x,\ y \mapsto -y,\ z \mapsto z$ defines an anti-Poisson involution if and only if $\gamma_n(t) = 0$ by Proposition 5.3.6. Note that we have $p = (-t_i^2, -\frac{\gamma_n}{t_i^2}, 0) = (-t_i^2, 0, 0)$ being fixed by θ_λ automatically. Therefore, (X_t, II, p) is an IST if and only if $\gamma_n(t) = 0$.

According to eq. (5.4.20) in the proof of Proposition 5.4.12, we have $\mathbb{C}[\widehat{X}_{t,p}] \simeq \mathbb{C}[[x, y, z]]/(z^2 + y^2 - x^{\mathrm{mult}_{g_t}(-t_i^2)})$, and one can compute that the restriction $\widehat{\theta}_{\lambda,p} \colon \mathbb{C}[\widehat{X}_{t,p}] \to \mathbb{C}[\widehat{X}_{t,p}]$ is given by

$$x \mapsto x,\ y \mapsto -y,\ z \mapsto z. \tag{5.4.24}$$

which is the Type I anti-Poisson involution of $A_{\mathrm{mult}_{g_t}(-t_i)-1}$ singularity by Lemma 5.4.10.

(2) Assume $\gamma_n = \delta_{2n-2} = 0,\ \delta_{2n-4} \neq 0$. Consider the singular point $q_+ = (0, \sqrt{\delta_{2n-4}}, 0)$.

(i) Type I. $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda]$ $x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ defines an anti-Poisson involution for any $\lambda \in \mathfrak{h}/W$ by Proposition 5.3.6. Moreover, it is clear that $q_+ = (0, \sqrt{\delta_{2n-4}}, 0)$ is being fixed by θ_λ. Thus, (X_t, I, q_+) is an IST by Definition 5.4.1.

According to eq. (5.4.21) in the proof of Proposition 5.4.12, we have $\mathbb{C}[\widehat{X}_{t,q_+}] \simeq \mathbb{C}[[x,y,z]]/(z^2+xy)$, and one can compute that the restriction $\widehat{\theta}_{\lambda,q_+}\colon \mathbb{C}[\widehat{X}_{t,q_+}] \to \mathbb{C}[\widehat{X}_{t,q_+}]$ is given by

$$x \mapsto x,\ y \mapsto y,\ z \mapsto -z. \tag{5.4.25}$$

which is the Type II, equivalently Type I, anti-Poisson involution of A_1 singularity by Proposition 3.3.1.

(ii) Type II. $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda]$ $x \mapsto x,\ y \mapsto -y,\ z \mapsto z$ defines an anti-Poisson involution if and only if $\gamma_n(t) = 0$ by Proposition 5.3.6. However, note that $\theta_\lambda(q_+) = q_-$, i.e., q_+ is not being fixed by θ_λ, thus (X_t, II, q_+) is not an IST, according to Definition 5.4.1.

One can prove the similar result for the singular point $q_- = (0, -\sqrt{\delta_{2n-4}}, 0)$, and we leave it to the reader.

(3) Assume $\gamma_n = \delta_{2n-2} = \delta_{2n-4} = 0$. Consider the singular point $(0,0,0)$.

(i) Type I. $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda]$ $x \mapsto x,\ y \mapsto y,\ z \mapsto -z$ defines an anti-Poisson involution for any $\lambda \in \mathfrak{h}/W$ by Proposition 5.3.6. Moreover, it is clear that $(0,0,0)$ is being fixed by θ_λ. Thus, $(X_t, \mathrm{I}, 0)$ is an IST by Definition 5.4.1.

According to eq. (5.4.22) in the proof of Proposition 5.4.12, we have $\mathbb{C}[\widehat{X}_{t,0}] \simeq \mathbb{C}[[x,y,z]]/((z^2+xy^2 - x^{\mathrm{mult}_{g_t}(0)-1}))$, is the Kleinian singularity of type $D_{\mathrm{mult}_{g_t}(0)}$. One can compute that the restriction $\widehat{\theta}_{\lambda,0}\colon \mathbb{C}[\widehat{X}_{t,0}] \to \mathbb{C}[\widehat{X}_{t,0}]$ is given by

$$x \mapsto x,\ y \mapsto y,\ z \mapsto -z. \tag{5.4.26}$$

which is the Type I anti-Poisson involution of $D_{\mathrm{mult}_{g_t}(0)}$ singularity by Propositions 3.3.5 and 3.3.8.

(ii) Type II. $\theta_\lambda\colon \mathbb{C}[X_\lambda] \to \mathbb{C}[X_\lambda]$ $x \mapsto x,\ y \mapsto -y,\ z \mapsto z$ defines an anti-Poisson involution if and only if $\gamma_n(t) = 0$ by Proposition 5.3.6. Moreover, it is clear that $(0,0,0)$ is being fixed by θ_λ. Thus, $(X_t, \mathrm{II}, 0)$ is an IST by Definition 5.4.1.

According to eq. (5.4.22) in the proof of Proposition 5.4.12, we have $\mathbb{C}[\widehat{X}_{t,0}] \simeq \mathbb{C}[[x,y,z]]/((z^2+xy^2 - x^{\mathrm{mult}_{g_t}(0)-1}))$, is the Kleinian singularity of type $D_{\mathrm{mult}_{g_t}(0)}$. One can compute that the restriction $\widehat{\theta}_{\lambda,0}\colon \mathbb{C}[\widehat{X}_{t,0}] \to \mathbb{C}[\widehat{X}_{t,0}]$ is given by

$$x \mapsto x,\ y \mapsto -y,\ z \mapsto z. \tag{5.4.27}$$

which is the Type II anti-Poisson involution of $D_{\mathrm{mult}_{g_t}(0)}$ singularity by Propositions 3.3.5 and 3.3.8.

$\square$

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